QUANTUM GEOMETRY OF MIRROR LANDAU-GINZBURG MODEL OF $(\mathbb{P}^2, E)$

PIERRICK BOUSSEAU, BERND SIEBERT, AND WANG YAO

Abstract. We prove the quantum closed string mirror symmetry conjecture of $K_{\mathbb{P}^2}$ proposed by Aganagic-Cheng-Dijkgraaf-Krefl-Vafa [ACDKV] about the equivalence between the higher-genus Nekrasov-Shatashvili limit of the refined Gromov-Witten theory and the quantum periods from the WKB approximation of the quantum mirror curve equation. Our proof is based on the intrinsic definition of quantum periods (Definition 3.13) via the deformation quantization theory developed by Kashiwara, Polesello, Schapira and D'Agnolo, and the Gross-Siebert construction of the proper Landau-Ginzburg model in [CPS]. Owing to the generality of these theories, our method can be directly generalized to all local toric del Pezzo surfaces. In addition, via quantum wall crossing of wavefunctions, we reformulate the quantum open string mirror symmetry conjecture of $K_{\mathbb{P}^2}$, or more generally of local toric del Pezzo surfaces, as the Punctured-Open Correspondence (Conjecture 7.9) in enumerative geometry.

Contents

1. Introduction 2
1.1. Background and Statements of Quantum Mirror Symmetry 2
1.2. Main Result 4
1.3. Methodology and Sketch of the Proof 5
1.4. Intrinsic Enumerative Mirror Symmetry 6
1.5. Outline of the Paper 7
1.6. Conventions and Notations 8
Acknowledgment 9
Declaration of Generative AI Use 9
2. Deformation Quantization Algebroids and Deformation Quantization Modules 9
2.1. Deformation Quantization of Holomorphic Symplectic Manifolds 9
2.2. Deformation Quantization of Holomorphic Lagrangian Submanifolds 15
2.3. Classification of Deformation Quantization Pairs and Relative Deligne-Fedosov Class 22
3. WKB Approximation, Quantum Periods and Deformation Quantization 25
3.1. WKB Approximation and the Original Definition of Quantum Periods 25
3.2. Deformation Quantization and the Intrinsic Definition of Quantum Periods 28
4. Dimensional Reduction of Local Mirror Symmetry: Takahashi's Log Mirror Symmetry 36
4.1. Local Mirror Symmetry of Toric Calabi-Yau Threefolds 36
4.2. Takahashi's Log Mirror Symmetry 37
5. Classical and Quantum Geometry of Mirror Landau-Ginzburg Model of Toric Degeneration of $(\mathbb{P}^2, E)$ 39
5.1. Classical Geometry of Mirror Landau-Ginzburg Model 39
5.2. Deformation Quantization of Mirror Landau-Ginzburg Model 43
5.3. Formal Quantum Geometry of the Asymptotic Chambers 49

Date: October 31, 2026.

6. Quantum Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$ 51
6.1. Computation of the Quantum Correction 1-Periods 51
6.2. Review of Classical Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$ 62
6.3. Proof of Quantum Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$ 64
7. Remarks on Quantum Open String Mirror Symmetry of $K_{\mathbb{P}^2}$ 66
7.1. Sheaf of WKB Wavefunctions and the Local Representations 66
7.2. Additional Data in the Open String Theory and Quantum Open String Mirror Symmetry 66
7.3. Quantum Flat Coordinates and Quantum Mirror Curve Under the Quantum Flat Coordinates 67
7.4. Hori-Vafa Picture and Gross-Siebert Picture of the Mirror Curve 68
7.5. Quantum Wall Crossing of Wavefunctions and Punctured-Open Correspondence 70
Appendix A. Categorical Characterization of Canonical Deformation Quantization 74
A.1. Categorical Characterization of Canonical DQ Algebroids 74
A.2. Categorical Characterization of Canonical DQ Pairs 78
References 80

1. Introduction

1.1. **Background and Statements of Quantum Mirror Symmetry.** Enumerative mirror symmetry relates curve counting invariants on the A-model side and period integrals on the mirror B-model side. For toric Calabi-Yau threefolds, this correspondence is known as *local mirror symmetry*. Its mathematical formulation was initiated by Chiang–Klemm–Yau–Zaslow [CKYZ], while a systematic physical framework was developed by Hori–Vafa [HV].

The classical, or genus zero, local mirror symmetry conjecture is an equivalence between the local genus zero Gromov-Witten theory of toric Calabi-Yau threefolds and the period integrals of the Calabi-Yau 3-forms on their mirror manifolds. Classical local mirror symmetry has been established in considerable generality, including a description of genus zero open Gromov–Witten invariants [FL].

At higher genus, several distinct formulations of local mirror symmetry arise. For ordinary higher-genus Gromov–Witten theory, the corresponding B-model is described by the Eynard–Orantin topological recursion [EO1]. This correspondence, known as the *remodeling conjecture* [BKMP], has likewise been established in broad generality [EO2; FLZ]; see also [FLYZ] for recent developments including Gromov–Witten invariants with descendants.

In contrast, the quantum mirror symmetry conjecture of Aganagic–Cheng–Dijkgraaf–Krefl–Vafa [ACDKV], which concerns the Nekrasov–Shatashvili (NS) limit of refined topological string theory, remains much less understood mathematically. Refined topological string theory depends on two parameters ϵ_1 and ϵ_2. The unrefined topological string theory, encoding ordinary higher-genus Gromov–Witten invariants, is recovered by specializing to $\epsilon_1 = \epsilon_2 = g_s$, whereas the NS limit is obtained by setting $\epsilon_1 = 0$ and $\epsilon_2 = \hbar$.

Mathematically, several conjecturally equivalent formulations of refined topological string theory and its NS limit have been proposed. These include a sheaf-theoretic approach based on refined stable pair invariants [CKK] and a formulation in terms of refined Gopakumar–Vafa invariants defined via moduli spaces of one-dimensional sheaves [MT]. For local del Pezzo surfaces, an alternative description of the NS limit of the refined topological string was proposed in terms of higher-genus relative/logarithmic Gromov–Witten invariants of the underlying del Pezzo surface relative to a smooth anticanonical divisor [Bo4]. Brini–Schuler [BS] subsequently extended this Gromov–Witten-theoretic approach to the full refined topological string theory of arbitrary toric Calabi–Yau threefolds, using the $(\mathbb{C}^\star)^2$-equivariant Gromov–Witten

theory of the Calabi–Yau fivefold $X \times \mathbb{C}^2$. Throughout this paper, we adopt this higher-genus relative Gromov–Witten-theoretic definition of the NS limit. In the case of the local projective plane, which will be our main focus, its equivalence with the formulation in terms of moduli spaces of one-dimensional sheaves follows from [Bo4].

On the B-model side, quantum mirror symmetry is formulated in terms of "quantum curves". The classical mirror curve of a toric Calabi–Yau threefold is defined by

$$C := \{(X, P) \in (\mathbb{C}^*)^2 \mid -1 + W(X, P) = 0\},$$

where W is a Laurent polynomial known as the Landau–Ginzburg superpotential. The classical B-model is described by period integrals of the multivalued 1-form $\log P \, d \log X$ along 1-cycles on C. To formulate quantum mirror symmetry, one quantizes the mirror curve, obtaining the quantum curve equation

$$\big(-1 + \widehat{W}(e^x, e^{\hbar \partial_x})\big)\psi(x) = 0,$$

where $\widehat{W}$ denotes the quantization of W, and the quantum operators $\widehat{X}$ and $\widehat{P}$ act on functions of x through

$$\widehat{X} = e^x, \qquad \widehat{P} = e^{\hbar \partial_x}.$$

The WKB approximation seeks a formal solution of the form

$$\psi(x) = \exp\left(\frac{1}{\hbar}\int^x \lambda_\hbar\right),$$
$$\lambda_\hbar = \sum_{i \geq 0} \lambda_i \hbar^i,$$

where the λ_i are 1-forms on the mirror curve C. Substituting this ansatz into the quantum curve equation determines the coefficients λ_i recursively, with leading term

$$\lambda_0 = \log P \, d \log X.$$

Thus, $\lambda_\hbar$ provides a formal deformation of the classical 1-form $\log P \, d \log X$. Its period integrals, known as "quantum periods", are formal deformations of the classical periods. The quantum open and closed string mirror symmetry conjectures proposed in [ACDKV] can then be formulated as follows.

Conjecture 1.1 (**Conjecture 4.2**) (Quantum Closed String Mirror Symmetry [ACDKV])**.** *For a toric Calabi–Yau threefold, the partition function of the NS limit of the refined closed topological string theory coincides, under the quantum closed mirror map, with the B-model partition function determined by the quantum special geometry relation for the quantum periods of the mirror quantum curve.*

Conjecture 1.2 (**Conjecture 7.1**) (Quantum Open String Mirror Symmetry [ACDKV])**.** *For a toric Calabi–Yau threefold, the partition function of the NS limit of the refined open topological string theory coincides, under the quantum open-closed mirror map, with the WKB wavefunction associated with the mirror quantum curve.*

In contrast to classical local mirror symmetry and the remodeling conjecture, quantum mirror symmetry remains largely open. Mathematical proofs are currently available only in a few simple cases, such as $\mathbb{C}^3$ and the resolved conifold $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, where the WKB approximation can be computed explicitly in closed form. While classical periods can be studied through Picard–Fuchs differential equations and the Eynard–Orantin topological recursion provides a recursive description of the B-model invariants appearing in the remodeling conjecture, the quantum periods arising from the WKB expansion are substantially more difficult to analyze. In general, the WKB expansion is only defined recursively as a formal

power series in $\hbar$, with no closed-form expression available. This makes the direct comparison of quantum periods with higher-genus Gromov–Witten invariants particularly challenging.

1.2. **Main Result.** Our main result is a proof of the quantum closed string mirror symmetry conjecture, Conjecture 1.1, for the local projective plane $K_{\mathbb{P}^2}$, namely, the total space of the canonical line bundle over $\mathbb{P}^2$. This is the simplest toric Calabi–Yau threefold containing a compact divisor. Unlike the cases of $\mathbb{C}^3$ and the resolved conifold, no closed-form expressions are known for the quantum periods or the NS limit of the refined topological string. Establishing the conjecture in this setting is therefore a highly non-trivial result.

We introduce some notation to state an explicit form of the result. For every $d \geq 1$ and $g \geq 0$, let N_d^g be the Gromov–Witten invariant of $\mathbb{P}^2$, relative to a smooth elliptic curve, with a single marked point with maximal tangency and insertion of the top lambda class λ_g – we refer to §4.2 for details. The generating series of the relative Gromov–Witten invariants N_d^g encodes the NS limit of the refined topological string on $K_{\mathbb{P}^2}$ [Bo4; BS].

Our main result expresses this generating series in terms of the quantum periods of the quantum mirror curve of $K_{\mathbb{P}^2}$. The classical mirror curve of $K_{\mathbb{P}^2}$ is the algebraic curve C defined by

$$C := \{(X,P) \in (\mathbb{C}^*)^2 \mid -1 + T\left(X + P + \frac{1}{XP}\right) = 0\},$$

where T is a complex-structure parameter. Its quantization produces the q-difference equation

$$(-1 + Te^x)\,\psi(x) + T\psi(x+\hbar) + Te^{\hbar/2}e^{-x}\psi(x-\hbar) = 0\,.$$

We consider a specific choice of A and B-cycles on the classical mirror curve, see Figure 11, and define the corresponding quantum periods $\Pi_A^\hbar(T)$ and $\Pi_B^\hbar(T)$ using the WKB approximation for this q-difference equation.

Theorem 1.3 (**Theorem 6.13**). *The quantum closed string mirror symmetry, that is, Conjecture 1.1, holds for the local projective plane $K_{\mathbb{P}^2}$. Explicitly, define the quantum closed string flat coordinate*

$$Q^\hbar(T) = \exp(\Pi_A^\hbar(T))\,.$$

Then, expressing $\Pi_B^\hbar(T)$ in terms of $Q^\hbar$, we have

$$\Pi_B^\hbar(T) = \frac{1}{2}\big(\log(-Q^\hbar)\big)^2 - 3\zeta(2) + \frac{\hbar^2}{8} + \sum_{d\geq 1}\sum_{g\geq 0}(-1)^g N_d^g(-Q^\hbar)^d\hbar^{2g}\,.$$

We prove the above theorem by using the deformation quantization theory developed by Kashiwara, Polesello, Schapira and D'Agnolo and the Gross-Siebert construction of the mirror proper Landau-Ginzburg model. Owing to the generality of these theories, our method should generalize to other local toric del Pezzo surfaces. Different from many previous work on this subject, our method is indeed conceptual rather than computational and reply on deeper understanding of the quantum geometry in the B-model.

In addition, using quantum wall crossing of wavefunctions, we reformulate the quantum open string mirror symmetry conjecture as the following Punctured-Open Correspondence conjecture in enumerative geometry.

Conjecture 1.4 (**Conjecture 7.9**) (Punctured-Open Correspondence). *Up to any finite T-order, the partition function of the NS-limit of refined open topological string theory A-model on $K_{\mathbb{P}^2}$ (more generally, on the canonical bundles of toric del Pezzo surfaces) with an internal Aganagic-Vafa A-brane is the same as*

$$\mathcal{P}\prod_{\mathfrak{d}}\exp(\epsilon_{\mathfrak{d}}\hat{H}_{\mathfrak{d}})\psi_\infty,\ \ \psi_\infty = 1,$$

where $\mathcal{P}\prod_{\mathfrak{d}}$ means the path-ordered product and $\hat{H}_{\mathfrak{d}}$ are the quantum Hamiltonians attached to the quantum walls. The wall crossing path is from the asymptotic chamber ∞ to the Hori-Vafa central chamber σ_0 in the quantum wall structure of the mirror Landau-Ginzburg model of the toric degeneration of $(\mathbb{P}^2, E)$ (more generally a toric del Pezzo surface together with a smooth anticanonical divisor). The asymptotic wavefunction ψ_∞ is just the constant function 1.

1.3. **Methodology and Sketch of the Proof.** To prove quantum mirror symmetry (Theorem 6.13), we need to understand mathematically both the A-model and the B-model of the NS-limit of refined topological string theory. A mathematical understanding of the A-model was obtained in the first author's paper [Bo4]. In short, the NS-limit of refined closed topological string theory A-model of $K_{\mathbb{P}^2}$ is the same as the logarithmic Gromov-Witten theory of $(\mathbb{P}^2, E)$ with a λ_g-class insertion. Logarithmic Gromov-Witten theory was developed by Abramovich-Chen [AC] and Gross-Siebert [GS3], which generalizes the relative Gromov-Witten theory developed by Jun Li [Li1; Li2]. This logarithmic interpretation motivates us to use the Gross-Siebert mirror construction of toric degenerations in [GS2; CPS; GS5] to study the quantum mirror symmetry conjecture.

For the mathematical understanding of the quantum B-model, we use the deformation quantization theory developed by Kashiwara, Polesello, Schapira and D'Agnolo. Based on the deformation quantization theory, we give an intrinsic interpretation of the quantum periods, which is crucial to our proof of the quantum mirror symmetry conjecture using the Gross-Siebert mirror construction.

Our proof is sketched as follows; see also the schematic in Figure 1.

(1) With the help of deformation quantization theory, we give an intrinsic definition of the quantum periods $\mathrm{QC}(\mathcal{M}_L)$ (Definition 3.13) of a simple holonomic DQ module $\mathcal{M}_L$ over the canonical DQ algebroid $\mathcal{A}_X$ supported on a holomorphic Lagrangian L. This is given by a lift of the relative Deligne-Fedosov class $\mathfrak{D}(\mathcal{A}_X, \mathcal{M}_L)$ (Definition 2.34), which classifies the pairs of DQ algebroids and simple holonomic DQ modules. The intrinsically defined quantum period $\mathrm{QC}(\mathcal{M}_L)$ also coincides with the monodromy of the Hom sheaf

$$\mathcal{H}om_{\mathcal{A}_X}(\mathcal{M}_L, \mathcal{M}_L^c)$$

between $\mathcal{M}_L$ and a quasi-canonical DQ module $\mathcal{M}_L^c$ (Definition 2.31).

(2) On the Hori-Vafa mirror Landau-Ginzburg model, let the DQ algebroid $\mathcal{A}_X$ be the Hori-Vafa quantum torus and consider the following Landau-Ginzburg DQ module

$$\mathcal{M}_{LG} := \mathcal{A}_X/\mathcal{A}_X(-1+\widehat{W}),$$

where $\widehat{W}$ is the quantum superpotential. The Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ is a simple holonomic DQ module supported on the mirror curve defined by $-1+W=0$. Let γ be a cycle on the mirror curve that is contractible on the ambient torus. Then we show in Theorem 3.15 that the pairing $<\mathrm{QC}(\mathcal{M}_{LG}), \gamma>$ of the intrinsically defined quantum period $\mathrm{QC}(\mathcal{M}_{LG})$ with γ is exactly the same as the monodromy of the WKB approximation along γ.

(3) Via the Gross-Siebert construction of the mirror Landau-Ginzburg model of the toric degeneration of $(\mathbb{P}^2, E)$ in [CPS] and its deformation quantization given by the quantum wall structure constructed in the first author's papers [Bo3; Bo2; Bo4], we obtain a partial compactification of the Hori-Vafa Landau-Ginzburg model and its deformation quantization, on which the superpotential W is proper and the mirror curve is a compact Riemann surface of genus 1.

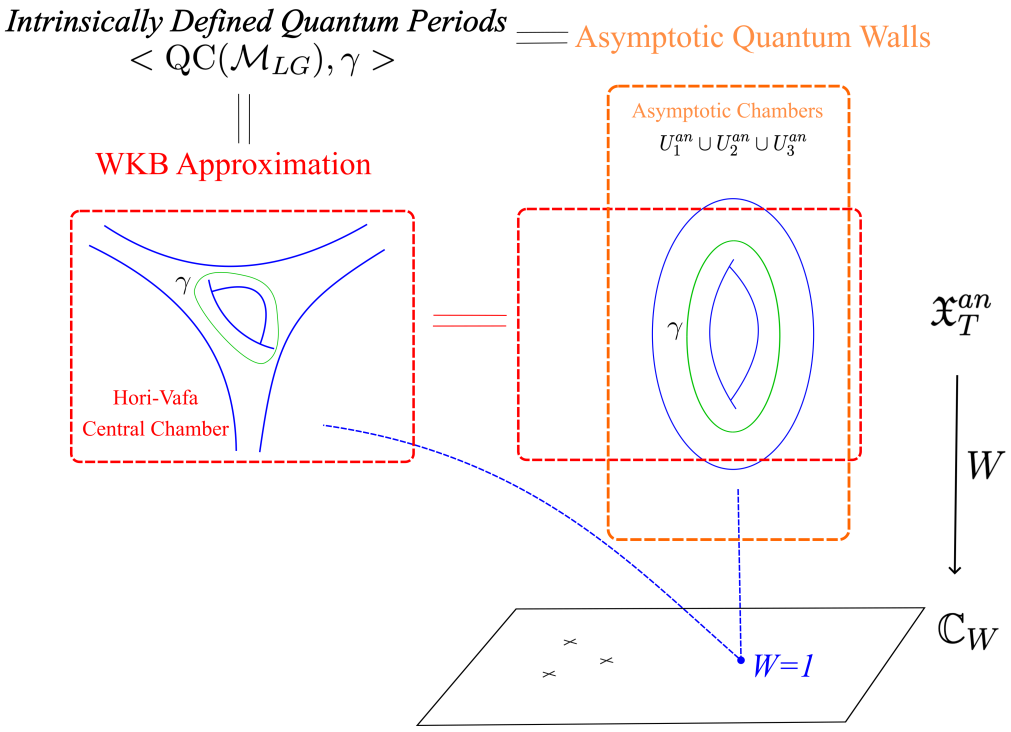


FIGURE 1. Sketch of the Proof of Quantum Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$. The key idea is to compute the pairing $< \mathrm{QC}(\mathcal{M}_{LG}), \gamma >$ of the intrinsically defined quantum period $\mathrm{QC}(\mathcal{M}_{LG})$ (Definition 3.13) with the cycle γ (green) on the mirror curve in different coordinates. More specifically, we compute the pairing $< \mathrm{QC}(\mathcal{M}_{LG}), \gamma >$ in the coordinates given by both the Hori-Vafa central chamber and the asymptotic chambers. On the Hori-Vafa central chamber (red), the restriction of the mirror curve is a genus 1 Riemann surface with three punctures, and the pairing is given by the WKB approximation by Theorem 3.15, while on the asymptotic chambers (orange), the pairing is given by the quantum Hamiltonians of the asymptotic quantum walls, whose enumerative meaning gives the desired NS-limit of refined topological string A-model.

(4) Owing to the intrinsic nature of the quantum period $\mathrm{QC}(\mathcal{M}_{LG})$, we are able to switch the computation of the pairing $< \mathrm{QC}(\mathcal{M}_{LG}), \gamma >$ to the coordinates given by the asymptotic chambers in the quantum wall structure in the Gross-Siebert program. Since the quantum Hamiltonians of the asymptotic quantum walls have the enumerative meaning of the NS-limit of refined closed topological string theory of local $\mathbb{P}^2$ by the first author's paper [Bo4], we obtain a clear enumerative meaning of the quantum period when computing in the asymptotic chambers, which gives a proof of the quantum closed string mirror symmetry conjecture (Conjecture 4.2).

1.4. **Intrinsic Enumerative Mirror Symmetry.** This paper fits into the large program of *Intrinsic Enumerative Mirror Symmetry*, which aims to establish enumerative mirror symmetry in great generality using the Gross-Siebert program (see Figure 2). The key ingredient is given by the tropicalization data of the A-side. On the one hand, the tropicalization encodes the enumerative information of the A-side in the wall structure via various tropical correspondences. On the other hand, the tropicalization also gives the gluing datum of the mirror

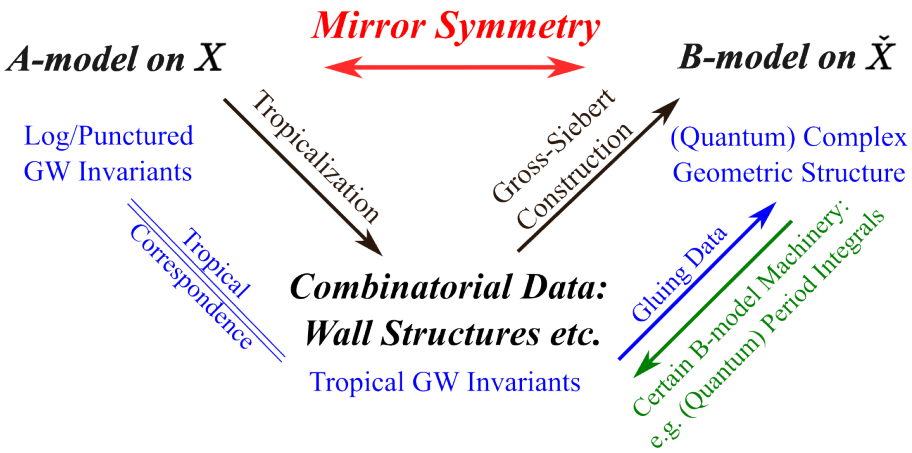


FIGURE 2. A "Mirror Triangle" Revealing the Mechanism of Enumerative Mirror Symmetry Via the Gross-Siebert Program. Under the machinery of the Gross-Siebert program, proving enumerative mirror symmetry amounts to establishing the green arrow, which is indeed what we do in proving the quantum closed string mirror symmetry (Conjecture 4.2).

B-side via the Gross-Siebert construction, and the information of the wall structure enters the complex structure of the B-side. Proving enumerative mirror symmetry then amounts to recovering and extracting the information of the wall structure, which is the same as the enumerative information of the A-side, by means of B-model machinery such as the period integrals and the WKB approximations. The second author's papers with Michel van Garrel and Helge Ruddat [GRS1; GS1] and the in-progress project [GRS2] are examples of this program.

In fact, the second author's paper [GRS1] reproves the classical local mirror symmetry of local $\mathbb{P}^2$ using the idea of Intrinsic Enumerative Mirror Symmetry. Our paper should be considered as a quantum enhancement of the classical case in [GRS1]. Unlike many other works proving enumerative mirror symmetry, our proof does not rely heavily on A-model computations, but rather on a deeper understanding of the quantum B-model. Once an intrinsic understanding of the quantum periods is established, enumerative mirror symmetry becomes very transparent via the Gross-Siebert program, which demonstrates the power of this program in proving enumerative mirror symmetry in great generality.

1.5. **Outline of the Paper.** In §2, we will first review the theory of deformation quantization developed by Kashiwara et al., especially the construction of the canonical DQ algebroid and the canonical DQ pair. After that, we will introduce a relative version of the Deligne-Fedosov class (Definition 2.34), which classifies the DQ pairs consisting of DQ algebroids and simple holonomic DQ modules.

In §3, we will review the WKB approximation of difference equations and give an intrinsic definition of the quantum periods (Definition 3.13). After that, we will show in Theorem 3.15 that the intrinsic definition of the quantum periods is exactly the same as the original definition from the WKB approximation.

In §4, we will review Takahashi's log mirror symmetry, which gives a dimensional reduction of the local mirror symmetry from 3d to 2d. This motivates the use of the Gross-Siebert construction of the mirror Landau-Ginzburg model of the toric degeneration of $(\mathbb{P}^2, E)$ to prove the quantum mirror symmetry of local $\mathbb{P}^2$.

In §5, we will review the Gross-Siebert construction of the mirror Landau-Ginzburg model in [CPS] and its deformation quantization given by the first author's papers [Bo2],[Bo4]. We will emphasize the geometry given by the asymptotic chambers since they play a key role in proving the quantum mirror symmetry.

In §6, we will give the mathematical proof of the quantum closed string mirror symmetry conjecture 4.2 proposed in [ACDKV]. We will first compute the intrinsically defined quantum periods (Definition 3.13) in the asymptotic chambers, which agree with those obtained from the WKB approximation by Theorem 3.15. Then, combining this with the results on classical mirror symmetry of local $\mathbb{P}^2$ shown in [GRS1], we prove quantum closed string mirror symmetry (Theorem 6.13).

In §7, we will discuss how to use the Gross-Siebert program to study the quantum open string mirror symmetry conjecture (Conjecture 7.1). We will study the quantum open-closed flat coordinates and describe the geometry under the quantum open-closed flat coordinates, which is the same as the Gross-Siebert picture of the quantum mirror curve. After that, via quantum wall crossing of the wavefunction, we reformulate quantum open mirror symmetry in terms of the Punctured-Open Correspondence conjecture (Conjecture 7.9) in enumerative geometry.

1.6. **Conventions and Notations.**

- In our paper, the analytification of the formal toric degeneration $\mathfrak{X}$ means a complex analytic degeneration $\mathfrak{X}^{an}$ of complex analytic spaces whose completion along the central fiber is isomorphic to the formal degeneration $\mathfrak{X}$. One should not confuse the analytification in our paper with the Berkovich analytification, which is an important notion in the non-Archimedean methods of mirror symmetry.
- In our paper, the word "formal" always refers to formal power series in the complex structure modulus T. Formal power series in the reduced Planck constant $\hbar$, which come from deformation quantization, are implicit in the word "deformation". We never consider convergence with respect to $\hbar$, which is known as resurgence in the literature.
- The reduced Planck constant $\hbar$ used in our paper should be regarded as $-i\hbar_{phy}$, where $\hbar_{phy}$ is the reduced Planck constant in the standard physics literature of quantum mechanics. The noncommutativity parameter of the quantum torus is given by $q = \exp(-\hbar) = \exp(i\hbar_{phy})$.
- As is standard in quantum mechanics, we use a hat $\widehat{\cdot}$ to denote the operator corresponding to a classical coordinate. For example, $\widehat{X}$ and $\widehat{P}$ are the quantum operators associated with X and P.
- Except in Appendix A, canonicity of a deformation quantization means the canonical isomorphism class of a certain deformation quantization structure, i.e. a special isomorphism class of deformation quantizations that does not depend on the choices made in the construction. The reason why we only consider this weak notion of canonicity is that we only want to extract certain intrinsic invariants, such as the Deligne-Fedosov classes, from certain deformation quantization structures. The canonical objects will have an upper index c in this paper. We will discuss the categorical characterization of the canonical deformation quantization Appendix A.
- The quantum periods in our paper refer to the quantum periods from WKB approximation considered in [ACDKV], which are different from the regularized quantum periods in [CCGGK]. The regularized quantum periods in [CCGGK] concern only classical genus-zero mirror symmetry and are in fact the logarithmic derivatives of the classical open mirror maps. There is also a q-refined version of these, which corresponds to the quantum open mirror map. See [GRZZ] for more details.

- The groupoid of contact manifolds considered in this paper is the groupoid of strict contact manifolds for the reason explained in Remark 2.9.

List of Notations and Definitions

- $\widehat{\mathcal{E}}_M^{\sqrt{v}}$: The sheaf of half-twisted formal microdifferentials of a complex manifold M.
- $\mathbb{C}_\hbar$: The algebra of formal power series $\mathbb{C}[\![\hbar]\!]$ with variable $\hbar$ and $\mathbb{C}$-coefficients.
- $\mathbb{C}_{w_2(M)}$: The gerbe of spin structures on M.
- $\mathcal{A}_X$: A DQ algebra (2.4) or DQ algebroid (2.5) on a holomorphic symplectic manifold X.
- $\mathcal{A}_X^c$: The canonical DQ algebroid (2.10) on a holomorphic symplectic manifold X.
- $\mathcal{M}_L$: A DQ module (2.14) supported on a holomorphic Lagrangian submanifold L.
- $\mathcal{M}_L^c$: A quasi-canonical DQ module (2.31) supported on a holomorphic Lagrangian submanifold L.
- $(\mathcal{A}_X^c, \mathcal{M}_L^c)$: The canonical DQ pair (2.27) of a holomorphic symplectic manifold X and a holomorphic Lagrangian submanifold L.
- $\mathfrak{D}(\mathcal{A}_X)$: The Deligne-Fedosov class (2.12) of the DQ algebroid $\mathcal{A}_X$.
- $\mathfrak{D}(\mathcal{A}_X, \mathcal{M}_L)$: The relative Deligne-Fedosov class (2.34) of the DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$.
- $\mathrm{QC}(\mathcal{M})$: The quantum correction 1-period (3.13) of the DQ module $\mathcal{M}$ over the canonical DQ algebroid.
- $\mathcal{M}_{LG}$: The Landau-Ginzburg DQ module defined by a quantum superpotential $\widehat{W}$.
- $\mathfrak{X}$: The formal toric degeneration constructed in the Gross-Siebert program [CPS].
- $\mathfrak{C}$: The formal toric degeneration of the mirror curve defined by the fiber of the proper superpotential W in [CPS].
- $\mathfrak{X}_T^{an}, \mathfrak{C}_T^{an}$: The generic fiber of the analytification $\mathfrak{X}^{an}, \mathfrak{C}^{an}$ of the formal toric degeneration $\mathfrak{X}, \mathfrak{C}$ at certain $T \neq 0$.
- $(B, \mathscr{P}, \varphi)$: The combinatorial data of tropicalization, where B is an affine manifold and topologically plays the role of the base of the SYZ fibration, and $\mathscr{P}$ is a polyhedral decomposition and φ is a multi-valued convex piecewise affine function.
- $\mathscr{S}, \mathscr{S}^\hbar, \widetilde{\mathscr{S}^\hbar}$: The classical wall structure, the quantum wall structure and the modified quantum wall structure (§6.1.1).

Acknowledgment. This paper is part of the PhD thesis of the third author at the University of Texas at Austin. We thank David Ben-Zvi, Sam Gunningham and Pavel Safronov for helpful discussions and for pointing out references on deformation quantization. We also thank Michel van Garrel for helpful discussions on [GRS1] and for his interest in our paper. Part of this paper was completed during the third author's visit to DPMMS at University of Cambridge, and the third author would like to express his gratitude to Mark Gross for his warm hospitality and helpful discussions. We also thank Diego Matessi, Helge Ruddat, Xuanchun Lu and Eric Zaslow for the opportunities to present this work and for helpful discussions.

Declaration of Generative AI Use. We only used Claude Opus 5.5 and ChatGPT 6.1 Sol to check the grammatical mistakes, typos and polish the language. All mathematical contents presented in this paper are the product of the human authors' own intellectual efforts, without the assistance of generative AI.

2. Deformation Quantization Algebroids and Deformation Quantization Modules

2.1. **Deformation Quantization of Holomorphic Symplectic Manifolds.** In this section, we will review the deformation quantization theory of holomorphic symplectic manifolds developed by Kashiwara, Polesello, Schapira and D'Agnolo. We recommend the book [KS]

for more details on this topic. The central problem of deformation quantization is to construct a deformation quantization (DQ) algebra $\mathcal{A}_X$ on any holomorphic symplectic manifold (X, ω), i.e. a sheaf of noncommutative algebras that is locally isomorphic to the canonical deformation quantization algebra $\mathcal{A}_{T^*M}$ of a cotangent bundle T^*M. However, for general holomorphic symplectic manifolds, a DQ algebra may not exist. Instead, one should consider categorifications of sheaves of algebras, which are called algebroids in the literature [KS]. First, let us briefly review the definitions of algebroids and DQ algebroids following [KS]. Throughout this section, we work in the complex analytic setting and every space is a complex manifold. Algebroids are just the once-categorifications of sheaves of algebras.

Definition 2.1 (Algebroids)**.** Let R be a commutative unital ring. An *R-algebroid* is a stack of R-linear categories such that it is locally non-empty and local objects are locally isomorphic.

For any R-algebra A, we can associate an R-linear category A^+ with a single object whose endomorphism algebra is A. However, for a sheaf of algebras $\mathcal{A}$, this construction only gives a presheaf of R-linear categories, and we need to stackify it. Let us denote the stackification by $\mathcal{A}^+$, which is called the associated algebroid of the sheaf of algebras $\mathcal{A}$. More explicitly, the section of $\mathcal{A}^+$ is just the category of right invertible $\mathcal{A}$-modules. Right invertible $\mathcal{A}$-modules are those right $\mathcal{A}$-modules which are locally isomorphic to $\mathcal{A}$ by definition. It is also worth pointing out that the isomorphism of associated algebroids is weaker than the isomorphism of sheaves of algebras in general. Please refer to [KS] for more details.

In this paper, we will mostly use the gluing datum of an R-algebroid. For a Hausdorff paracompact topological space, an R-algebroid $\mathcal{A}$ can be given by the following gluing datum on an open cover $\{U_i\}$

$$(\mathcal{A}_i, \Psi_{ij} : \mathcal{A}_j|_{U_{ij}} \longrightarrow \mathcal{A}_i|_{U_{ij}}, a_{ijk} \in \Gamma(U_{ijk}, \mathcal{A}_i^*)), \tag{2.2}$$

where $\mathcal{A}_i$ is a sheaf of R-algebras on U_i, $\Psi_{ij} : \mathcal{A}_j|_{U_{ij}} \longrightarrow \mathcal{A}_i|_{U_{ij}}$ is an isomorphism of sheaves of R-algebras, and $a_{ijk} \in \Gamma(U_{ijk}, \mathcal{A}_i^*)$ is an invertible section on the triple intersection U_{ijk} corresponding to the natural transformations, or 2-morphisms, in the gluing datum. The gluing datum (2.2) above must satisfy the following consistency conditions

$$\Psi_{ij} \circ \Psi_{jk} \circ \Psi_{ki} = \mathrm{Ad}_{a_{ijk}} \text{ on } U_{ijk}$$
$$a_{ijk} a_{ikl} = \Psi_{ij}(a_{jkl}) a_{ijl} \text{ on } U_{ijkl}.$$

In the context of deformation quantization, the base ring R is usually taken to be the algebra of formal power series $\mathbb{C}_\hbar := \mathbb{C}[\![\hbar]\!]$. On a complex manifold X, the local model of deformation quantization is given by the star-products, or Moyal-products, defined below.

Definition 2.3 (Star-Algebras)**.** A $\mathbb{C}_\hbar$-bilinear associative product $\star$ on $\mathcal{O}_X[\![\hbar]\!]$ is called a *star-product* if the product of $f, g \in \mathcal{O}_X$ can be expressed as

$$f \star g = fg + \sum_{i \geq 1} Q_i(f, g)\hbar^i,$$

where Q_i are bidifferential operators, i.e. differential operators with respect to both variables, and satisfy

$$Q_i(f, 1) = Q_i(1, f) = 0, \ \forall f \in \mathcal{O}_X.$$

$(\mathcal{O}_X[\![\hbar]\!], \star)$ is called a *star-algebra*.

It is clear that the constant function 1 is the unit element in a star-algebra, and every star algebra $(\mathcal{O}_X[\![\hbar]\!], \star)$ is $\hbar$-adic complete and flat over $\mathbb{C}_\hbar$. Also, every star algebra is equipped with a classical limit homomorphism $\mathcal{O}_X[\![\hbar]\!] \longrightarrow \mathcal{O}_X$ by taking the quotient by $\hbar$. The leading order of the commutator of any star-algebra defines a Poisson bracket $\{-, -\}$ on $\mathcal{O}_X$.

Hence a star algebra $(\mathcal{O}_X[\![\hbar]\!], \star)$ gives a quantization of the corresponding Poisson manifold $(\mathcal{O}_X, \{-,-\})$.

Definition 2.4 (Deformation Quantization Algebras)**.** A *deformation quantization (DQ) algebra* is a sheaf of $\mathbb{C}_\hbar$-algebras such that it is locally isomorphic to a star-algebra $(\mathcal{O}_X[\![\hbar]\!], \star)$.

Just as the star-algebra $(\mathcal{O}_X[\![\hbar]\!], \star)$, any DQ algebra $\mathcal{A}$ is also $\hbar$-adic complete and flat over $\mathbb{C}_\hbar$. Also, there is a unique well-defined $\mathbb{C}$-algebra isomorphism $\mathcal{A}/\hbar\mathcal{A} \cong \mathcal{O}_X$ standing for the classical limit. The uniqueness and well-definedness come from the fact that the only $\mathbb{C}$-algebra automorphism of $\mathcal{O}_X$ is the identity.

Definition 2.5 (Deformation Quantization Algebroids)**.** A *deformation quantization (DQ) algebroid* is a $\mathbb{C}_\hbar$-algebroid such that for any local object σ, the endomorphism sheaf $\mathcal{E}nd(\sigma)$ is a DQ algebra.

Any DQ algebroid will give a Poisson structure on $\mathcal{O}_X$ and can be thought of as a deformation quantization of the corresponding Poisson manifold. If we consider DQ algebroids instead, then it is well known that there always exists a canonical DQ algebroid, unique up to (non-unique) isomorphisms, on any holomorphic symplectic manifold, or equivalently a nondegenerate Poisson manifold.

Theorem 2.6 ([PS, Theorem 9.1])**.** *For any holomorphic symplectic manifold X, there exists a canonical deformation quantization algebroid $\mathcal{A}_X^c$, which is unique up to (non-unique) isomorphisms.*

The upper index c refers to the canonical objects in this paper. In order to review the construction of the canonical DQ algebroid, let us first introduce the sheaf of formal microdifferentials defined in [SKK], which will give the local model of the DQ algebroids on a holomorphic symplectic manifold.

Construction 2.7 (Sheaf of Microdifferentials and Local Model of Deformation Quantization [PS])**.** For any complex manifold M, we consider the sheaf of formal microdifferential operators $\widehat{\mathcal{E}}_M(m)$ up to order $m \in \mathbb{Z}$ on T^*M. For each open subset $U \subseteq T^*M$, the sections of $\widehat{\mathcal{E}}_M(m)$ in the local Darboux coordinates (x, ξ) are formal differential operators

$$P(\hat{x}, \partial_x) := \sum_{k=-\infty}^{m} P_k(\hat{x}, \partial_x),$$

where $P_k(x,\xi)$ is a homogeneous holomorphic function in the fiber direction of T^*M of degree k, i.e. a holomorphic function satisfying $\sum_i \xi_i \partial_{\xi_i} P_k(x,\xi) = kP_k(x,\xi)$, and $P_k(\hat{x}, \partial_x)$ refers to an operator whose Taylor expansion in terms of the normally ordered operator $f(\hat{x})\partial_x^\alpha$ has the same coefficients as the Taylor expansion of $P_k(x,\xi)$ in terms of $f(x)\xi^\alpha$, where $f(x)$ is an analytic function of x. Following the standard convention in quantum mechanics, a hat $\widehat{\cdot}$ denotes the operator corresponding to a classical coordinate. Define $\widehat{\mathcal{E}}_M$ to be the direct limit $\varinjlim_{m\in\mathbb{Z}} \widehat{\mathcal{E}}_M(m)$.

More precisely, one can equivalently define the local sections of $\widehat{\mathcal{E}}_M(m)$ to be the collections of analytic functions

$$\{P_k(x,\xi)\}_{k\le m},$$

where each component $P_k(x,\xi)$ is a homogeneous holomorphic function on T^*M of degree k. The homogeneous component function $R_k(x,\xi)$ of the product $R = PQ$ of two operators $P, Q \in \widehat{\mathcal{E}}_M$ is defined to be

$$R_k(x,\xi) = \sum_{k=i+j-|\alpha|} \frac{1}{\alpha!} \left(\partial_\xi^\alpha P_i(x,\xi)\right) \left(\partial_x^\alpha Q_j(x,\xi)\right),$$

which makes sense when P_k and Q_k are holomorphic. One can show that the above local coordinate description glues to a sheaf of noncommutative $\mathbb{C}$-algebras on T^*M. This sheaf of formal microdifferentials provides a way to geometrize differential and difference equations and plays a key role in this paper.

Let K_M be the canonical bundle of M and $K_M^{1/2}$ be the twisted module of the square root of K_M, which always exists as a module over the $\mathbb{C}$-algebroid $\mathcal{O}_M \otimes_{\mathbb{C}} \mathbb{C}_{w_2(M)}$, where $\mathbb{C}_{w_2(M)}$ is the invertible $\mathbb{C}$-algebroid given by the second Stiefel-Whitney class of M. Here an invertible $\mathbb{C}$-algebroid is defined to be a $\mathbb{C}$-algebroid satisfying that $\mathbb{C} \to \mathcal{E}nd(\sigma)$ is an isomorphism for any local object σ. All invertible $\mathbb{C}$-algebroids on M are classified by $H^2(M;\mathbb{C})$ and $\mathbb{C}_{w_2(M)}$ means the invertible $\mathbb{C}$-algebroid corresponding to $w_2(M)$. Canonically, $\mathbb{C}_{w_2(M)}$ can be defined as the associated $\mathbb{C}$-algebroid of the gerbe of spin structures on M.

Consider the sheaf of half-twisted formal microdifferentials

$$\widehat{\mathcal{E}}_M^{\sqrt{v}} := \pi^{-1}K_M^{1/2} \otimes_{\pi^{-1}\mathcal{O}_M} \widehat{\mathcal{E}}_M \otimes_{\pi^{-1}\mathcal{O}_M} \pi^{-1}K_M^{-1/2},$$

where $\pi : T^*M \to M$ is the projection. This is always a sheaf of algebras since the gerbe part of $K_M^{\pm 1/2}$ will cancel out.

One can also define the formal adjoint operator $*$ on the sheaf of half-twisted formal microdifferentials $\widehat{\mathcal{E}}_M^{\sqrt{v}}$ as follows

$$\begin{aligned} * : \widehat{\mathcal{E}}_M^{\sqrt{v}} &\longrightarrow (a^{-1}\widehat{\mathcal{E}}_M^{\sqrt{v}})^{op} \\ (dx)^{1/2} \otimes P \otimes (dx)^{-1/2} &\longmapsto (dx)^{-1/2} \otimes P^* \otimes (dx)^{1/2}, \end{aligned}$$

where for any operator $P = \sum_k P_k \in \widehat{\mathcal{E}}_M$, P^* is defined to be

$$P^*(\hat{x}, \partial_x) := \sum_k P_k^*(\hat{x}, \partial_x), \;\; P_k^*(x, -\xi) := \sum_{\substack{j=l-|\alpha| \\ \alpha \in N^n}} \frac{(-1)^{|\alpha|}}{\alpha!} \partial_\xi^\alpha \partial_x^\alpha P_j(x, \xi),$$

and $a : T^*M \to T^*M$ is the antipodal map, i.e. multiplication by -1. The formal adjoint operator $*$ above does not depend on the choice of coordinates on M as an operator on the sheaf of half-twisted formal microdifferentials.

The local model of the DQ algebroid comes from the following setup. Let M be a complex manifold. Denote the Darboux coordinate of T^*M by (x, p) and denote the Darboux coordinates of the cotangent bundle $T^*(M \times \mathbb{C})$ by (x, t, ξ, τ), where (x, t) parametrizes the base direction and (ξ, τ) parametrizes the fiber direction. Let $\dot{T}^*(M \times \mathbb{C})$ be the complement of $\{\tau = 0\}$ in $T^*(M \times \mathbb{C})$. Let $P^*(M \times \mathbb{C})$ be the projectivization of the cotangent bundle $T^*(M \times \mathbb{C})$ and $\dot{P}^*(M \times \mathbb{C})$ be the complement of $\{\tau = 0\}$ in $P^*(M \times \mathbb{C})$. There are natural symplectic structures on T^*M and $T^*(M \times \mathbb{C})$ given by $\sum_i dx^i \wedge d\xi_i$ and $\sum_i dx^i \wedge d\xi_i + dt \wedge d\tau$ respectively. $P^*(M \times \mathbb{C})$ carries a natural contact structure given by the contact 1-form $dt + \sum_i \xi_i dx^i$. It is easy to see that $\dot{T}^*(M \times \mathbb{C})$ is isomorphic to $T^*M \times \mathbb{C}_t \times \mathbb{C}_\tau^*$ and $\dot{P}^*(M \times \mathbb{C})$ is isomorphic to $T^*M \times \mathbb{C}_t$, which is the same as the total space of the first jet of M. There is a diagram

$$\begin{aligned} &\dot{T}^*(M \times \mathbb{C}) \xrightarrow{\sigma} \dot{P}^*(M \times \mathbb{C}) \xrightarrow{\rho} T^*M. \\ &(x, t, \xi, \tau) \longmapsto (x, t; \xi : \tau) \longmapsto (x, p := \xi/\tau), \end{aligned} \tag{2.8}$$

where σ is a trivial $\mathbb{C}^*$-bundle and ρ is a trivial $\mathbb{C}$-bundle. The additional $\mathbb{C}$-factor is related to the $\hbar$-parameter in the deformation quantization.

Remark 2.9. The groupoid of contact manifolds considered in this paper is the groupoid of strict contact manifolds whose objects are complex manifolds with global contact 1-forms, and isomorphisms are biholomorphisms preserving the contact 1-forms. The reason is that for

every contact manifold arising in the deformation quantization of symplectic manifolds, we always consider a nowhere vanishing section τ of the contact line bundle and the τ-preserving contact transformations. The corresponding groupoid is indeed equivalent to the groupoid of strict contact manifolds by dividing the line-bundle-valued contact 1-form by τ.

Consider the sheaf of half-twisted formal microdifferential operators $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$ on $\dot{T}^*(M\times\mathbb{C})$. One could consider the pushforward $\sigma_*\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$ to the projective bundle $\dot{P}^*(M\times\mathbb{C})$, which we also denote by $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$ in this paper if there is no ambiguity. Consider the subalgebra $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}$ of $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$ consisting of operators commuting with ∂_t, or equivalently not depending on t. Define the DQ algebra on T^*M to be $\mathcal{A}_{T^*M} := \rho_*\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}(0)$. In $\mathcal{A}_{T^*M}$, we identify ∂_t^{-1} with $\hbar$ and hence get a sheaf of algebras over $\mathbb{C}_\hbar := \mathbb{C}[\![\hbar]\!]$. This DQ algebra $\mathcal{A}_{T^*M}$, which is also called the WKB algebra in [Po2], gives the local model of deformation quantization. We point out that, by tracing the definition of the grading back to $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$, the grading in $\rho_*\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}$ is indeed given by $deg(\hat{x}) = 0,\ deg(\hat{p}) = 0,\ deg(\hbar) = deg(\hat{\tau}^{-1}) = -1$. Therefore, $\rho_*\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}(0)$ just means that we only keep nonnegative $\hbar$-order terms. It is also a well-known fact that for any holomorphic symplectic manifold, the local model of any DQ algebra and hence DQ algebroid is given by the DQ algebra $\mathcal{A}_{T^*M}$ defined above.

Construction 2.10 (Canonical DQ Algebroid $\mathcal{A}^c_X$)**.** Now we are in a position to glue the local model $\mathcal{A}_{T^*M}$ into the canonical DQ algebroid $\mathcal{A}^c_X$ for any holomorphic symplectic manifold X. Take a Darboux cover $(U_i, \varphi_i : U_i \hookrightarrow T^*M_i)$ of X, where $U_i \subseteq X$ is an open subset and φ_i is an open symplectic embedding. Consider the symplectic transformation $\Psi_{ij} : \varphi_i(U_{ij}) \to \varphi_j(U_{ij})$ on $U_{ij} := U_i \cap U_j$ and a lift to the contact transformation $\tilde{\Psi}_{ij} : \rho_i^{-1}(\varphi_i(U_{ij})) \to \rho_j^{-1}(\varphi_j(U_{ij}))$ satisfying the following commutative diagram

$$\begin{array}{ccccccc}
\dot{P}^*(M_i\times\mathbb{C}) & \longleftarrow & \rho_i^{-1}(\varphi_i(U_{ij})) & \xrightarrow{\tilde{\Psi}_{ij}} & \rho_j^{-1}(\varphi_j(U_{ij})) & \longrightarrow & \dot{P}^*(M_j\times\mathbb{C}) \\
\downarrow{\scriptstyle \rho_i} & & \downarrow{\scriptstyle \rho_i} & & \downarrow{\scriptstyle \rho_j} & & \downarrow{\scriptstyle \rho_j} \\
T^*M_i & \longleftarrow & \varphi_i(U_{ij}) & \xrightarrow{\Psi_{ij}} & \varphi_j(U_{ij}) & \longrightarrow & T^*M_j
\end{array}.$$

In fact, after shrinking the open cover $\{U_i\}$, $\tilde{\Psi}_{ij}$ always takes the form $\tilde{\Psi}_{ij}(x_i, p_i, t_i) = (\Psi_{ij}(x_i, p_i), t_i + \rho^* f_{ij})$, where $f_{ij} \in \mathcal{O}_X(U_{ij})$ and df_{ij} gives the *Čech* cocycle of the holomorphic symplectic form $[\omega] \in H^1(X, d\mathcal{O}_X)$. Take a $\hat{\tau}^{-1}, *$-preserving *quantized contact transformation* (QCT), i.e. an isomorphism between the sheaves of microdifferentials

$$\tilde{\Psi}^\hbar_{ij} : \tilde{\Psi}^{-1}_{ij}\widehat{\mathcal{E}}^{\sqrt{v}}_{M_j\times\mathbb{C}}(0) \longrightarrow \widehat{\mathcal{E}}^{\sqrt{v}}_{M_i\times\mathbb{C}}(0)$$

which is $\hat{\tau}^{-1}, *$-preserving and the induced isomorphism on the principal symbol σ_0 is $\tilde{\Psi}_{ij}$. Such a quantized contact transformation always exists (see [PS, Lemma 8.7]) but is not unique. Since the quantized contact transformation $\tilde{\Psi}^\hbar_{ij}$ is $\hat{\tau}^{-1}$-preserving, it will induce a *quantized symplectic transformation* (QST)

$$\Psi^\hbar_{ij} : \Psi^{-1}_{ij}\widehat{\mathcal{E}}^{\sqrt{v}}_{M_j\times\mathbb{C},\partial_t}(0) \longrightarrow \widehat{\mathcal{E}}^{\sqrt{v}}_{M_i\times\mathbb{C},\partial_t}(0).$$

For the triple intersection U_{ijk}, we could shrink the open cover such that $\tilde{\Psi}_{ki}\tilde{\Psi}_{jk}\tilde{\Psi}_{ij}(x_i, p_i, t_i) = (x_i, p_i, t_i + c_{ijk})$, where $c_{ijk} = f_{ij} + f_{jk} + f_{ki} \in \mathbb{C}$. Indeed, c_{ijk} defines the *Čech* class of $[\omega] \in H^2(X, \mathbb{C})$. To get the canonical DQ algebroid, we need the following uniqueness result of the quantized contact transformations on triple intersections U_{ijk}.

Proposition 2.11 ([Ka1, Lemma 1], [PS, Lemma 8.11]). *Consider the classical contact transformation $\tilde{\Psi}(x,p,t) = (x,p,t+c)$, where $c \in \mathbb{C}$ is a constant. Let $\tilde{\Psi}^\hbar : \tilde{\Psi}^{-1}\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0) \longrightarrow \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$ be a quantized contact transformation on an open subset $\rho^{-1}(U) \subseteq \dot{P}^*(M\times\mathbb{C})$, which is $\hat{\tau}^{-1}, *$-preserving with classical limit $\tilde{\Psi}$. Then there exists a unique $P \in \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}(0)$ such that*

(1) P is of order zero with principal symbol $\sigma_0(P) = 1$,
*(2) P is unitary under *, i.e. $PP^* = 1$,*
(3) $\tilde{\Psi}^\hbar = \mathrm{Ad}_{\exp(c\partial_t)} \circ \mathrm{Ad}_P : \tilde{\Psi}^{-1}\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0) \longrightarrow \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$,

where Ad *means conjugation and we define $\mathrm{Ad}_{\exp(c\partial_t)}(Q(x,t,\partial_x,\partial_t)) := Q(x,t+c,\partial_x,\partial_t)$ for any $Q \in \tilde{\Psi}^{-1}\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$.*

We emphasize that the uniqueness of P above holds only when considering the quantized contact transformation $\tilde{\Psi}^\hbar$ on the one-dimensional higher contactification $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$, rather than $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}(0)$ because of the nontrivial center in $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}(0)$. The operator $\mathrm{Ad}_{\exp(c\partial_t)}$ represents the classical limit $\tilde{\Psi}$ and P represents the quantum correction. In this paper, we will use P^c to refer to the unique operator in Proposition 2.11, and the upper index c refers to canonicity. Using the uniqueness result in Proposition 2.11, we get the gluing datum for the canonical DQ algebroid

$$\{(\mathcal{A}_{T^*M_i}|_{U_i})^+; (\Psi^\hbar_{ij})^+; P^c_{ijk} : Id \implies (\Psi^\hbar_{ij}\Psi^\hbar_{jk}\Psi^\hbar_{ki})^+\},$$

where $(\mathcal{A}_{T^*M_i}|_{U_i})^+$ is the associated algebroid of the sheaf of algebra

$$\mathcal{A}_{T^*M_i}|_{U_i} := \rho_{i*}\widehat{\mathcal{E}}^{\sqrt{v}}_{M_i\times\mathbb{C},\partial_t}(0),$$

i.e. the stack of the categories of right invertible $\mathcal{A}_{T^*M_i}|_{U_i}$ modules, and $(\Psi^\hbar_{ij})^+$ is the associated transition functor of the quantized symplectic transformation $\Psi^\hbar_{ij}$, and P^c_{ijk} is the natural transformation between functors on $(\mathcal{A}_{T^*M_i}|_{U_{ijk}})^+$ induced by P^c_{ijk}. As shown in [PS, Theorem 9.1], one can verify that the gluing datum above glues to a DQ-algebroid, and that this DQ-algebroid does not depend on the choices made in the construction because of the uniqueness result in Proposition 2.11. Hence, we get a canonical DQ-algebroid $\mathcal{A}^c_X$ on every holomorphic symplectic manifold (X,ω).

According to [Po2, Theorem 5.3] and [BR, Theorem 7.2] in a more general setup, the DQ algebroids on a given holomorphic symplectic manifold (X,ω) are classified by the second cohomology $H^2(X;\mathbb{C}^*_\hbar)$. In fact, by adding $a_{ijk} \in \mathbb{C}^*_\hbar$ in front of the P^c_{ijk} above, one would get a different DQ algebroid, and all DQ algebroids can be obtained in this way. In other words, all DQ algebroids can be written as the tensor product of the canonical DQ algebroid with an invertible $\mathbb{C}_\hbar$-algebroid. For any DQ algebroid $\mathcal{A}_X$, the corresponding second cohomology class is sometimes called the Deligne-Fedosov class in the literature, which we will denote by $\mathfrak{D}(\mathcal{A}_X) \in H^2(X;\mathbb{C}^*_\hbar)$ in this paper. The canonical DQ algebroid is exactly the DQ algebroid with trivial Deligne-Fedosov class.

Theorem 2.12 (Classification of DQ Algebroids and Deligne-Fedosov Class [Po2, Theorem 5.3], [BR, Theorem 7.2]). *There is a canonical bijection $\mathfrak{D}$, which is called the Deligne-Fedosov class, from the collection of isomorphism classes of DQ algebroids $\mathfrak{DQ}(X)$ to the second cohomology group $H^2(X;\mathbb{C}^*_\hbar)$ defined by*

$$\mathfrak{D} : \mathfrak{DQ}(X) \longrightarrow H^2(X;\mathbb{C}^*_\hbar)$$
$$(\mathcal{A}_{T^*M_i}, \Psi^\hbar_{ij}, a_{ijk}P^c_{ijk}) \mapsto [a_{ijk}],$$

where $[a_{ijk}]$ is the corresponding second Čech cohomology class of X with $\mathbb{C}^*_\hbar$-coefficients. In other words, the Deligne-Fedosov class $\mathfrak{D}$ gives the classification of all DQ algebroids on a given holomorphic symplectic manifold X by comparing with the canonical DQ algebroid $\mathcal{A}^c_X$.

Remark 2.13. Note that [Po2] classifies DQ algebroids with the extra data of the formal adjoint $*$ ([Po2, Definition 3.3, item $(ii)^+$]) and the classical limit isomorphism to $\mathcal{O}^+_X$, which is the trivial invertible $\mathcal{O}_X$-algebroid ([Po2, Definition 3.3, item $(i)^+$]). Such DQ algebroids are classified by $H^2(X; \mathbb{C}^{*,1,u}_\hbar)$. The coefficient group $\mathbb{C}^{*,1,u}_\hbar$ is defined to be

$$\mathbb{C}^{*,1,u}_\hbar := \{a(\hbar) \in \mathbb{C}[\![\hbar]\!] | a = 1 + O(\hbar),\ a(\hbar)a(-\hbar) = 1\},$$

where the upper index 1 refers to the leading term condition and the upper index u refers to the unitary condition. In fact, the canonical DQ algebroid has a well-defined formal adjoint $*$ and classical limit isomorphism to $\mathcal{O}^+_X$ from Construction 2.10 and all other DQ algebroids with formal adjoint $*$ and classical limit isomorphism to $\mathcal{O}^+_X$ are constructed by adding $a_{ijk} \in \mathbb{C}^{*,1,u}$ in front of the P^c_{ijk} above. In the literature of deformation quantization in the algebraic context such as [BK], the Deligne-Fedosov class, which is also called the non-commutative period, is defined in $H^2(X, \hbar\mathbb{C}_\hbar)$. The relation between the different definitions in the algebraic context and the analytic context is given by the exponential map (see [BR, §7.3] for more details).

2.2. Deformation Quantization of Holomorphic Lagrangian Submanifolds.

2.2.1. *Deformation Quantization Modules.* The deformation quantization of a holomorphic Lagrangian submanifold L should be regarded as a left simple holonomic deformation quantization (DQ) module over a DQ algebroid supported on L. Deformation quantization modules are defined categorically as follows.

Definition 2.14 (Deformation Quantization (DQ) Module)**.** A *deformation quantization (DQ) module* over a DQ algebroid $\mathcal{A}_X$ is a $\mathbb{C}_\hbar$-linear functor from the algebroid $\mathcal{A}_X$ to the stack of sheaves of $\mathbb{C}_\hbar$-modules.

Note that when the DQ algebroid is the associated algebroid of a DQ algebra $\mathcal{A}_X$, the definition above is the same as the usual definition of $\mathcal{A}_X$-module since the $\mathbb{C}_\hbar$-linear functor can be induced by the tensor product. In this paper, we will mostly use the gluing datum of a DQ-module. Based on the gluing datum of a DQ algebroid $\mathcal{A}_X$, the gluing datum of a DQ module $\mathcal{M}$ over $\mathcal{A}_X$ is given by

$$(\mathcal{M}_i, \xi_{ij}),$$

where $\mathcal{M}_i$ are $\mathcal{A}_i$-modules and $\xi_{ij} : \mathcal{M}_j|_{U_{ij}} \longrightarrow \mathcal{M}_i|_{U_{ij}}$ are module isomorphisms under the algebra isomorphisms Ψ_{ij}. ξ_{ij} should satisfy the following consistency condition

$$\xi_{ij} \circ \xi_{jk} \circ \xi_{ki} = a_{ijk} \text{ on } U_{ijk}.$$

Definition 2.15 (Simple Holonomic DQ Module on L)**.** A coherent DQ module $\mathcal{M}_L$ supported on L over a DQ algebroid $\mathcal{A}_X$ is called a *simple holonomic DQ module* if it is $\hbar$-torsion free and the classical limit $\mathcal{M}_L/\hbar\mathcal{M}_L$ is an invertible module over $\mathcal{A}_X/\hbar\mathcal{A}_X \otimes_{\mathcal{O}_X} \mathcal{O}_L$.

Remark 2.16. Consider the functor

$$gr_\hbar(-) := \mathcal{A}_X/\hbar\mathcal{A}_X \otimes^L_{\mathcal{A}_X} (-) = \mathbb{C} \otimes^L_{\mathbb{C}_\hbar} (-)$$

on the bounded derived category of coherent $\mathcal{A}_X$-modules. In [KS, Definition 2.3.10], a simple holonomic DQ module $\mathcal{M}_L$ supported on L is defined to be a coherent DQ module satisfying that $gr_\hbar(\mathcal{M}_L)$ is concentrated in degree 0 and $H^0(gr_\hbar(\mathcal{M}_L))$ is an invertible $\mathcal{A}_X/\hbar\mathcal{A}_X \otimes_{\mathcal{O}_X} \mathcal{O}_L$-module. This definition is equivalent to Definition 2.15 since a free $\mathbb{C}_\hbar$-resolution of $\mathbb{C}$ can be given by the multiplication map $\hbar : \mathbb{C}_\hbar \longrightarrow \mathbb{C}_\hbar$ (also see [KS, Proposition 1.4.5]).

Proposition 2.17 ([KS, Lemma 6.2.1])**.** *A DQ module $\mathcal{M}_L$ supported on L is a simple holonomic DQ module if and only if it is locally isomorphic to $\mathcal{O}_M[\![\hbar]\!]$ as an $\mathcal{A}_{T^*M}$-module.*

Given a holomorphic symplectic manifold X and a holomorphic Lagrangian closed submanifold L, in this paper we study the deformation quantization of the pair (X, L).

Definition 2.18 (Deformation Quantization (DQ) Pair)**.** A *deformation quantization (DQ) pair* on (X, L) is defined to be a pair $(\mathcal{A}_X, \mathcal{M}_L)$, where $\mathcal{A}_X$ is a DQ algebroid and $\mathcal{M}_L$ is a simple holonomic DQ module over $\mathcal{A}_X \otimes \mathbb{C}_{w_2(L)}$ supported on L, where $\mathbb{C}_{w_2(L)}$ is the invertible $\mathbb{C}$-algebroid associated with the second Stiefel-Whitney class of L. The property of being simple holonomic over $\mathcal{A}_X \otimes \mathbb{C}_{w_2(L)}$ is defined similarly to Definition 2.15 since it is a local property.

Definition 2.19. An *isomorphism* between the DQ pairs $(\mathcal{A}_1, \mathcal{M}_1)$ and $(\mathcal{A}_2, \mathcal{M}_2)$ is given by a pair

$$(\Phi : \mathcal{A}_1 \longrightarrow \mathcal{A}_2, \varphi : \mathcal{M}_1 \longrightarrow {}_\Phi\mathcal{M}_2),$$

where Φ is an equivalence of algebroids, φ is an isomorphism of modules over algebroids, and ${}_\Phi\mathcal{M}_2$ means the base change of $\mathcal{M}_2$ from $\mathcal{A}_2$ to $\mathcal{A}_1$ by Φ.

Given a Lagrangian submanifold L and a spin structure $K_L^{1/2}$,[1] a natural question is whether there exists a canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ on (X, L) such that the underlying DQ algebroid $\mathcal{A}_X^c$ is isomorphic to the canonical DQ algebroid. In fact, following the DQ module construction by D'Agnolo-Schapira in [DS, Theorem 6.1], we can give the definition, or gluing construction, of the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$, whose uniqueness will be shown later in Proposition 2.29. Before giving the construction of the canonical DQ pair, we need some notions concerning microdifferential modules.

2.2.2. *Simple Holonomic Microdifferential Modules.*

Definition 2.20. (Simple Holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module) A coherent $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$ supported on a Legendrian submanifold L inside $\dot{P}^*(M \times \mathbb{C})$ is a *simple holonomic* $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-*module* if it is τ^{-1}-torsion free and the classical limit $\mathcal{M}_L/\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(-1)\mathcal{M}$ is a line bundle on L.

For any simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$, by Nakayama's Lemma (see for example [Ka2, Proposition 7.9]), any lift of a local generator of the classical limit $\mathcal{M}_L/\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(-1)\mathcal{M}$ is a local generator of $\mathcal{M}_L$. Hence a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module locally always admits a generator.

Remark 2.21. For any simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$, one can consider the base change

$$\widetilde{\mathcal{M}}_L := \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}} \otimes_{\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)} \mathcal{M}_L,$$

which is the localization of $\mathcal{M}_L$ with respect to τ^{-1} (note that τ^{-1} is not invertible in $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$). Since $\mathcal{M}_L$ is τ^{-1}-torsion free, we have an embedding of $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-modules

$$\mathcal{M}_L \hookrightarrow \widetilde{\mathcal{M}}_L.$$

Hence $\widetilde{\mathcal{M}}_L$ is a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-module in [DS, Definition 4.1 (a)]. Any local generator of $\mathcal{M}_L$ will give a simple local generator of $\widetilde{\mathcal{M}}_L$ [DS, Definition 4.1 (b)].

[1] This spin structure is a complex-geometric avatar of the metalinear structure on a real Lagrangian submanifold, or equivalently the metaplectic structure on a tubular neighborhood of a real Lagrangian submanifold, in the context of geometric quantization ([BW, §7.2]). One should distinguish this with the notion of spin in quantum mechanics.

Example 2.22 (Local Model of Simple Holonomic $\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$-modules)**.** Consider the Legendrian $L \subseteq \dot{P}^*(M\times\mathbb{C}) \cong T^*M\times\mathbb{C}_t$ given by $M\times\{0\}$, where $M \hookrightarrow T^*M$ is regarded as the zero section. For every $\lambda\in\mathbb{C}$, we can define a simple holonomic $\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$-module supported on L denoted by

$$\tau^\lambda \mathcal{O}_M[\![\tau^{-1}]\!],$$

where the sections are formal products $\tau^\lambda f(x,\tau^{-1})$. The action of $\hat{t}$ is given by $-\partial_\tau$ as

$$\hat{t}\cdot\tau^\lambda f(x,\tau^{-1}) := -\lambda\tau^{\lambda-1}f(x,\tau^{-1}) - \tau^\lambda\partial_\tau f(x,\tau^{-1}),$$

which is well defined in $\tau^\lambda\mathcal{O}_M[\![\tau^{-1}]\!]$. The action of $\hat{\xi}$ is given by ∂_x and the actions of $\hat{x},\hat{\tau}^{-1}$ are given by multiplications. This simple holonomic $\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$-module is generated by τ^λ and is isomorphic to the quotient module

$$\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)/\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)(\hat{t}+\lambda\hat{\tau}^{-1},\tau^{-1}\hat{\xi}_1,\cdots,\tau^{-1}\hat{\xi}_n).$$

In the case of $\lambda=0$, this simple holonomic $\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$-module comes from the restriction of the microlocalization of the $\mathcal{D}_{M\times\mathbb{C}}$-module $\mathcal{B}_{M\times\{0\}|M\times\mathbb{C}}$, which is called the sheaf of microfunctions (see [Ka2, Formula 3.30]), on the locus $\tau\neq 0$. If one chooses a spin structure $K_M^{1/2}$ on M, it induces a natural spin structure on $M\times\mathbb{C}_t$ by using the spin structure on $\mathbb{C}_t$ induced by dt, and then we can regard

$$\tau^\lambda K_M^{1/2}[\![\tau^{-1}]\!],$$

as a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module similarly.

Proposition 2.23. *Any simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module is locally isomorphic to*

$$\tau^\lambda K_M^{1/2}[\![\tau^{-1}]\!]$$

under a certain $\hat{\tau}^{-1},*$-preserving quantized contact transformation for some $\lambda\in\mathbb{C}$.

Proof. Every Legendrian L is locally isomorphic to $M\times\{0\}\hookrightarrow T^*M\times\mathbb{C}$ via a contact transformation, which always has a $\hat{\tau}^{-1},*$-preserving quantized contact transformation lift by [PS, Lemma 8.7]. Thus we only need to consider the case $L=M\times\{0\}$. The proposition then follows from the usual treatment of the classification of simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-modules in [Ka2, Theorem 8.20]. Since this is a local question, we can ignore the spin structure $K_M^{1/2}$. For any simple holonomic $\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$, consider the base change to $\widehat{\mathcal{E}}_{M\times\mathbb{C}}$ and denote it by $\widetilde{\mathcal{M}}_L$. Take a local generator $u\in\mathcal{M}_L$, which will give a simple local generator in $\widetilde{\mathcal{M}}_L$. By [Ka2, Theorem 8.20], there exists locally an invertible operator $P\in\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$ such that $\mathrm{Ann}_{\widehat{\mathcal{E}}_{M\times\mathbb{C}}}(Pu)$ is generated by $\hat{t}\hat{\tau}+\lambda,\hat{\xi}_1,\cdots,\hat{\xi}_n$. Since Pu is also a generator of the $\widehat{\mathcal{E}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$, we know that $\mathcal{M}_L$ is isomorphic to $\tau^\lambda\mathcal{O}_M[\![\tau^{-1}]\!]$. □

We now recall the definition of the principal symbol of a generator of a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module supported on the Legendrian L following [DS, Definition 4.3] (also see [Ka2, §8.6]). Let $\widetilde{L}$ be the preimage of L under the projection σ in (2.8), which can be identified with $L\times\mathbb{C}^*_\tau$. $\widetilde{L}$ is a conic Lagrangian submanifold of $\dot{T}(M\times\mathbb{C})$.

Consider the sheaf of algebras $\widehat{\mathcal{E}}^{\sqrt{v}}_L$ generated by the sections

$$\{P\in\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(1)\mid \sigma_1(P)|_{\widetilde{L}}=0\}. \tag{2.24}$$

We can view $\widehat{\mathcal{E}}^{\sqrt{v}}_L$ as a subalgebra of $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$ on $\dot{P}^*(M\times\mathbb{C})$ by the projection σ.

Take any spin structure $K_L^{1/2}$ on the Legendrian L. It induces a spin structure on $\widetilde{L} = L \times \mathbb{C}_\tau^*$ by taking the spin structure on $\mathbb{C}_\tau^*$ to be the one given by $d\tau$. Consider the algebra homomorphism

$$\mathbf{L} : \widehat{\mathcal{E}}_L^{\sqrt{v}} \longrightarrow \sigma_* \mathcal{D}_{\widetilde{L}}^{\sqrt{v}}$$

defined by

$$\mathbf{L}(P) := \mathcal{L}^{1/2}_{H_{\sigma_1(P)|_{\widetilde{L}}}} + \sigma_0'(P)|_{\widetilde{L}}$$

for any P in (2.24). Here $\mathcal{D}_{\widetilde{L}}^{\sqrt{v}}$ is the sheaf of half-twisted differential operators on $\widetilde{L}$, and $\mathcal{L}^{1/2}_{H_{\sigma_1(P)|_{\widetilde{L}}}}$ is the half-twist Lie derivative of the Hamiltonian vector field $H_{\sigma_1(P)|_{\widetilde{L}}}$. Note that the Hamiltonian vector field is tangent to the Lagrangian $\widetilde{L}$ since $\sigma_1(P)|_{\widetilde{L}} = 0$. Here $\sigma_0'(P)$ is the subprincipal symbol of P defined as $\sigma_0'(P) := \frac{1}{2}(P + P^*)$.

Any simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$ carries a natural action of the larger algebra $\widehat{\mathcal{E}}_{\widetilde{L}}^{\sqrt{v}}$. In fact, for any local generator u, every operator P in (2.24) can be decomposed as $\tau P' + Q$, satisfying $P', Q \in \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$ and $P'u = 0$. Then for any $A \in \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$, the action of P can be defined as $P(Au) := ([\tau P', A] + Q)u$; note that $[\tau P', A]$ is at most of order 0.

For any local generator $u \in \Gamma(U, \mathcal{M}_L)$, one can consider the following subsheaf of $K_{\widetilde{L}}^{1/2}$

$$\{s \in K_{\widetilde{L}}^{1/2} | \mathbf{L}(P)s = 0, \forall P \in \mathrm{Ann}_{\widehat{\mathcal{E}}_{\widetilde{L}}^{\sqrt{v}}}(u)\},$$

which is a rank 1 $\mathbb{C}$-local system. This local system gives a section of the quotient sheaf $\sigma_L(u) \in \Gamma(U \times \mathbb{C}_\tau^*, K_{\widetilde{L}}^{1/2}/\mathbb{C}^*)$, which is called the principal symbol of the generator u. If $\sigma_L(u)$ has the eigenvalue $\lambda \in \mathbb{C}$ under the Euler vector field, then we say that the generator u has order $\lambda \in \mathbb{C}$.

Note that the order of a local generator of a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module $\mathcal{M}_L$ does not depend on the choice of generator. In fact, any two generators $u, u' \in \mathcal{M}_L$ are related by $u' = Pu$, where $P \in \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$ is invertible. Since P is of order zero, u', u have the same order. Therefore, we define the order of a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-module to be the order of its local generators, which is a constant in $\mathbb{C}$ on each connected component of M.

Remark 2.25. If one only considers a simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-module, then the order of a module only makes sense in $\mathbb{C}/\mathbb{Z}$ since one can shift any simple generator by a power of $\hat{\tau}$, which is invertible in $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$.

Example 2.26. For the local model $\tau^\lambda K_M^{1/2}[\![\tau^{-1}]\!]$ in Example 2.22, take a nowhere vanishing section $\sqrt{\eta}$ of the spin structure $K_M^{1/2}$. Then $\tau^\lambda\sqrt{\eta}$ is a generator and the principal symbol is

$$\sigma_L(\tau^\lambda\sqrt{\eta}) = [\tau^\lambda\sqrt{\eta \wedge d\tau}] \in \Gamma(L \times \mathbb{C}_\tau^*, K^{1/2}_{L\times\mathbb{C}_\tau^*}/\mathbb{C}^*),$$

which has order $\lambda + 1/2$. Therefore, the order of $\tau^\lambda K_M^{1/2}[\![\tau^{-1}]\!]$ is $\lambda + 1/2$.

Since the principal symbol σ_L is preserved under $\hat{\tau}^{-1}, *$-preserving quantized contact transformations, we know that simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-modules are locally fully classified by their orders under $\hat{\tau}^{-1}, *$-preserving quantized contact transformations.

2.2.3. *Canonical Deformation Quantization Pair.* Let us describe the gluing datum of a DQ pair. Given a sufficiently refined open cover $\{U_i\}$, we can always take the local model of the classical pair (X, L) on an open subset U_i near L to be (T^*M_i, M_i), where M_i is regarded as the zero section of T^*M_i and gives the local model of the Lagrangian L on the open subset U_i whose intersection with L is non-empty. Also, the classical symplectic transformations Ψ_{ij} between the local models (T^*M_i, M_i) should preserve the zero section M_i. The local model of any DQ pair on (X, L) is given by $(\mathcal{A}_{T^*M_i}, K_{M_i}^{1/2}[\![\hbar]\!])$, where $K_{M_i}^{1/2}[\![\hbar]\!]$ has a natural structure of simple holonomic DQ module. Therefore, the gluing datum of any DQ pair can be written as the tuple

$$(\mathcal{A}_{T^*M_i}, \Psi_{ij}^{\hbar}, P_{ijk}, K_{M_i}^{1/2}[\![\hbar]\!], \xi_{ij}),$$

where $\Psi_{ij}^{\hbar}$ are quantized symplectic transformation lifts of the classical symplectic transformations Ψ_{ij}, P_{ijk} give the natural transformation part of the gluing datum of the DQ algebroid, and ξ_{ij} are the gluing morphisms between the DQ modules $K_{M_i}^{1/2}[\![\hbar]\!]$.

The canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ is characterized by the following properties on the quantized contact transformation lifts of the quantized symplectic transformations in the gluing datum.

Definition 2.27 (Canonical DQ Pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$)**.** Given a choice of the spin structure $K_L^{1/2}$, the *canonical DQ pair* $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ with respect to $K_L^{1/2}$ is the DQ pair which can be constructed by the gluing datum

$$(\mathcal{A}_{T^*M_i}, \Psi_{ij}^{\hbar}, P_{ijk}^c, K_{M_i}^{1/2}[\![\hbar]\!], \xi_{ij}^c)$$

such that there exist $\hat{\tau}^{-1}, *$-preserving quantized contact transformation lifts $\tilde{\Psi}_{ij}^{\hbar}$ of the quantized symplectic transformations $\Psi_{ij}^{\hbar}$ satisfying the following conditions:

(1) On every triple intersection U_{ijk} of the whole open cover $\{U_i\}$, we have

$$\tilde{\Psi}_{ij}^{\hbar}\tilde{\Psi}_{jk}^{\hbar}\tilde{\Psi}_{ki}^{\hbar} = \mathrm{Ad}_{\exp(c_{ijk}\partial_t)P_{ijk}^c} \text{ on } U_{ijk},\ c_{ijk} \in \mathbb{C},$$

where P_{ijk}^c is the unique operator satisfying the conditions in Proposition 2.11. Also, the corresponding classical contact transformations $\tilde{\Psi}_{ij}$ should map the locus $t_i = 0, p_i = 0$ to the locus $t_j = 0, p_j = 0$ in the local model (2.8) near the Lagrangian L. This indeed implies $c_{ijk} = 0$ near L.

(2) On every open subset U_i whose intersection with L is non-empty and viewing $K_{M_i}^{1/2}[\![\hbar]\!]$ as a simple holonomic microdifferential $\widehat{\mathcal{E}}_{M_i\times\mathbb{C}}^{\sqrt{v}}(0)$-module of order $\frac{1}{2}$ (see Example 2.26), the gluing morphism between the DQ modules

$$\xi_{ij}^c : \tilde{\Psi}_{ij}^{-1}(K_{M_j}^{1/2}[\![\hbar]\!]) \longrightarrow K_{M_i}^{1/2}[\![\hbar]\!]$$

should be a microdifferential module isomorphism under $\tilde{\Psi}_{ij}^{\hbar}$.

(3) The classical limits of ξ_{ij}^c should be the transition functions of the spin structure $K_L^{1/2}$. To be more precise, there exists a generator u_i of the simple holonomic microdifferential $\widehat{\mathcal{E}}_{M_i\times\mathbb{C}}^{\sqrt{v}}(0)$-module $K_{M_i}^{1/2}[\![\hbar]\!]$ on every U_i near L_i, $\sqrt{\eta_i} \in \Gamma(M_i, K_{M_i}^{1/2,*})$ and $h_{ij} \in \Gamma(U_{ij}, \widehat{\mathcal{E}}_{M_i\times\mathbb{C}}^{\sqrt{v}}(0))$ such that

$$\begin{aligned} \sigma_L(u_i) &= [\sqrt{\eta_i \wedge d\tau}] \in \Gamma(M_i \times \mathbb{C}_\tau^*, K_{M_i\times\mathbb{C}_\tau^*}^{1/2}/\mathbb{C}^*) \\ \xi_{ij}^c(u_j) &= h_{ij}u_i, \\ \sigma_0(h_{ij}) &= \frac{\sqrt{\eta_j}}{\sqrt{\eta_i}}, \end{aligned} \tag{2.28}$$

where $\sigma_0(h_{ij})$ means the principal symbol of the operator h_{ij}.

Note that given a choice of quantized contact transformation lifts $\tilde{\Psi}_{ij}^{\hbar}$ near L, the transition morphisms ξ_{ij}^c are uniquely determined by the conditions (2) and (3) above. In fact, the collection of all transition morphisms between two simple holonomic DQ modules forms a $\mathbb{C}_{\hbar}^*$-torsor (see [KS, Lemma 6.2.1]). The constraint of being a microdifferential module isomorphism fixes the degree of freedom of $\mathbb{C}_{\hbar}^{*,1}$, and the constraint on the classical limit fixes the residual $\mathbb{C}^*$ degree of freedom. We also emphasize that the quantized contact transformations $\tilde{\Psi}_{ij}^{\hbar}$ in the definition above only impose a condition that the canonical DQ pair should satisfy; they are not part of the gluing datum of the canonical DQ pair in our definition.

The existence of the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ is already contained in the proof of [DS, Theorem 6.1]. In fact, one can first take any quantized symplectic transformations $\Psi_{ij}^{\hbar}$ and any quantized contact transformation lifts $\tilde{\Psi}_{ij}^{\hbar}$ between the local models $\mathcal{A}_{T^*M_i}$ which preserve the Lagrangian L. Then there exist uniquely determined P_{ijk}^c and ξ_{ij}^c satisfying the conditions (1),(2) and (3) in Definition 2.27 above. The only property that one needs to check to prove the existence is the compatibility of the gluing datum $(\Psi_{ij}^{\hbar}, P_{ijk}^c, \xi_{ij}^c)$, which was already contained in the proof of [DS, Theorem 6.1].

In fact, the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ defined in Definition 2.27 is also unique up to isomorphisms and gives the canonical isomorphism class of DQ pairs.

Proposition 2.29 (Uniqueness of the Canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$)**.** *The canonical DQ pair defined in Definition 2.27 is unique up to (non-unique) isomorphisms.*

Proof. Consider another gluing datum

$$(\mathcal{A}_{T^*M_i'}, \Psi_{ij}'^{\hbar}, P_{ijk}'^c, K_{M_i'}^{1/2}[\![\hbar]\!], \xi_{ij}'^c)$$

of a DQ pair $(\mathcal{A}'^c_X, \mathcal{M}'^c_L)$ satisfying the conditions in Definition 2.27. By taking the common refinement of the open covers, we may assume that the gluing datum above is on the same open cover U_i. Let Φ_i be the classical symplectic transformation from T^*M_i' to T^*M_i and choose a contact transformation lift $\tilde{\Phi}_i$. Assume that the pullback of the coordinate t_i in $\dot{P}^*(M_i \times \mathbb{C})$ under the contact transformation lift $\tilde{\Phi}_i$ is $t_i' + g_i$, where g_i is a function on U_i. Since $\tilde{\Phi}_i$ preserves the contact 1-form, we have $dg_i = p_i' dx_i' - p_i dx_i$. Hence we have $df_{ij} + dg_i - df_{ij}' - dg_j = 0$, where f_{ij} and f_{ij}' are the transition functions of the contactifications in Construction 2.10. Assume that $f_{ij} + g_i - f_{ij}' - g_j = b_{ij} \in \mathbb{C}$. Take a $\tau, *$-preserving quantized contact transformation lift $\tilde{\Phi}_i^{\hbar}$ of $\tilde{\Phi}_i$. According to Proposition 2.11, there exists a unique operator $Q_{ij} \in \mathcal{A}_{T^*M_i'}$ of order zero satisfying $Q_{ij}Q_{ij}^* = 1$, $\sigma_0(Q_{ij}) = 1$ and

$$\mathrm{Ad}_{\exp(b_{ij}\partial_t)Q_{ij}} \circ \tilde{\Psi}'^{\hbar}_{ij} \circ \tilde{\Phi}_j^{\hbar} = \tilde{\Phi}_i^{\hbar} \circ \tilde{\Psi}_{ij}^{\hbar}.$$

Also, according to the uniqueness in Proposition 2.11, we have the following identities

$$\begin{aligned} Q_{ij}\Psi_{ij}'^{\hbar}(Q_{jk})P_{ijk}'^c &= \Phi_i^{\hbar}(P_{ijk}^c)Q_{ik} \in \mathcal{A}_{T^*M_i'}, \\ b_{ij} + b_{jk} + c_{ijk}' &= c_{ijk} + b_{ik}. \end{aligned}$$

Therefore, the datum $(\Phi_i^\hbar, Q_{ij})$ shown in the following diagram will give the gluing datum of an equivalence Φ between the DQ algebroids $\mathcal{A}_X$ and $\mathcal{A}'_X$.

$$\begin{array}{ccc} \Psi_{ij}^{-1}\mathcal{A}_{T^*M_j} & \xrightarrow{\Phi_j^\hbar} & \Psi_{ij}'^{-1}\mathcal{A}_{T^*M'_j} \\ \downarrow{\scriptstyle \Psi_{ij}^\hbar} & \swarrow{\scriptstyle Q_{ij}} & \downarrow{\scriptstyle \Psi_{ij}'^\hbar} \\ \mathcal{A}_{T^*M_i} & \xrightarrow[\Phi_i^\hbar]{} & \mathcal{A}_{T^*M'_i} \end{array}$$

Now we need to construct the isomorphism between DQ modules $\mathcal{M}_L^c$ and $\mathcal{M}_L'^c$ under the DQ algebroid equivalence Φ. Consider the unique DQ module isomorphism φ_i satisfying the conditions that it is a microdifferential module isomorphism under $\tilde{\Phi}_i^\hbar$ and satisfies

$$\varphi_i(u_i) = d_i u'_i, d_i \in \widehat{\mathcal{E}}^{\sqrt{v}}_{M'_i\times\mathbb{C}}(0)$$
$$\sigma_0(d_i) = \frac{\sqrt{\eta_i}}{\sqrt{\eta'_i}},$$

where the generators u_i, u'_i and sections $\sqrt{\eta_i}, \sqrt{\eta'_i}$ come from condition (3) in the definition of the canonical DQ pair (Definition 2.27). Without loss of generality, we may further assume that the contact transformation Φ_i maps the locus $\{t'_i = 0, p'_i = 0\}$ to the locus $\{t_i = 0, p_i = 0\}$. Then from the uniqueness of the DQ module isomorphism satisfying conditions (2) and (3) in Definition 2.27 and $\sigma_0(Q_{ij}) = 1$, we have

$$Q_{ij} \circ \xi_{ij}'^c \circ \varphi_j = \varphi_i \circ \xi_{ij}^c$$

as shown in the diagram

$$\begin{array}{ccc} \Psi_{ij}^{-1}K_{M_j}^{1/2}[\![\hbar]\!] & \xrightarrow{\varphi_j} & \Psi_{ij}'^{-1}K_{M'_j}^{1/2}[\![\hbar]\!] \\ \downarrow{\scriptstyle \xi_{ij}^c} & \swarrow{\scriptstyle Q_{ij}} & \downarrow{\scriptstyle \xi_{ij}'^c} \\ K_{M_i}^{1/2}[\![\hbar]\!] & \xrightarrow[\varphi_i]{} & K_{M'_i}^{1/2}[\![\hbar]\!] \end{array} .$$

Therefore, φ_i gives the gluing datum of a DQ module isomorphism φ under the DQ algebroid equivalence Φ, and the pair (Φ, φ) gives an isomorphism between the canonical pairs. □

Finally, we have the existence and uniqueness of the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ just as in the case of the canonical DQ algebroid.

Theorem 2.30. *Given a holomorphic symplectic manifold (X, ω), a holomorphic Lagrangian submanifold L and a choice of spin structure $K_L^{1/2}$, there exists a canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$, where $\mathcal{A}_X^c$ is isomorphic to the canonical DQ algebroid constructed in Construction 2.10 and $\mathcal{M}_L^c$ is a simple holonomic DQ module supported on L over $\mathcal{A}_X^c \otimes_{\mathbb{C}} \mathbb{C}_{w_2(L)}$. Moreover, the canonical deformation quantization pair is unique up to (non-unique) isomorphism.*

From the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$, we can define a good class of simple holonomic DQ modules supported on L with certain canonicity.

Definition 2.31 (Quasi-canonical Deformation Quantization (DQ) Module)**.** Given a certain choice of spin structure $K_L^{1/2}$, a simple holonomic DQ module $\mathcal{M}_L$ supported on L over $\mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(L)}$ is called a *quasi-canonical DQ module* with respect to (X, L) if the DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$ is isomorphic to the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$.

Note that the notion of quasi-canonical DQ modules is sensitive to the ambient symplectic manifold X. Over a canonical DQ algebroid, the quasi-canonical DQ modules defined above are not unique and form a torsor over

$$\mathrm{Im}\left(i^* : H^1(X;\mathbb{C}_\hbar^*) \longrightarrow H^1(L;\mathbb{C}_\hbar^*)\right),$$

where $i : L \hookrightarrow X$ is the closed embedding of L into X. According to [DS, Corollary 9.2], the collection of all simple holonomic DQ modules supported on L over any DQ algebroid forms a torsor over $H^1(L;\mathbb{C}_\hbar^*)$. In fact, the gluing data of other simple holonomic DQ modules are given by adding factors $l_{ij} \in \mathbb{C}_\hbar^*$ in front of the transition morphisms ξ_{ij} of a given simple holonomic DQ module satisfying the cycle condition $l_{ij}l_{jk}l_{ki} = 1$. The global ambiguity of $H^1(X;\mathbb{C}_\hbar^*)$ of the quasi-canonical DQ modules defined above comes from the autoequivalences of the canonical DQ algebroid, which are canonically isomorphic to the group $H^1(X;\mathbb{C}_\hbar^*)$.

Remark 2.32. This remark is due to a discussion with Pavel Safronov. The canonicity we have been using in this paper refers to the canonical isomorphism class of certain deformation quantization structures. The reason why we only consider this weak notion of canonicity is that we only care about the invariants, such as the Deligne-Fedosov classes, of certain deformation quantization structures. As in this paper, if one only considers the DQ algebroid structure without additional data, the canonical DQ algebroid is only unique up to a non-unique equivalence, but not unique up to a canonical equivalence. This is due to the fact that when one tries to construct the canonical equivalence, there are various choices of quantized contact transformation lifts of quantized symplectic transformations and this degree of freedom spoils the canonicity of the equivalence. The way to fix this issue and get a more canonical notion of the canonical DQ algebroid in the categorical sense is to encode some additional pieces of data to rigidify the canonical DQ algebroid (see [DK, Appendix A] and Appendix A for more details). In fact, the most important additional datum is a $\mathbb{C}$-linear derivation preserving $\hbar$ and $*$, which is also called a dilation equivariance structure (DES) in [BR]. This piece of data indeed encodes what happens in the t-direction in the contactification and fixes the degree of freedom of quantized contact transformation. By considering these additional data, one can also make sense of the unique canonical DQ module on a given Lagrangian since the degree of freedom of the equivalences of DQ algebroids will be fixed. For example, [GS6] works in this setup.

We will put this categorical characterization of the canonical deformation quantization in Appendix A. For other parts of the paper, we will not keep track of these additional data. Without them, we can only make sense of the quasi-canonical DQ modules defined in Definition 2.31, since there is only a unique canonical DQ pair (Definition 2.27).

2.3. **Classification of Deformation Quantization Pairs and Relative Deligne-Fedosov Class.** In this section, we classify DQ pairs of DQ algebroids and simple holonomic DQ modules and define a relative version of the Deligne-Fedosov class. The gluing datum of any DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$ takes the form

$$(\mathcal{A}_{T^*M_i}, \Psi_{ij}^\hbar, a_{ijk}P_{ijk}^c, K_{M_i}^{1/2}[\![\hbar]\!], l_{ij}\xi_{ij}^c),\ a_{ijk}, l_{ij} \in \mathbb{C}_\hbar^*,$$

where the canonical gluing datum P_{ijk}^c and ξ_{ij}^c satisfy the conditions in the definition of the canonical DQ pair (Definition 2.27). Also, the factors a_{ijk} and l_{ij} need to satisfy the compatibility conditions

$$a_{ijk}a_{ikl} = a_{jkl}a_{ijl},$$
$$l_{ij}l_{jk}l_{ki} = a_{ijk} \text{ near } L.$$

Without loss of generality, we may further assume that $l_{ij} = 1$ near L. In fact, consider any extension $\tilde{l}_{ij}$ of the *Čech* 1-cochain l_{ij} in the above gluing datum to the whole X. We

can modify a_{ijk} to $\tilde{a}_{ijk}$ by

$$\tilde{a}_{ijk} := a_{ijk}\tilde{l}_{ij}^{-1}\tilde{l}_{jk}^{-1}\tilde{l}_{ki}^{-1}.$$

Then the DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$ is isomorphic to the DQ pair constructed by the following gluing datum

$$(\mathcal{A}_{T^*M_i}, \Psi_{ij}^{\hbar}, \tilde{a}_{ijk}P_{ijk}^c, K_{M_i}^{1/2}[\![\hbar]\!], \xi_{ij}^c).$$

Now we can define the relative Deligne-Fedosov class by comparing the DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$ with the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$.

Proposition 2.33. *Given a choice of spin structure $K_L^{1/2}$, there is a canonical bijection $\mathfrak{D}$ from the collection of isomorphism classes of DQ pairs $\mathfrak{DQ}(X, L)$ to the second relative cohomology group $H^2(X, L; \mathbb{C}_\hbar^*)$ defined by*

$$\mathfrak{D} : \mathfrak{DQ}(X, L) \longrightarrow H^2(X, L; \mathbb{C}_\hbar^*) = H^2(X, j_!\underline{\mathbb{C}_\hbar^*}_{X\setminus L})$$

$$(\mathcal{A}_{T^*M_i}, \Psi_{ij}^{\hbar}, \tilde{a}_{ijk}P_{ijk}^c, K_{M_i}^{1/2}[\![\hbar]\!], \xi_{ij}^c) \mapsto [\tilde{a}_{ijk}],$$

where $[\tilde{a}_{ijk}]$ is regarded as a class in the second Čech cohomology of the constructible sheaf $j_!\underline{\mathbb{C}_\hbar^*}_{X\setminus L}$, which is isomorphic to the second relative cohomology group $H^2(X, L; \mathbb{C}_\hbar^*)$. The map j refers to the open embedding $j : X \setminus L \hookrightarrow X$ and $j_!$ is the proper pushforward induced by j. To be more explicit, the sections of $j_!\underline{\mathbb{C}_\hbar^*}_{X\setminus L}$ are locally constant functions valued in $\mathbb{C}_\hbar^*$ whose restriction to L is 1.

Proof. Well-definedness of $\mathfrak{D}$: Assume that we have another gluing datum of the DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$ in the form

$$(\mathcal{A}_{T^*M_i'}, \Psi_{ij}'^{\hbar}, \tilde{a}'_{ijk}P_{ijk}'^c, K_{M_i'}^{1/2}[\![\hbar]\!], \xi_{ij}'^c).$$

Without loss of generality, we may assume that the gluing datum of the isomorphism between the two gluing data is given by $(\Phi_i^{\hbar}, \tilde{Q}_{ij})$, where $\Phi_i^{\hbar}$ is the same as the $\Phi_i^{\hbar}$ between two canonical DQ algebroids constructed in the proof of Proposition 2.29. In fact, since every automorphism of $\mathcal{A}_{T^*M_i}$ is inner, we can always make the quantized symplectic transformation part the same as $\Phi_i^{\hbar}$ by modifying the natural transformation part Q_{ij}. By the 2-commutative diagram of the isomorphism $(\Phi_i^{\hbar}, \tilde{Q}_{ij})$, we have

$$\mathrm{Ad}_{\exp(b_{ij}\partial_t)\tilde{Q}_{ij}} \circ \Psi_{ij}'^{\hbar} \circ \Phi_j^{\hbar} = \Phi_i^{\hbar} \circ \Psi_{ij}^{\hbar},$$
$$\Phi_i^{\hbar}(\tilde{a}_{ijk}P_{ijk}^c)\tilde{Q}_{ij}\Psi_{ij}'^{\hbar}(\tilde{Q}_{jk}) = \tilde{Q}_{ik}\tilde{a}'_{ijk}P_{ijk}'^c \in \mathcal{A}_{T^*M_i'}.$$

As a result, we know that $\mathrm{Ad}_{\tilde{Q}_{ij}} = \mathrm{Ad}_{Q_{ij}}$ and hence $\tilde{Q}_{ij} = e_{ij}Q_{ij}$ for some $e_{ij} \in \mathbb{C}_\hbar^*$. Also, by the compatibility condition with the DQ module gluing morphisms $\xi_{ij}^c, \xi_{ij}'^c$, we know that $e_{ij} = 1$ near L. Therefore, by comparing with the compatibility condition of the natural transformations in the proof of Proposition 2.29, we get

$$\tilde{a}_{ijk}e_{ij}e_{jk} = \tilde{a}'_{ijk}e_{ik}$$

and hence the *Čech* 2-cochain $\tilde{a}'_{ijk}$ is cohomologous to $\tilde{a}_{ijk}$.

Injectivity of $\mathfrak{D}$: Assume $\tilde{a}_{ijk}e_{ij}e_{jk} = \tilde{a}'_{ijk}e_{ik}$ for some e_{ij}. Then we modify the gluing data of the isomorphism in the proof of Proposition 2.29 to be $(\Phi_i^{\hbar}, e_{ij}Q_{ij}, \varphi_i)$. This gluing datum of an isomorphism is compatible with the gluing data of the DQ pairs and gives an isomorphism between them.

Surjectivity of $\mathfrak{D}$: Consider the DQ pair with gluing datum

$$(\mathcal{A}_{T^*M_i}, \Psi_{ij}^{\hbar}, \tilde{a}_{ijk}P_{ijk}^c, K_{M_i}^{1/2}[\![\hbar]\!], \xi_{ij}^c).$$

for any given class $[\tilde{a}_{ijk}]$. □

Definition 2.34 (Relative Deligne-Fedosov Class)**.** For any DQ pair $(\mathcal{A}_X, \mathcal{M}_L)$, the *relative Deligne-Fedosov class* $\mathfrak{D}(\mathcal{A}_X, \mathcal{M}_L) \in H^2(X, L; \mathbb{C}^*_\hbar)$ with respect to a given spin structure $K_L^{1/2}$ is defined to be the corresponding class under the bijection $\mathfrak{D}$ in Proposition 2.33.

The relative Deligne-Fedosov class gives a classification of, and a complete invariant for, DQ pairs. In fact, as we will see in §3, the relative Deligne-Fedosov class gives the intrinsic definition of quantum periods.

Remark 2.35. One can also express the classification of the DQ pairs $(\mathcal{A}_X, \mathcal{M}_L)$ in terms of the cohomology of stacks of 2-groups used in the classification of DQ algebroids in [Po2, §4]. Consider the sheaf of automorphisms $\mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c)$ of the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ which is in fact a stack of 2-groups for reasons similar to those in the classification of DQ algebroids in [Po2]. There is a natural morphism between the stacks of 2-groups

$$j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L}[1] \longrightarrow \mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c) \tag{2.36}$$

given by regarding sections of $j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L}$ as a natural transformation of the identity 1-morphism of $\mathcal{A}_X^c$ by multiplication, where $j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L}[1]$ means the shift of the stack of 2-groups $j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L}$ by 1, i.e. the stack of 2-groups corresponding to $[j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L} \to 1]$. The fact that the restriction of the sections of $j_!\underline{\mathbb{C}^*_\hbar}$ near L is 1 corresponds to the compatibility with the identity morphism of the DQ module $\mathcal{M}_L^c$. On the local model, the stack of 2-groups $\mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c)$ is isomorphic to the associated stack of 2-groups of the sheaf of crossed modules

$$[\mathcal{A}^*_{T^*M} \longrightarrow \mathcal{A}ut(\mathcal{A}_{T^*M}, K_M^{1/2}[\![\hbar]\!])],$$

where the arrow is given by conjugation on the DQ algebra $\mathcal{A}_{T^*M}$ and multiplication on the DQ module $K_M^{1/2}[\![\hbar]\!]$. It is a well-known fact in microlocal analysis that the above arrow is a surjective morphism of sheaves. Hence there is a short exact sequence

$$1 \longrightarrow j_!\underline{\mathbb{C}^*_\hbar}_{T^*M\setminus M} \longrightarrow \mathcal{A}^*_{T^*M} \longrightarrow \mathcal{A}ut(\mathcal{A}_{T^*M}, K_M^{1/2}[\![\hbar]\!]) \longrightarrow 1$$

on the local model. Therefore, the natural morphism (2.36) is indeed an isomorphism. Since the set of isomorphism classes of DQ pairs is in one-to-one correspondence with the first cohomology of the stack of 2-groups $\mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c)$, we know that the classification of DQ pairs is given by a class in

$$H^1(X, \mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c)) = H^1(X, j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L}[1]) = H^2(X, j_!\underline{\mathbb{C}^*_\hbar}_{X\setminus L}) = H^2(X, L; \mathbb{C}^*_\hbar).$$

Also, there is a natural sequence of morphisms

$$\mathfrak{Aut}_{\mathcal{A}_X^c}(\mathcal{M}_L^c) \longrightarrow \mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c) \longrightarrow \mathfrak{Aut}(\mathcal{A}_X^c)$$

among the stacks of automorphisms of the DQ module $\mathcal{M}_L^c$, the DQ pair $(\mathcal{A}_X^c, \mathcal{M}_L^c)$ and the DQ algebroid $\mathcal{A}_X^c$, where the first arrow is given by putting the identity as the autoequivalence of the DQ algebroid $\mathcal{A}_X^c$ and the second arrow is given by forgetting the isomorphism of the DQ module. This sequence of morphisms behaves as an exact triangle and induces a long

exact sequence of cohomology

$$\begin{array}{ccccc} H^0(X, \mathfrak{Aut}_{\mathcal{A}_X^c}(\mathcal{M}_L^c)) & \longrightarrow & H^0(X, \mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c)) & \longrightarrow & H^0(X, \mathfrak{Aut}(\mathcal{A}_X^c)) \\ \cong\downarrow & & \cong\downarrow & & \cong\downarrow \\ H^0(L; \mathbb{C}_\hbar^*) & \xrightarrow{\delta} & H^1(X, L; \mathbb{C}_\hbar^*) & \longrightarrow & H^1(X; \mathbb{C}_\hbar^*) \end{array},$$

$$\begin{array}{ccccc} \longrightarrow H^1(X, \mathfrak{Aut}_{\mathcal{A}_X^c}(\mathcal{M}_L^c)) & \longrightarrow & H^1(X, \mathfrak{Aut}(\mathcal{A}_X^c, \mathcal{M}_L^c)) & \longrightarrow & H^1(X, \mathfrak{Aut}(\mathcal{A}_X^c)) \\ \cong\downarrow & & \mathfrak{D}\cong\downarrow & & \mathfrak{D}\cong\downarrow \\ \longrightarrow H^1(L; \mathbb{C}_\hbar^*) & \xrightarrow{\delta} & H^2(X, L; \mathbb{C}_\hbar^*) & \longrightarrow & H^2(X; \mathbb{C}_\hbar^*) \end{array}$$

which is isomorphic to the long exact sequence of relative cohomology with $\mathbb{C}_\hbar^*$-coefficients. The last two vertical arrows refer to the relative Deligne-Fedosov class and the usual Deligne-Fedosov class respectively. From this long exact sequence, it is also easy to see that if the underlying DQ algebroid of a DQ pair is canonical, then the relative Deligne-Fedosov class has a lift to $H^1(L; \mathbb{C}_\hbar^*)$ up to the ambiguity of $H^1(X; \mathbb{C}_\hbar^*)$, which corresponds to autoequivalences of the DQ algebroid. This is exactly where the ambiguity of the quasi-canonical DQ modules defined in Definition 2.31 comes from.

3. WKB Approximation, Quantum Periods and Deformation Quantization

3.1. **WKB Approximation and the Original Definition of Quantum Periods.** In the context of quantum curves, the key notion is the WKB approximation of a differential or difference equation. For the Hori-Vafa mirror curve, the quantum curve equation is given by the difference equation

$$(-1 + \widehat{W}_{HV}(\exp(x), \exp(\hbar\partial_x))\psi(x) = 0, \tag{3.1}$$

where the operator $-1 + \widehat{W}_{HV}$ is a quantization of the classical defining function $-1 + W_{HV}$ of the mirror curve C in the torus $\operatorname{Spec}\mathbb{C}[M]$, where M is a rank 2 symplectic lattice. In fact, we can make sense of $\widehat{W}_{HV}$ as an element in the so-called quantum torus $\mathbb{C}_\hbar[\widehat{M}]$, whose sheafification gives a canonical deformation quantization of the classical torus $\operatorname{Spec}\mathbb{C}[M]$. The quantum torus $\mathbb{C}_\hbar[\widehat{M}]$ is defined as follows. First, we consider the noncommutative algebra $\mathbb{C}[q^{\pm\frac{1}{2}}][\widehat{M}]$ over $\mathbb{C}[q^{\pm\frac{1}{2}}]$ whose $\mathbb{C}[q^{\pm\frac{1}{2}}]$-basis is given by $\hat{z}^m$, $m \in M$ and whose relations are given by

$$\hat{z}^m \cdot \hat{z}^{m'} = q^{\frac{1}{2}\omega(m,m')}\hat{z}^{m+m'}, \tag{3.2}$$

where ω is the symplectic form on the lattice. The quantum torus $\mathbb{C}_\hbar[\widehat{M}]$ is defined by the complete tensor product with $\mathbb{C}_\hbar$

$$\mathbb{C}_\hbar[\widehat{M}] := \mathbb{C}[q^{\pm\frac{1}{2}}][\widehat{M}]\widehat{\otimes}_{\mathbb{C}[q^{\pm\frac{1}{2}}]}\mathbb{C}_\hbar := \varprojlim_k \mathbb{C}[q^{\pm\frac{1}{2}}][\widehat{M}] \otimes_{\mathbb{C}[q^{\pm\frac{1}{2}}]} \mathbb{C}_\hbar/\hbar^k,$$

where we view $\mathbb{C}_\hbar$ as a $\mathbb{C}[q^{\pm\frac{1}{2}}]$-algebra via the identification

$$q = \exp(-\hbar).$$

The sheafification process of a noncommutative algebra in algebraic geometry is also called the Ore localization. The Ore localization works for any 1-parameter noncommutative deformation of a commutative algebra and hence works for the quantum torus $\mathbb{C}_\hbar[\widehat{M}]$ and any noncommutative algebra in the context of deformation quantization. See for example [Bo2, §2.4] for more details on Ore localization.

In the Hori-Vafa mirror Landau-Ginzburg model, if the classical superpotential W_{HV} is given by

$$W_{HV} := \sum_i a_i z^{m_i} \in \mathbb{C}[M],\ m_i \in M,\ a_i \in \mathbb{C}, \tag{3.3}$$

then the Hori-Vafa quantum superpotential $\widehat{W}$ is given by

$$\widehat{W}_{HV} := \sum_i a_i \hat{z}^{m_i} \in \mathbb{C}_\hbar[\widehat{M}],\ m_i \in M,\ a_i \in \mathbb{C} \tag{3.4}$$

and should be regarded as an element in the quantum torus $\mathbb{C}_\hbar[\widehat{M}]$ or equivalently a global section of the corresponding DQ algebra.

After choosing a symplectic basis of the lattice M and denoting the corresponding toric coordinates by $\widehat{X}, \widehat{P}$, we can represent the quantum superpotential $\widehat{W}_{HV}$ as an operator on functions with variable x by representing $\widehat{X}$ as $\exp(x)$ and $\widehat{P}$ as $\exp(\hbar\partial_x)$ to get the quantum curve equation 3.1. Note that for the quantum torus case in the Hori-Vafa mirror, the quantum curve equation 3.1 is indeed a difference equation since $\exp(\hbar\partial_x)$ is the translation operator by $\hbar$. For example, in the case of the quantum mirror curve of $K_{\mathbb{P}^2}$ explained in §4, the difference equation is given by

$$(-1+Te^x)\psi(x) + T\psi(x+\hbar) + Te^{\frac{1}{2}\hbar}e^{-x}\psi(x-\hbar) = 0,$$

where T is the complex structure modulus. The WKB approximation is an attempt to solve the difference equation 3.1 by assuming that the local expression of the wavefunction $\psi(x)$ takes the form

$$\exp(\frac{1}{\hbar}\int^x f_\hbar(x)dx),$$
$$f_\hbar(x) = \sum_{i\geq 0} f_i(x)\hbar^i.$$

and treating $\hbar$ as a formal parameter. Substituting this expression into the difference equation 3.1, one obtains rather complicated recursive relations for the $f_i(x)$, which allows one to compute out the WKB approximation order by order.

In fact, one is able to put the story of WKB approximation into the framework of mircolocal analysis and deformation quantization. Please refer to [AKKT] for more details on such a treatment. To be more specific, away from the branch locus of the projection to the x-direction, on an open subset U of the curve C, the *local WKB approximation* ψ is a section of

$$\mathcal{O}_U[\![\hbar]\!]\exp(\frac{1}{\hbar}\int^x f_0(x)dx), \tag{3.5}$$

where $p = f_0(x)$ is the implicit function obtained from the classical mirror curve equation $-1+W=0$ of a classical superpotential W. The sections of (3.5) are just the formal products. The sheaf (3.5) naturally has the structure of a DQ module over the DQ algebra $\mathcal{A}_{T_{HV}}$ defined by the quantum torus on the Hori-Vafa torus T_{HV}. In fact, the action of $\hat{x}$ is given by the usual multiplication and the action of $\hat{p}$ is given by the Leibniz rule

$$\hat{p}\cdot\left(g(x,\hbar)\exp(\frac{1}{\hbar}\int^x f_0(x)dx)\right) := \hbar\partial_x\cdot\left(g(x,\hbar)\exp(\frac{1}{\hbar}\int^x f_0(x)dx)\right)$$
$$= \hbar(\partial_x g(x,\hbar))\exp(\frac{1}{\hbar}\int^x f_0(x)dx) + g(x,\hbar)f_0(x)\exp(\frac{1}{\hbar}\int^x f_0(x)dx).$$

In fact, this DQ module (3.5) is a simple holonomic DQ module supported on the open subset U of the curve C generated by $\exp(\frac{1}{\hbar}\int^x f_0(x)dx)$, whose annihilator is generated by $\hat{p}-f_0(\hat{x})$.

Now consider any global section $\widehat{W}$ of the quantum torus $\mathcal{A}_{T_{HV}}$ whose classical limit is the superpotential W. For example the Hori-Vafa quantum superpotential (3.4) W_{HV} can be such a choice. Then collections of WKB approximations is given by a subsheaf of (3.5) satisfying the equation $(-1+\widehat{W})\psi = 0$. There is a rather complicated recursive relation deciding ψ order by order ([AKKT, Equation (4.6)]) and hence the solution locally exists as a section in (3.5) up to a constant in $\mathbb{C}^*_\hbar$ ([AKKT, Theorem 4.1]).[2]

Remark 3.6. Since there are different branches of the branched cover $\widetilde{C} \longrightarrow \mathbb{C}_x$, where $\widetilde{C}$ is the preimage of the mirror curve C under the coordinate-wise exponential map $\mathrm{Exp} : \mathbb{C}^2 \longrightarrow (\mathbb{C}^*)^2$, there are also different branches of the WKB approximation by applying the recursive relation to different classical branches $f_0(x)$. Therefore, although the original aim of the WKB approximation is to obtain a local analytic function on $\mathbb{C}_x$ solving the difference equation 3.1, the wavefunction ψ we finally get is not naturally a local function on $\mathbb{C}_x$, but a local formal product on the curve $\widetilde{C}$.

Remark 3.7. In the WKB approximation, $\hbar$ should always be regarded as a formal parameter and the convergence with respect to $\hbar$ usually fails. This is also reasonable from the perspective of mirror symmetry since in the A-model, $\hbar$ is the genus parameter and the Gromov-Witten invariants grow very fast with respect to the genus, which means that the partition function of the A-model never converges with respect to the genus parameter.

In order to get a well-defined local analytic function on $\mathbb{C}_x$ solving the difference equation 3.1, which is also convergent with respect to $\hbar$, we need a further process, called resurgence in the literature. After resurgence, we might obtain a global, non-perturbative interpretation of the topological string theory. However, for the quantum mirror symmetry conjecture in [ACDKV], everything should be considered before resurgence since in the A-model, $\hbar$ should be the genus parameter and the partition function is divergent with respect to $\hbar$.

Always from the branch locus, we can write the WKB approximation as

$$\begin{aligned} \psi &= \exp\left(\frac{1}{\hbar}\int^x \lambda_\hbar\right), \\ \lambda_\hbar &= \lambda_{Cl} + \lambda_{QC}, \end{aligned} \tag{3.8}$$

where $\lambda_\hbar$ is a local analytic 1-form on the mirror curve C, λ_{Cl} is the $\hbar^0$-term in $\lambda_\hbar$, and λ_{QC} is the sum of $O(\hbar)$-terms in $\lambda_\hbar$. The classical part is the pullback of the 1-form $pdx = \log P d\log X$, which is multivalued on C, and the quantum correction part λ_{QC} is indeed single-valued on C for second-order difference equations ([ML, §2.5, Theorem 1], [Ba, §1.2.2]). Hence, one can regard $\lambda_\hbar$ as the quantum version of the classical 1-form $\log P d\log X$.

Take a loop on the curve C away from the branch locus, the local WKB approximation ψ can be analytically continued along the loop becuase of the local existence of the WKB solutions. Therefore, one can define the *quantum periods* by the monodromy of the WKB approximation along the loop

$$\oint \lambda_\hbar,$$

which is the original definition of the quantum periods in the physics literature such as [ACDKV].

In closed string theory, the relevant quantum periods are the *big quantum periods*

$$\int_{\partial\Gamma} \lambda_\hbar,$$

[2]In fact, the subsheaf of WKB approximations can be identified with the Hom sheaf between the Landau-Ginzburg DQ module (Definition 3.10) and the DQ module (3.5), which is a rank-1 $\mathbb{C}_\hbar$-local system since the DQ modules involved are both simple holonomic (Proposition 3.12).

which are the periods along the boundary of a relative 2-cycle Γ to the mirror curve C shown in Figure 3. The quantum special geometry relation of these big quantum periods gives the partition function of the B-model, and the quantum closed string mirror symmetry conjecture in [ACDKV] predicts that this partition function is the same as the partition function of the mirror A-model.

Motivating Observation 3.9. However, the original definition of the big quantum periods is unsatisfactory in the following sense. In order to define the big quantum periods, we need to first choose a symplectic basis of the lattice M, or Darboux coordinates X, P of the torus, and represent the quantum superpotential $\widehat{W}$ as an operator on functions on $\mathbb{C}_x$ to define the WKB approximation. If we switch to another choice of the symplectic basis, we would get a different WKB approximation. But the big quantum periods, or monodromy of the WKB approximations, are the same! This means that the big quantum periods do not depend on the choice of the Darboux coordinates and should have an intrinsic meaning based on the noncommutative geometry of the quantum torus and the quantum superpotential $\widehat{W}$. This motivates us to find an intrinsic way to define the big quantum periods just from the data of deformation quantization. From the mirror symmetry perspective, this intrinsic feature of the quantum periods also makes sense since the big quantum periods should correspond to the closed string theory and the choices of Darboux coordinates, as we will explain in §7, are additional data coming from open string theory.

3.2. **Deformation Quantization and the Intrinsic Definition of Quantum Periods.** Via the theory of deformation quantization of holomorphic symplectic manifolds and holomorphic Lagrangian submanifolds, we are able to obtain an intrinsic description of the quantum periods arising from the WKB approximation in §3.1. In the quantum curve setting, we should consider a holomorphic symplectic surface (X, ω) together with a closed embedded curve C inside X. Because of the dimension of X, this curve C is a Lagrangian submanifold of X. From §2, we know that given a spin structure $K_C^{1/2}$, we have the canonical DQ pair $(\mathcal{A}_X^c, \mathcal{M}_C^c)$, where $\mathcal{A}_X^c$ is the canonical DQ algebroid and $\mathcal{M}_C^c$ is a simple holonomic DQ module supported on C over $\mathcal{A}_X^c \otimes_{\mathbb{C}} \mathbb{C}_{w_2(C)}$. Since, in the context of quantum curves, the curve C is a smooth algebraic curve, the second Stiefel-Whitney class is always zero and the square root of K_C always exists as a line bundle. Hence the invertible $\mathbb{C}$-algebroid $\mathbb{C}_{w_2(C)}$ is indeed trivial. Therefore, we can omit the $\mathbb{C}_{w_2(C)}$-factor in the study of quantum curves. The canonical deformation quantization structure comes entirely from the classical geometry of (X, C) and has no quantum information.

In order to quantize the curve C and get nontrivial quantum corrections, such as the quantum periods in the quantum curve story [ACDKV], we need an additional piece of data, which is another simple holonomic DQ module supported on C given by the quantum superpotential $\widehat{W}$.

Definition 3.10. (Quantum Landau-Ginzburg Model and Landau-Ginzburg DQ Module) Consider a *classical Landau-Ginzburg model* (X, W), where X is a holomorphic symplectic surface and W is a global holomorphic function on X such that $\mathcal{O}_X/(-1+W)$ is the structure sheaf of a smooth curve C. Then a *quantum Landau-Ginzburg model* over (X, W) is a pair $(\mathcal{A}_X, \widehat{W})$, where $\mathcal{A}_X$ is a DQ algebra on X and $\widehat{W}$ is a global section of $\mathcal{A}_X$ such that the classical limit of $\widehat{W}$ is W. The global section $\widehat{W}$ is called the *quantum superpotential* of the quantum Landau-Ginzburg model. From a quantum Landau-Ginzburg model $(\mathcal{A}_X, \widehat{W})$, we call the left DQ-module supported on C

$$\mathcal{M}_{LG} := \mathcal{A}_X/\mathcal{A}_X(-1+\widehat{W}),$$

by the *Landau-Ginzburg DQ module* of the quantum Landau-Ginzburg model. This Landau-Ginzburg DQ module provides an intrinsic description of the quantum curve.

Remark 3.11. One can also generalize the definition of quantum Landau-Ginzburg models and Landau-Ginzburg DQ modules using the language of DQ algebroids. In this case, one should replace the global section $-1+\widehat{W}$ with an injective morphism of DQ-modules

$$\mathcal{A}_X \longrightarrow \mathcal{A}_X,$$

where $\mathcal{A}_X$ is just a DQ algebroid on X. The Landau-Ginzburg DQ module is defined as the cokernel of the injective morphism above.

Proposition 3.12. *Under the setup in Definition 3.10, the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ is a simple holonomic DQ module supported on C with global generator* 1.

Proof. By Definition 2.15, we only need to show that $\mathcal{M}_{LG}$ is $\hbar$-torsion free. Suppose that there is a local section $\widehat{A} \in \mathcal{A}_X$ such that $\hbar\widehat{A} = \widehat{Q}(-1+\widehat{W}), \widehat{Q} \in \mathcal{A}_X$. Putting the equality in the classical limit $\mathcal{O}_X$, we have $\sigma_0(\widehat{Q})(-1+W) = 0$. Since X is a complex manifold, we know that locally $(-1+W)$ is a non-zero-divisor in $\mathcal{O}_X$, which implies $\sigma_0(\widehat{Q}) = 0$. Then we are able to write $\widehat{Q} = \hbar\widehat{Q}', \widehat{Q}' \in \mathcal{A}_X$. Since $\mathcal{A}_X$ is $\hbar$-torsion free, we have $\widehat{A} = \widehat{Q}'(-1+\widehat{W})$, which implies $\widehat{A}$ will induce the zero element in $\mathcal{M}_{LG}$ and finishes the proof. □

On a quantum Landau-Ginzburg model, we get the following *Landau-Ginzburg DQ pair*

$$(\mathcal{A}_X, \mathcal{M}_{LG})$$

as a quantization of the classical pair (X, C). As we will see in the intrinsic definition of quantum periods (Definition 3.13), this Landau-Ginzburg DQ pair is exactly the content of the quantization of the curve C, or the so-called quantum curve, in the physics literature [ACDKV].

In the Hori-Vafa mirror symmetry [HV], the ambient symplectic surface X is just an algebraic torus $T_{HV} = \operatorname{Spec}\mathbb{C}[M]$, where M is a rank 2 symplectic lattice and the DQ algebra is just the quantum torus $\mathbb{C}_\hbar[\widehat{M}]$, which is in fact a canonical DQ algebra in the sense that the underlying DQ algebroid, or the Deligne-Fedosov class, is trivial (see the computation on the triple intersections $N_{i,j,k}$ in (3.18)). From the quantum superpotential (3.4), one can define the Landau-Ginzburg DQ-Module $\mathcal{M}_{LG}$ which provides the quantization of the curve in the closed string case. As we will review later in §5, by the construction of the Gross-Siebert program in [CPS], we get a partial compactification X of the algebraic torus $\operatorname{Spec}\mathbb{C}[M]$, a proper Landau-Ginzburg superpotential W on X together with a deformation quantization given by the quantum wall structure in [Bo2; Bo4]. The Gross-Siebert program provides a compactification of the quantum Landau-Ginzburg model in the Hori-Vafa mirror symmetry.

Defined by an additional piece of data $\widehat{W}$, this Landau-Ginzburg DQ pair is usually different from the canonical DQ pair. In fact, as we will show later, the difference, which is just the relative Deligne-Fedosov class defined in Definition 2.34, is exactly the quantum correction of the classical period integral, which contains all the information of the quantum closed string B-model. The relative Deligne-Fedosov class (Definition 2.34) motivates the following intrinsic definition of the big quantum 1-periods arising in the quantum curve theory.

Definition 3.13 (Intrinsic Definition of Big Quantum 1-Periods)**.** Let X be a smooth holomorphic symplectic surface and C be a closed embedded smooth holomorphic curve through the closed embedding $i : C \hookrightarrow X$. Consider a DQ pair $(\mathcal{A}_X, \mathcal{M})$ of a DQ algebroid on X and a simple holonomic DQ module $\mathcal{M}$ over $\mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(C)}$ supported on C. Also, assume that the DQ algebroid $\mathcal{A}_X$ is canonical, i.e. $\mathfrak{D}(\mathcal{A}_X) = 1$. Define the *quantum correction 1-period*

$$\mathrm{QC}(\mathcal{M}) \in H^1(C; \mathbb{C}_\hbar^*)/i^*H^1(X; \mathbb{C}_\hbar^*)$$

of $\mathcal{M}$ to be the lift of the relative Deligne-Fedosov class $\mathfrak{D}(\mathcal{A}_X, \mathcal{M}) \in H^2(X, C; \mathbb{C}_\hbar^*)$ to $H^1(C; \mathbb{C}_\hbar^*)$ under $-\delta$, where $\delta : H^1(C; \mathbb{C}_\hbar^*) \longrightarrow H^2(X, C; \mathbb{C}_\hbar^*)$ is the connecting homomorphism. The quantum correction 1-period $\mathrm{QC}(\mathcal{M})$ is well-defined in this case since the image

of the relative Deligne-Fedosov class $\mathfrak{D}(\mathcal{A}_X, \mathcal{M})$, which is exactly the Deligne-Fedosov class $\mathfrak{D}(\mathcal{A}_X)$ of the DQ algebroid $\mathcal{A}_X$, is trivial by assumption.

Equivalently, the quantum correction 1-period $\mathrm{QC}(\mathcal{M})$ is also given by the monodromy of the rank 1 $\mathbb{C}^*_\hbar$-local system

$$\mathcal{H}om_{\mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(C)}}(\mathcal{M}, \mathcal{M}^c_C),$$

where $\mathcal{M}^c_C$ is a quasi-canonical DQ module supported on C defined in Definition 2.31. Since the choice of quasi-canonical DQ module has an ambiguity of $i^*H^1(X; \mathbb{C}^*_\hbar)$, the monodromy of the Hom sheaf above defines a class in $H^1(C; \mathbb{C}^*_\hbar)/i^*H^1(X; \mathbb{C}^*_\hbar)$.

Remark 3.14. In Definition 3.13 of the quantum correction 1-period above, we need to take the lift of the relative Deligne-Fedosov class under $-\delta$. The negative sign here matches the order of $\mathcal{M}$ and $\mathcal{M}^c_C$ in the Hom sheaf. In addition, this negative sign is also compatible with the fact that the classical 1-form is defined to be $\log P d \log X$ in the context of quantum curves, while $-\log P d \log X$ is a primitive of the holomorphic symplectic form $d \log X \wedge d \log P$.

In the Hori-Vafa mirror Landau-Ginzburg model where the ambient symplectic surface is the Hori-Vafa torus $T_{HV} := \operatorname{Spec} \mathbb{C}[M]$, pick a symplectic basis of the symplectic lattice M and denote the corresponding coordinates by X, P, where $X := \exp(x)$, $P = \exp(p)$. The coordinates x, p represent the position and the momentum respectively in quantum mechanics. The symplectic form ω takes the form $\omega = d \log X \wedge d \log P = dx \wedge dp$ on the algebraic torus. Assume that $W = 1$ defines a smooth curve C inside X. Also assume $\frac{\partial W}{\partial P} \not\equiv 0$ and consider the projection to the X coordinate $\pi : (\mathbb{C}^*)^2 \to \mathbb{C}^*$. To study the quantum curve and WKB approximation in the Landau-Ginzburg model, consider the following exponential commutative diagram

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{\mathrm{Exp}} & (\mathbb{C}^*)^2 \\ \uparrow & & \uparrow \\ \widetilde{C} & \xrightarrow{\mathrm{Exp}} & C \\ \downarrow{\scriptstyle \widetilde{\pi}} & & \downarrow{\scriptstyle \pi} \\ \mathbb{C} & \xrightarrow{\exp} & \mathbb{C}^* \end{array},$$

where $\mathrm{Exp} = \exp \times \exp : (\mathbb{C})^2 \longrightarrow (\mathbb{C}^*)^2$ is the coordinate-wise exponential map and the top square is a fiber product. Let $\Delta \subseteq \mathbb{C}^*$ be the branch locus of $\pi|_C$. Away from the branch locus, both

$$\pi : C \setminus \pi^{-1}(\Delta) \longrightarrow \mathbb{C}^* \setminus \Delta$$

and

$$\widetilde{\pi} : \widetilde{C} \setminus \widetilde{\pi}^{-1}(\exp^{-1}(\Delta)) \longrightarrow \mathbb{C} \setminus \exp^{-1}(\Delta)$$

are local isomorphisms. As a result, the restriction of K_C to $C \setminus \pi^{-1}(\Delta)$ is trivial. Therefore, we can always choose the spin structure $K^{1/2}_{C \setminus \pi^{-1}(\Delta)}$ to be trivial via the 1-form $\frac{dX}{X} = dx$. In the Hori-Vafa Landau-Ginzburg model, we have the following equality between the quantum periods in the WKB approximation and the quantum correction 1-periods in Definition 3.13, which justifies our intrinsic definition of quantum periods (Definition 3.13). Indeed, the WKB approximation should be regarded as a method to compute the intrinsically defined quantum periods (Definition 3.13) in the case of the Hori-Vafa mirror curve.

Theorem 3.15. *In the Hori-Vafa Landau-Ginzburg model with superpotential W on the Hori-Vafa torus T_{HV} in (3.3), choose the coordinates X, P and assume that $W = 1$ defines a*

smooth curve C and $\frac{\partial W}{\partial P}$ is not identically zero. Denote $T_{HV}\backslash\pi|_C^{-1}(\Delta)$ by T°_{HV} and $C\backslash\pi|_C^{-1}(\Delta)$ by C°. Also, choose the spin structure $K_{C^\circ}^{1/2}$ to be

$$(\mathcal{O}_{C^\circ}, \mathcal{O}_{C^\circ}^{\otimes 2} \xrightarrow{m} \mathcal{O}_{C^\circ} \xrightarrow{\frac{dX}{X}=dx} K_{C^\circ}) \ ,$$

where m means the multiplication of $\mathcal{O}_{C^\circ}$ and the second arrow is defined by multiplying by the nowhere vanishing 1-form $\frac{dX}{X} = dx$. Let $\mathcal{M}_{LG}$ be the Landau-Ginzburg DQ module defined by a quantum superpotential $\widehat{W}$ with classical limit W (which can indeed be any lift of the classical superpotential W and does not need to be the specific one (3.4)). Then there exists a quasi-canonical DQ module $\mathcal{M}^c_{C^\circ}$ with respect to (T°_{HV}, C°) supported on C° over the quantum torus $\mathcal{A}_{T^\circ_{HV}}$ such that for any continuous loop

$$\gamma : [0,1] \longrightarrow C^\circ := C \setminus \pi^{-1}(\Delta)$$

on the curve C° and any lift of γ to

$$\widetilde{\gamma} : [0,1] \to \widetilde{C}^\circ := \widetilde{C} \setminus \widetilde{\pi}^{-1}(\exp^{-1}(\Delta)),$$

we have

$$\mathrm{Mono}_\gamma \left(\mathcal{H}om_{\mathcal{A}_{T^\circ_{HV}}}(\mathcal{M}_{LG}|_{C^\circ}, \mathcal{M}^c_{C^\circ}) \right) = \exp \frac{1}{\hbar} \int_{\widetilde{\gamma}} \lambda_{QC} \in \mathbb{C}^*_\hbar,$$

where Mono_γ means the monodromy of the local system along γ and $\lambda_{QC} = O(\hbar)$ is the quantum correction 1-form obtained from the WKB approximation in (3.8).

Proof. Construction of Quasi-canonical DQ Module $\mathcal{M}^c_{C^\circ}$: Via the nowhere vanishing 1-form $\frac{dX}{X} = dx$, we can trivialize the spin structure and hence ignore the square-root issue in this construction and computation. Cover the Hori-Vafa torus T°_{HV} and C° with the fundamental domains of the exponential covering map Exp. Divide the cover into two parts $\{N_i\}, \{V_i\}$, where the intersection of each N_i with C° is empty, while the intersection of each V_i with C° is non-empty. In order to adapt to the local models of the curve C°, we will choose different local models of the DQ algebra on $\{N_i\}$ and $\{V_i\}$.

For N_i, just choose the local model of the DQ algebra to be the standard DQ algebra $\mathcal{A}_{\widetilde{N}_i}$ of the cotangent bundle $T^*\mathbb{C} = \mathbb{C}^2$, where $\widetilde{N}_i$ is a branch of $\mathrm{Exp}^{-1}(N_i)$. For V_i, shrink V_i such that the implicit function theorem for the curve holds and take a branch $\widetilde{V}_i$ under the Exp-map. Then choose the local model of the DQ algebra to be the standard DQ algebra $\mathcal{A}_{\widetilde{Q}_i}$, where $\widetilde{Q}_i$ is the image of $\widetilde{V}_i$ under the map

$$\begin{aligned} \Phi_i : \widetilde{V}_i &\longrightarrow \widetilde{Q}_i \subseteq \mathbb{C}^2 \\ (x,p) &\mapsto (x, p - f_i(x)), \end{aligned}$$

and $f_i(x)$ is the implicit function solved from the equation of the curve $\widetilde{C}$ on $\widetilde{V}_i$. Denote the image of $\widetilde{C} \cap \widetilde{V}_i$ under Φ_i by M_i, which is just the zero section of $\widetilde{Q}_i$. Then the local model of the quasi-canonical DQ module is just given by $\mathcal{O}_{M_i}[\![\hbar]\!]$.

Now we investigate the gluing morphisms of the local models of the DQ algebra and the quasi-canonical DQ module. The gluing morphism $\Psi^\hbar_{N_i,N_j}$ of local models of the DQ algebra between N_i and N_j is given by a deck transformation of the form

$$D^\hbar_{ij} := \mathrm{Ad}_{\exp \frac{1}{\hbar}(-2\pi i m_p^{ij} \hat{x} + 2\pi i m_x^{ij} \hat{p})}, \ (m_x^{ij}, m_p^{ij}) \in \mathbb{Z}^2.$$

The gluing morphism $\Psi^\hbar_{N_i,V_j}$ between N_i and V_j takes the form

$$D^\hbar_{ij} \circ \mathrm{Ad}_{\exp \frac{1}{\hbar} \int^{\hat{x}} f_j(x) dx},$$

The gluing morphism $\Psi^{\hbar}_{V_i,V_j}$ between V_i and V_j is given by

$$\mathrm{Ad}_{\exp(-\frac{1}{\hbar}\int^{\hat{x}} f_i(x)dx)} \circ D^{\hbar}_{ij} \circ \mathrm{Ad}_{\exp\frac{1}{\hbar}\int^{\hat{x}} f_j(x)dx}.$$

As a quantized symplectic transformation, it is just the same as the translation transformation

$$T^{\hbar}_{ij} := \mathrm{Ad}_{\exp\frac{1}{\hbar}(2\pi i m^{ij}_x \hat{p})}$$

by computation. Under the choice of quantized symplectic transformations $\Psi^{\hbar}_{V_i,V_j}$ above, take the gluing morphism ξ^c_{ij} of the local models of the DQ module to be the translation morphism

$$\begin{aligned} \xi^c_{ij} : \mathcal{O}_{M_j}[\![\hbar]\!] &\longrightarrow T_*\mathcal{O}_{M_i}[\![\hbar]\!]. \\ f(x) &\mapsto f(x + 2\pi i m^{ij}_x). \end{aligned} \tag{3.16}$$

It is easy to check that ξ^c_{ij} is a DQ module isomorphism under the quantized symplectic transformation $\Psi^{\hbar}_{V_i,V_j}$. Taking the natural transformation part to be the identity, we get the following gluing datum of a DQ pair on $(T^{\circ}_{HV}, C^{\circ})$

$$(\{\mathcal{A}_{\tilde{N}_i}, \mathcal{A}_{\tilde{Q}_i}\}; \{\Psi^{\hbar}_{N_i,N_j}, \Psi^{\hbar}_{N_i,V_j}, \Psi^{\hbar}_{V_i,V_j}\}; Id; \mathcal{O}_{M_i}[\![\hbar]\!]; \xi^c_{ij}). \tag{3.17}$$

Now we would like to show that there exist quantized contact transformation lifts of the quantized symplectic transformations above satisfying conditions (1), (2) and (3) in the definition of the canonical DQ pair (Definition 2.27). The quantized contact transformation lifts of $\Psi^{\hbar}_{N_i,N_j}$ are taken to be

$$\tilde{\Psi}^{\hbar}_{N_i,N_j} := \tilde{D}^{\hbar}_{ij} = \mathrm{Ad}_{\exp\hat{\tau}(-2\pi i m^{ij}_p \hat{x} + 2\pi i m^{ij}_x \hat{p})}, \ (m^{ij}_x, m^{ij}_p) \in \mathbb{Z}^2.$$

The quantized contact transformation lifts of $\Psi^{\hbar}_{N_i,V_j}$ are taken to be

$$\tilde{\Psi}^{\hbar}_{N_i,V_j} := \tilde{D}^{\hbar}_{ij} \circ \mathrm{Ad}_{\exp\hat{\tau}\int^{\hat{x}} f_j(x)dx}.$$

Finally, for $\Psi^{\hbar}_{V_i,V_j}$, we take the quantized contact transformation lifts to be

$$\tilde{\Psi}^{\hbar}_{V_i,V_j} := \tilde{T}^{\hbar}_{ij} = \mathrm{Ad}_{\exp\hat{\tau}(2\pi i m^{ij}_x \hat{p})}$$

To check that these choices of quantized contact transformation lifts satisfy condition (1), we make the following computations

$$\begin{aligned} &\tilde{\Psi}^{\hbar}_{V_i,V_j} \circ \tilde{\Psi}^{\hbar}_{V_j,V_k} \circ \tilde{\Psi}^{\hbar}_{V_k,V_i} = Id, \\ &\tilde{\Psi}^{\hbar}_{V_i,V_j} \circ \tilde{\Psi}^{\hbar}_{V_j,N_k} \circ \tilde{\Psi}^{\hbar}_{N_k,V_i} = \mathrm{Ad}_{\exp\hat{\tau}(C + 4\pi^2(m^{jk}_p m^{ij}_x + m^{ki}_p m^{ij}_x + m^{ki}_p m^{jk}_x))}, \\ &\tilde{\Psi}^{\hbar}_{V_i,N_j} \circ \tilde{\Psi}^{\hbar}_{N_j,N_k} \circ \tilde{\Psi}^{\hbar}_{N_k,V_i} = \mathrm{Ad}_{\exp\hat{\tau}(C + 4\pi^2(m^{jk}_p m^{ij}_x + m^{ki}_p m^{ij}_x + m^{ki}_p m^{jk}_x))}, \\ &\tilde{\Psi}^{\hbar}_{N_i,N_j} \circ \tilde{\Psi}^{\hbar}_{N_j,N_k} \circ \tilde{\Psi}^{\hbar}_{N_k,N_i} = \mathrm{Ad}_{\exp\hat{\tau}4\pi^2(m^{jk}_p m^{ij}_x + m^{ki}_p m^{ij}_x + m^{ki}_p m^{jk}_x)}, \end{aligned} \tag{3.18}$$

where $C \in \mathbb{C}$ is a constant coming from different choices of the primitive functions of the implicit functions $f_i(x)$. Since there is no quantum correction term in those compositions, it follows that the gluing datum (3.17) satisfies condition (1) in Definition 2.27. For conditions (2) and (3), it is easy to check that the gluing morphism ξ^c_{ij} is a microdifferential isomorphism under the quantized contact transformation $\tilde{\Psi}^{\hbar}_{V_i,V_j}$ and its classical limit is the transition function of $K^{1/2}_{C^{\circ}}$. Therefore, the gluing datum (3.17) gives the gluing datum of the canonical DQ pair. Since the underlying DQ algebroid of the DQ pair is just the associated DQ algebroid of the quantum torus $\mathcal{A}_{T^{\circ}_{HV}}$, the gluing datum $(\mathcal{O}_{M_i}[\![\hbar]\!], \xi^c_{ij})$ will give a quasi-canonical DQ module $\mathcal{M}^c_{C^{\circ}}$ over the quantum torus $\mathcal{A}_{T^{\circ}_{HV}}$ with respect to $(T^{\circ}_{HV}, C^{\circ})$.

Computation of the WKB Approximation: Now we would like to compute the monodromy of the Hom sheaf $\mathcal{H}om_{\mathcal{A}_{T^\circ_{HV}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{C^\circ})$ along γ and compare it with the WKB approximation. Subdivide $[0,1]$ into $\{[(i-1)/n, i/n]\}_{1\leq i\leq n}$ such that $\gamma([i/n,(i+1)/n])$ is contained in some V_i defined above. Denote $\widetilde{\gamma}(i/n)$ and $\gamma(i/n)$ by $\widetilde{q}_i = (x_i, p_i)$ and $q_i = (X_i, P_i)$ respectively. Let $\{\widetilde{V}_i\}$ be the corresponding lifts which give an open cover of $\widetilde{\gamma}$. Consider the WKB approximation of the difference equation $(-1+\widehat{W})\psi = 0$ on each $\widetilde{V}_i$

$$\psi_{WKB,i}(x) = \psi_{Cl,i}(x)\cdot\psi_{QC,i}(x), x\in\widetilde{\pi}(\widetilde{V}_i\cap C^\circ).$$

The classical part $\psi_{Cl,i}$ takes the form

$$\psi_{Cl,i} = \exp\frac{1}{\hbar}\int_{\widetilde{q}_0}^{x} f_i(x)dx,$$

where $p = f_i(x)$ is the implicit function solved from the classical equation $-1+W=0$ on $\widetilde{V}_i$. The quantum correction part $\psi_{QC,i}$ takes the form

$$\psi_{QC,i} = \exp\frac{1}{\hbar}\int_{\widetilde{q}_0}^{x}\lambda_{QC}.$$

The integral is along the cover $\{\widetilde{V}_i\}$ of the path $\widetilde{\gamma}$ and ψ_{WKB} is also analytically continued along the path $\widetilde{\gamma}$. From the definition of the WKB approximation, the quantum correction 1-period is given by

$$\exp\int_{\widetilde{\gamma}}\frac{1}{\hbar}\lambda_{QC}\in\mathbb{C}^*_\hbar = \psi_{QC,n}(\tilde{x}_n).$$

Gluing Datum of the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$: In order to compute the monodromy of the Hom sheaf $\mathcal{H}om_{\mathcal{A}_{T^\circ_{HV}}}(\mathcal{M}_{LG}|_{C^\circ}, \mathcal{M}^c_{C^\circ})$ along γ, we also need to keep track of the transformation of the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ under the choice of trivialization of the DQ algebra in the gluing datum 3.17 and then compute the transition functions $c_{i,i+1}$ of the trivialization. The product of the transition functions $c_{i,i+1}$ gives the monodromy.

The pullback of the quantum superpotential $\widehat{W}$ under the exponential map Exp is just given by replacing $\widehat{X}$ by $\exp(\hat{x})$ and $\widehat{P}$ by $\exp(\hat{p})$, which we will denote by $\widetilde{\widehat{W}}$. Let $\widetilde{\mathcal{M}}_{LG}$ be the Landau-Ginzburg DQ module corresponding to $\widetilde{\widehat{W}}$. The local trivialization of the DQ algebra $\mathcal{A}_{T^\circ_{HV}}$ in (3.17) induces the corresponding local trivialization of the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ as follows

$$\mathcal{M}_{LG}|_{V_i}\xrightarrow{\mathrm{Exp}^\hbar}\mathrm{Exp}_*\,\widetilde{\mathcal{M}}_{LG}|_{\widetilde{V}_i}\xrightarrow{\mathrm{Ad}_{\hat{\psi}^{-1}_{Cl,i}}}\mathrm{Exp}_*\,\Phi_i^{-1}\widetilde{\mathcal{M}}^{QC}_{LG,i}\ ,$$

where $\hat{\psi}_{Cl,i}(\hat{x})$ is the associated operator of the function $\psi_{Cl,i}(x)$, $\mathrm{Ad}_{\hat{\psi}_{CL,i}}$ is a quantized symplectic transformation lift of Φ_i, and the DQ module $\widetilde{M}^{QC}_{LG,i}$ is the Landau-Ginzburg DQ module supported on M_i defined by the quantum superpotential

$$\widehat{W}^{QC} := \mathrm{Ad}_{\hat{\psi}^{-1}_{Cl,i}}(\widetilde{\widehat{W}}).$$

From the local trivialization above, we get the following gluing morphism of the Landau-Ginzburg DQ module

$$\mathrm{Ad}_{\widehat{\psi}^{-1}_{Cl,i+1}}\circ\mathrm{Ad}_{\widehat{\psi}_{Cl,i}} = Id : \widetilde{M}^{QC}_{LG,i}\longrightarrow\widetilde{M}^{QC}_{LG,i+1},\ 1\leq i\leq n-1,$$
$$T^\hbar : \widetilde{M}^{QC}_{LG,n}\longrightarrow T_*\widetilde{M}^{QC}_{LG,1},$$

where the first row is the identity since the WKB approximation ψ_{WKB} is analytically continued along $\widetilde{\gamma}$ and the second row is the translation morphism

$$T^{\hbar} = \mathrm{Ad}_{\exp \frac{1}{\hbar}(2\pi i m_x \hat{p})},\ m_x \in \mathbb{Z}.$$

induced by the deck transformation D between $\widetilde{V}_n$ and $\widetilde{V}_1$.

Computation of the Transition Functions and Monodromy: Using the local trivialization isomorphisms for the DQ algebra and DQ modules above, we can define a local trivialization Ξ_i of the Hom sheaf as follows

$$\Xi_i : Hom_{\mathcal{A}_{V_i}}(\mathcal{M}_{LG}|_{V_i}, \mathcal{M}^c_{C^\circ}|_{V_i}) \longrightarrow Hom_{\mathcal{A}_{\widehat{Q}_i}}(\widetilde{\mathcal{M}}^{QC}_{LG,i}, \mathcal{O}_{M_i}[\![\hbar]\!]) \tag{3.19}$$

$$\xrightarrow{Sol_i} \{\varphi \in \Gamma(M_i, \mathcal{O}_{M_i}[\![\hbar]\!]) | (-1 + \widehat{W}_i^{QC})\varphi = 0\} \xrightarrow{\lambda\psi_{QC,i} \longmapsto \lambda, \lambda \in \mathbb{C}_\hbar} \mathbb{C}_\hbar$$

where the map Sol_i means taking the solution of the equation defined by $-1 + \widehat{W}_i^{QC}$.

Now we investigate the transition functions under the local trivializations Ξ_i. First, we look at the transition function $c_{i,i+1}$ for $1 \leq i \leq n-1$. The computation is shown in the following diagram

$$\begin{array}{ccc} \widetilde{\mathcal{M}}^{QC}_{LG,i} & \xrightarrow{Sol_i^{-1}(\psi_{QC,i})} & \mathcal{O}_{M_i}[\![\hbar]\!] \\ \downarrow{\scriptstyle \mathrm{Ad}^{-1}_{\widehat{\psi}_{Cl,i+1}} \circ \mathrm{Ad}_{\widehat{\psi}_{Cl,i}} = Id} & \nearrow{\scriptstyle c_{i,i+1}} & \downarrow{\scriptstyle \xi^c_{i+1,i} = Id} \\ \widetilde{\mathcal{M}}^{QC}_{LG,i+1} & \xrightarrow{Sol_{i+1}^{-1}(\psi_{QC,i+1})} & \mathcal{O}_{M_{i+1}}[\![\hbar]\!] \end{array}$$

The reason for $\xi^c_{i+1,i} = Id$ is the choice of the lifts $\widetilde{V}_i$. Since the WKB approximation is analytically continued along $\widetilde{\gamma}$, the transition functions are computed to be $c_{i,i+1} = 1, 1 \leq i \leq n-1$. Therefore, the computation of the monodromy boils down to the last transition function $c_{n,1}$.

Next, we compute the last transition function $c_{n,1}$, which is shown in the following diagram

$$\begin{array}{ccc} \widetilde{\mathcal{M}}^{QC}_{LG,n} & \xrightarrow{Sol_n^{-1}(\psi_{QC,n})} & \mathcal{O}_{M_n}[\![\hbar]\!] \\ \downarrow{\scriptstyle T^{\hbar}} & \nearrow{\scriptstyle c_{n,1}} & \downarrow{\scriptstyle \xi^c_{1,n}} \\ T_*\widetilde{\mathcal{M}}^{QC}_{LG,1} & \xrightarrow{Sol_1^{-1}(\psi_{QC,1})} & T_*(\mathcal{O}_{M_1}[\![\hbar]\!]) \end{array}$$

$\xi^c_{1,n}$ is a translation morphism of the form (3.16). Therefore, the last transition function $c_{n,1}$ is just given by the comparison between $\psi_{QC,n}$ and $\psi_{QC,1}$

$$c_{n,1} = \frac{\psi_{QC,n}(\widetilde{q}_n)}{\psi_{QC,1}(\widetilde{q}_0)} = \exp \frac{1}{\hbar} \int_{\widetilde{\gamma}} \lambda_{QC}.$$

This finishes the proof since we have

$$\mathrm{Mono}_\gamma \left(\mathcal{H}om_{\mathcal{A}_{T^\circ_{HV}}}(\mathcal{M}_{LG}|_{C^\circ}, \mathcal{M}^c_{C^\circ}) \right) = c_{1,2}c_{2,3}\cdots c_{n-1,n}c_{n,1} = c_{n,1} = \exp \frac{1}{\hbar} \int_{\widetilde{\gamma}} \lambda_{QC} \in \mathbb{C}^*_\hbar.$$

□

Remark 3.20. In Theorem 3.15, one can see that the quantum correction 1-period

$$\exp \frac{1}{\hbar} \int_{\widetilde{\gamma}} \lambda_{QC}$$

does not depend on the choice of the lift $\widetilde{\gamma}$. This is exactly what happens in the WKB approximation since all of the logarithmic multi-valuedness of the quantum 1-form $\lambda_\hbar$ is in the classical part λ_{Cl} and the quantum correction part λ_{QC} is a well-defined 1-form on $C \setminus \pi^{-1}(\Delta)$ ([ML, §2.5, Theorem 1], [Ba, §1.2.2]). Our treatment indeed gives a general explanation of this fact since the quantum correction 1-period corresponds to the monodromy of a local system. The reason why this Hom sheaf only captures the quantum correction part lies in the idea of microlocalization. Note that the local model of the quasi-canonical DQ module $\mathcal{M}^c_{C^\circ}$ is the module of functions on the zero section of a cotangent bundle, which automatically eliminates the classical part.

Remark 3.21. If the ambient symplectic surface is $\mathbb{C}^2$ rather than $(\mathbb{C}^*)^2$, the same proof as that of Theorem 3.15 shows that the monodromy of the Hom sheaf $\mathcal{H}om(\mathcal{M}_{LG}, \mathcal{M}^c_C)$ is still the same as the quantum correction part of the quantum periods from the WKB approximation. The proof is even easier since there is no deck transformation in this case. This case arises in the context of quantum curves where the quantum curve equation is a differential equation rather than a difference equation. Also, since $\mathbb{C}^2$ is simply-connected, the quasi-canonical DQ module in Theorem 3.15 is indeed unique in this case.

Remark 3.22. From our intrinsic definition of quantum periods (Definition 3.13), we can also deduce some properties of the residues of λ_{QC} at the ramification points in $\pi^{-1}(\Delta)$. Note that the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ is well-defined on the whole curve C. Choose any spin structure $\widetilde{K}_C^{1/2}$ on the whole curve C, which always exists as a line bundle on the smooth algebraic curve C. We have the following relation with the trivial spin structure $K_{C^\circ}^{1/2}$ chosen in Theorem 3.15

$$\widetilde{K}_C^{1/2}|_{C^\circ} = K_{C^\circ}^{1/2} \otimes_{\mathbb{C}} \mathcal{L},$$

where $\mathcal{L}$ is a rank 1 local system on C° with structure group $\{\pm 1\}$. Consider the DQ module

$$\mathcal{M}^c_{C^\circ} \otimes_{\mathbb{C}} \mathcal{L},$$

which is a quasi-canonical DQ module on C° with respect to the spin structure $\widetilde{K}_C^{1/2}|_{C^\circ}$. This DQ module above can indeed be extended to a quasi-canonical DQ module $\widetilde{\mathcal{M}^c_C}$ on the whole space (T_{HV}, C). In fact, recall that the collection of quasi-canonical DQ modules with respect to the pair (T_{HV}, C) is a torsor over

$$i^*_{T_{HV}} H^1(T_{HV}; \mathbb{C}^*_\hbar),$$

while the collection of quasi-canonical DQ modules with respect to the pair (T°_{HV}, C°) is a torsor over

$$i^*_{T^\circ_{HV}} H^1(T^\circ_{HV}; \mathbb{C}^*_\hbar),$$

where $i_{T_{HV}} : C \hookrightarrow T_{HV}$ and $i^*_{T^\circ_{HV}} : C^\circ \hookrightarrow T^\circ_{HV}$ are the corresponding closed embeddings. According to the Mayer-Vietoris sequence, we have an isomorphism of first cohomology groups

$$H^1(T_{HV}; \mathbb{C}^*_\hbar) \cong H^1(T^\circ_{HV}; \mathbb{C}^*_\hbar)$$

since $\pi|_C^{-1}(\Delta)$ is just a discrete set of points. This implies that the natural homomorphism

$$i^*_{T_{HV}} H^1(T_{HV}; \mathbb{C}^*_\hbar) \longrightarrow i^*_{T^\circ_{HV}} H^1(T^\circ_{HV}; \mathbb{C}^*_\hbar)$$

is surjective, which implies the existence of the quasi-canonical DQ module $\widetilde{\mathcal{M}^c_C}$.

Now, consider the following monodromy

$$\mathrm{Mono}_\alpha\left(\mathcal{H}om_{\mathcal{A}_{T_{HV}}}(\mathcal{M}_{LG},\widetilde{\mathcal{M}}^c_C)\right),$$

where α is a small loop around a ramification point. Since $[\alpha]$ is a contractible cycle on the whole C, the monodromy above is just 1. On the other hand, from Theorem 3.15, we know

$$\begin{aligned}&\mathrm{Mono}_\alpha\left(\mathcal{H}om_{\mathcal{A}_{T_{HV}}}(\mathcal{M}_{LG},\widetilde{\mathcal{M}}^c_C)\right)=\mathrm{Mono}_\alpha\left(\mathcal{H}om_{\mathcal{A}_{T^\circ_{HV}}}(\mathcal{M}_{LG}|_{C^\circ},\mathcal{M}^c_{C^\circ}\otimes_{\mathbb{C}}\mathcal{L})\right)\\&=\exp\frac{1}{\hbar}\int_\alpha\lambda_{QC}\cdot\mathrm{Mono}_\alpha(\mathcal{L})=\pm\exp\frac{1}{\hbar}\int_\alpha\lambda_{QC}\end{aligned}$$

As a result, the residues of $O(\hbar^2)$-terms in λ_{QC} at a ramification point are always zero and the residue of the $\hbar^1$-term gives an overall ±1-factor in the quantum periods due to the spin structure. This matches the WKB computations in the literature (for example [Ba, p. 21], [HN, p. 142]).

As an obvious corollary of Theorem 3.15, we know that for big quantum periods, i.e. quantum periods along the cycles that are contractible on the ambient torus T°_{HV}, the intrinsic definition of quantum periods (Definition 3.13) agrees with what one obtains from the WKB approximation.

Corollary 3.23. *In the setup of Theorem 3.15, assume in addition that the cycle γ is contractible in the torus T°_{HV}, i.e. $\gamma=\partial\Gamma$ for some relative 2-cycle Γ to the curve C°. We have the following equalities*

$$<\mathrm{QC}(\mathcal{M}_{LG}|_{C^\circ}),\gamma>=-<\mathfrak{D}(\mathcal{A}_{T^\circ_{HV}},\mathcal{M}_{LG}|_{C^\circ}),\Gamma>=\exp\frac{1}{\hbar}\int_\gamma\lambda_{QC},$$

where we note that the pairing of the quantum correction 1-period (Definition 3.13) with γ is well-defined since γ is contractible in the ambient torus.

4. Dimensional Reduction of Local Mirror Symmetry: Takahashi's Log Mirror Symmetry

4.1. **Local Mirror Symmetry of Toric Calabi-Yau Threefolds.** The mirror symmetry conjecture that we will deal with in this paper is the local mirror symmetry of toric Calabi-Yau threefolds proposed by Hori-Vafa in their physics paper [HV]. For a toric Calabi-Yau threefold $\check{\mathcal{X}}$, the Calabi-Yau threefold mirror $\mathcal{X}$ is predicted to be

$$\mathcal{X}:=\{(u,v,X,P)\in\mathbb{C}^2_{u,v}\times(\mathbb{C}^*)^2_{X,P}|uv=W(X,P)\},$$

where the Calabi-Yau form is

$$\Omega_{\mathcal{X}}=\mathrm{Res}_{\mathcal{X}}\frac{du\wedge dv\wedge d\log X\wedge d\log P}{uv-W(X,P)}. \tag{4.1}$$

The holomorphic function $W:(\mathbb{C}^*)^2\longrightarrow\mathbb{C}$ is called the Landau-Ginzburg superpotential and plays a key role in the construction of the mirror $\mathcal{X}$. In the case of $\check{\mathcal{X}}=K_{\mathbb{P}^2}$, i.e. the total space of the canonical bundle of $\mathbb{P}^2$, the Landau-Ginzburg superpotential of the mirror $\mathcal{X}$ is given by [3]

$$W_{K_{\mathbb{P}^2}}=-1+T(X+P+\frac{1}{XP}),$$

where T is the complex structure parameter of $\mathcal{X}$. In general, the Newton polygon of the superpotential W is the intersection of the toric fan of $\check{\mathcal{X}}$ with a hyperplane of height 1. The mirror $\mathcal{X}$ is a conic bundle over $(\mathbb{C}^*)^2_{X,P}$ by projection and the discriminant locus of the

[3] In the physics literature such as [ACDKV], the superpotential takes the form $W_{K_{\mathbb{P}^2}}=-1+X+P+z/XP$. It is easy to see that this superpotential is equivalent to the one above by a change of coordinates.

bundle is given by the vanishing locus of the superpotential W. The geometry of the mirror Calabi-Yau threefold of $K_{\mathbb{P}^2}$ is shown in Figure 3.

In the topological string B-model, the key objects are the period integrals of the Calabi-Yau form along certain 3-cycles determined by Hodge theory. For local mirror symmetry, one can reduce the 3-period integrals of the Calabi-Yau 3-form to the 2-period integrals of the holomorphic symplectic form $d\log X \wedge d\log P$ on the Hori-Vafa torus along the relative 2-cycles to the mirror curve. The picture of 3-cycles is shown in Figure 3, and the reduction is given by integrating out the fiber direction. Hence, the B-model can be reduced to the period integrals on the 2-dimensional Landau-Ginzburg model $((\mathbb{C}^*)^2_{X,P}, W)$.

The classical, or genus zero, closed local string mirror symmetry in [HV] says that the A-model partition function of local Gromov-Witten invariants of a toric Calabi-Yau threefold is the same as the B-model partition function determined by the special geometry relation of the period integrals of the Calabi-Yau 3-form (4.1) along 3-cycles after imposing the classical closed mirror map. This conjecture was proven long ago (see, for example, [FL]). For higher-genus mirror symmetry, there are many versions of mirror symmetry statements in the literature. The version we are interested in is the quantum mirror symmetry conjecture introduced by Aganagic-Cheng-Dijkgraaf-Krefl-Vafa in [ACDKV] about the Nekrasov-Shatashvili limit of refined topological string theory. The A-model of the quantum mirror symmetry conjecture in [ACDKV] is physically defined from the M-theoretic point of view and can be computed from the refined topological vertex [IKV]. For the corresponding B-model, we need to add quantum corrections to the corresponding period integrals via the WKB approximation of the difference equation introduced in §3.1. The B-model partition function is defined by the quantum special geometry relation of the big quantum periods.

Conjecture 4.2 (Quantum Closed String Mirror Symmetry [ACDKV])**.** *The A-model partition function of the NS-limit of refined closed topological string theory on a toric Calabi-Yau threefold is the same as the B-model partition function obtained from the quantum special geometry relation of the big quantum periods from the WKB approximation of the quantum mirror curve after imposing the quantum closed mirror map.*

In §6, we will give the first mathematical proof of the quantum closed string mirror symmetry conjecture (Conjecture 4.2) for $K_{\mathbb{P}^2}$, and our method could be generalized to other local toric del Pezzo surfaces. In order to do so, we need a better mathematical understanding of the A-model counting of the NS-limit of refined topological string theory.

4.2. **Takahashi's Log Mirror Symmetry.** We have already seen that the B-model can be reduced to period integrals along relative 2-cycles by integrating out the conic fiber direction of the Calabi-Yau threefold (Figure 3). In fact, the A-model counting relevant to local mirror symmetry can also be reduced to curve counting on a log surface, following an idea of Takahashi in [Ta]. To be more explicit, Takahashi conjectured that, in genus zero, the local mirror symmetry of the total space of the canonical bundle of the del Pezzo surface is equivalent to the log mirror symmetry of the del Pezzo surface with a smooth anticanonical divisor. The genus zero Takahashi's conjecture, which is also known as the log-local principle in the literature, was proven in [GGR]; see [BBG] for more general cases.

The higher-genus Takahashi conjecture was established by the first author in [Bo4]. To be more explicit, the first author proposed a rigorous mathematical definition of the NS-limit of refined topological string theory A-model of local del Pezzo surfaces by relating the refined topological string theory to a certain refined Donaldson-Thomas theory on the del Pezzo surfaces. Also, by comparing the wall structures in the space of stability conditions and the SYZ base from [CPS], the first author found that the NS-limit of refined topological string theory on $K_{\mathbb{P}^2}$ is the same as the higher-genus relative or logarithmic Gromov-Witten theory of $(\mathbb{P}^2, E)$ with a lambda class insertion. E is a smooth elliptic curve, or equivalently

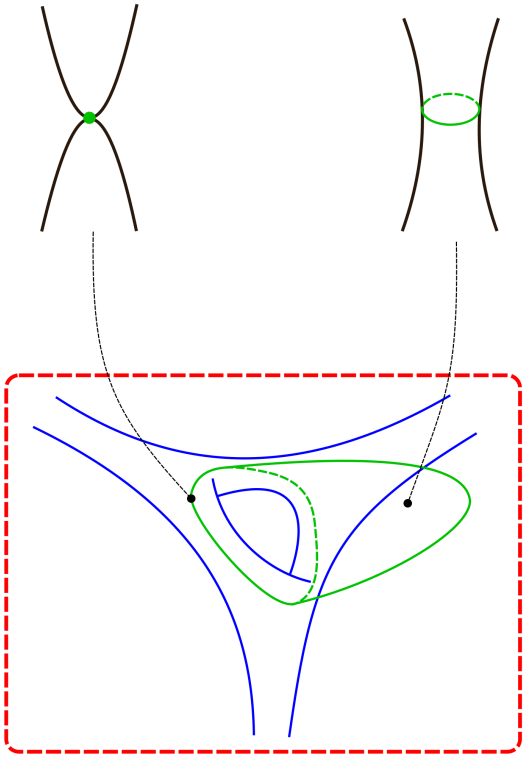


FIGURE 3. Calabi-Yau Threefold Mirror of $K_{\mathbb{P}^2}$ and Reduction of Cycles. The discriminant locus of the conic bundle is the mirror curve (blue) inside the ambient Hori-Vafa torus (red). Each 3-cycle (green) considered in the B-model is a fibration over a relative 2-cycle on the Hori-Vafa torus. The fiber of the 3-cycle is S^1 generically and degenerates to a point at the boundary of the relative 2-cycle. The 3-period integrals can be reduced to the 2-period integrals along the relative 2-cycle by integrating out the fiber direction.

smooth anticanonical divisor, in $\mathbb{P}^2$. Therefore, we can reduce the A-model of interest to the higher-genus logarithmic Gromov-Witten invariants. A general mathematical theory of the refined topological string theory was established in Brini-Schuler [BS].

Logarithmic Gromov-Witten theory is a generalization of Jun Li's relative Gromov-Witten theory [Li1; Li2] and was established by Abramovich-Chen [AC] and Gross-Siebert [GS3]. The higher genus logarithmic Gromov-Witten invariants of $(\mathbb{P}^2, E)$ corresponding to the NS-limit of refined topological string theory of $K_{\mathbb{P}^2}$ are defined as follows. Let $\overline{\mathcal{M}}_{g,1}(\mathbb{P}^2/E, d)$ be the moduli stack of stable maps from a genus g closed Riemann surface to $(\mathbb{P}^2, E)$ of degree d such that the image of the curve intersects E at only one point with intersection number $3d$. This moduli stack is a proper Deligne-Mumford stack and carries a virtual fundamental class of degree g. Let $\pi : \mathcal{C} \longrightarrow \overline{\mathcal{M}}_{g,1}(\mathbb{P}^2/E, d)$ be the universal curve and ω_π be the relative dualizing sheaf. The lambda class λ_g is defined to be the g-th Chern class of the Hodge bundle $\pi_*\omega_\pi$. The higher-genus logarithmic Gromov-Witten invariants with λ_g-class insertion are defined to be

$$N_{3d}^g := \int_{[\overline{\mathcal{M}}_{g,1}(\mathbb{P}^2/E,d)]^{vir}} (-1)^g \lambda_g.$$

The invariants N_{3d}^g are the A-model counts of the NS-limit of refined topological string theory of $K_{\mathbb{P}^2}$ that we consider in this paper. The importance of the λ_g-class insertion is that the resulting invariants satisfy the quantum Kontsevich-Soibelman scattering rule and form a consistent quantum wall structure according to [Bo1; Bo3; Bo2; Bo4]. This point is crucial for proving the quantum mirror symmetry conjecture.

Therefore, we can reduce quantum local mirror symmetry (Conjecture 4.2) for $K_{\mathbb{P}^2}$ to the quantum log mirror symmetry of $(\mathbb{P}^2, E)$, where E is a smooth elliptic curve inside $\mathbb{P}^2$. In order to study the quantum log mirror symmetry, we need to use the Gross-Siebert construction of mirrors of toric degenerations. To be more explicit, we will use the toric degeneration of $(\mathbb{P}^2, E)$ and its mirror proper Landau-Ginzburg model developed in [CPS] and its deformation quantization constructed by the first author in [Bo2; Bo4].

5. Classical and Quantum Geometry of Mirror Landau-Ginzburg Model of Toric Degeneration of $(\mathbb{P}^2, E)$

5.1. **Classical Geometry of Mirror Landau-Ginzburg Model.**

5.1.1. *Construction of Toric Degeneration of Mirror Landau-Ginzburg Model* $\mathfrak{X} \to \operatorname{Spf} \mathbb{C}[\![T]\!]$. Let us consider the toric degeneration of $(\mathbb{P}^2, E)$ and its mirror Landau-Ginzburg model in the framework of the Gross-Siebert program [CPS]. The toric degeneration of $(\mathbb{P}^2, E)$ denoted by $(\check{\pi} : \check{\mathfrak{X}} \to \operatorname{Spf} \mathbb{C}[\![\check{T}]\!], \check{\mathfrak{D}})$ is defined by the tropicalization data $(\check{B}, \check{\mathscr{P}}, \check{\varphi}, \check{\mathscr{S}})$, where $\check{B}$ is an affine manifold with singularity, $\check{\mathscr{P}}$ is a polyhedral decomposition and $\check{\varphi}$ is a multivalued piecewise affine function on B, and $\check{\mathscr{S}}$ is a consistent wall structure. After taking the discrete Legendre dual, we get the mirror toric degeneration defined by $(B, \mathscr{P}, \varphi, \mathscr{S})$. The figures of $(B, \mathscr{P}, \varphi)$ and $(\check{B}, \check{\mathscr{P}}, \check{\varphi})$ are shown in Figure 4. The upper left figure in Figure 4 shows the affine manifold $(\check{B}, \check{\mathscr{P}}, \check{\varphi})$, where there are three focus-focus affine singularities on the internal edges of the polyhedral decomposition $\check{\mathscr{P}}$. Under the affine structure of $\check{B}$, the boundary $\partial\check{B}$ is indeed straight or flat due to the focus-focus singularities, which is different from the Mumford degeneration in the pure toric case. The kinks of the multivalued piecewise affine function $\check{\varphi}$ on the three internal edges are all 1. After putting a canonical wall structure $\check{\mathscr{S}}$ on $(\check{B}, \check{\mathscr{P}}, \check{\varphi})$ and running the Gross-Siebert construction of toric degeneration in [GS2], we obtain the following algebraic toric degeneration of $(\mathbb{P}^2, E)$

$$\check{\mathfrak{X}}^{an} = \operatorname{Proj} \mathbb{C}[\check{X}, \check{Y}, \check{Z}, \check{U}, \check{T}]/(\check{X}\check{Y}\check{Z} - \check{T}(\check{U} + f(\check{X}, \check{Y}, \check{Z}))) \longrightarrow \operatorname{Spec} \mathbb{C}[\check{T}]$$
$$\check{\mathfrak{D}}^{an} = V(\check{U}),$$

where $f(\check{X}, \check{Y}, \check{Z})$ is a generic homogeneous polynomial of degree 3 and the grading is given by $deg(\check{X}) = deg(\check{Y}) = deg(\check{Z}) = 1, deg(\check{U}) = 3, deg(\check{T}) = 0$. On the generic fiber, $(\check{\mathfrak{X}}^{an}_{\check{T}}, \check{\mathfrak{D}}^{an}_{\check{T}})$ are just $(\mathbb{P}^2, E)$, while on the central fiber, $\mathbb{P}^2$ degenerates to a union of 3 copies of weighted projective planes $\mathbb{P}(1, 1, 3)$ and E degenerates to a union of 3 copies of $\mathbb{P}^1$ shown as the boundary $\partial\check{B}$.

By taking the discrete Legendre dual following the machinery in [GS2], we obtain the affine geometry $(B, \mathscr{P}, \varphi)$, whose two affine charts are shown in the upper right and bottom figures in Figure 4. There are three focus-focus singularities on the bounded edges in $\mathscr{P}$. The kinks of φ on the three bounded edges of $\mathscr{P}$ are all 1 and the kinks on the three rays of $\mathscr{P}$ are all 3. The wall structure $\mathscr{S}$ is obtained from the simple initial wall structure $\mathscr{S}_0$ on the three bounded edges, which are also called slabs in [GS2]. More explicitly, on each piece of the bounded slab, the initial wall function is given by

$$f_0 = 1 + z^m, \tag{5.1}$$

where m is the primitive integral vector pointing from the vertex v_i to the focus-focus singularity. With the simple initial wall structure $\mathscr{S}_0$ at hand, there is an algorithm to generate

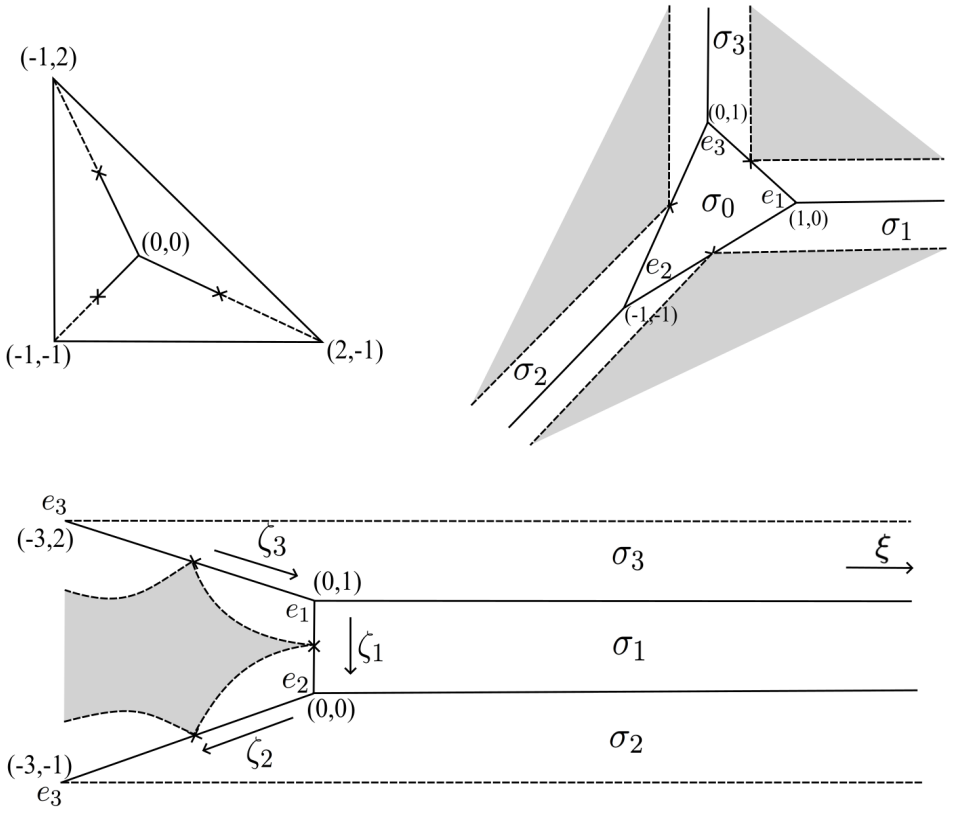


FIGURE 4. Affine Charts of the Affine Manifolds $(\check{B}, \check{\mathscr{P}}, \check{\varphi})$ and $(B, \mathscr{P}, \varphi)$ from [CPS]. The upper left figure shows an affine chart of $\check{B}$, while the upper right and the bottom figures show two affine charts of B. In the upper right figure, the three shaded regions are removed and the two boundary rays of each shadow region are glued affinely to give a focus-focus affine singularity. Note that the three rays in $\mathscr{P}$ are parallel to the asymptotic direction ξ.

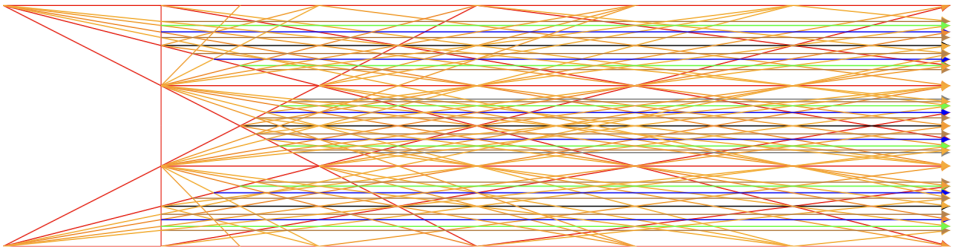


FIGURE 5. Wall Structure $\mathscr{S}$ up to Order 18 from [Gr1]. This was generated by Tim Gräfnitz's Sage code.

a consistent wall structure $\mathscr{S}$ up to each T-order due to Kontsevich-Soibelman. The picture of the wall structure $\mathscr{S}$ up to order 18 is shown in Figure 5, which was generated by Tim Gräfnitz's Sage code in [Gr1].

Since $(B, \mathscr{P}, \varphi, \mathscr{S})$ is asymptotically cylindrical in the sense of [CPS], there is a well-defined asymptotic direction ξ and one can define a formal Landau-Ginzburg potential $W : \mathfrak{X} \to \mathbb{C}$ on

the total space of the mirror toric degeneration given by the theta function of the asymptotic direction ξ in the sense of [GHS]. Hence on the mirror side, we get a formal toric degeneration of the Landau-Ginzburg model $(\pi : \mathfrak{X} \to \operatorname{Spf} \mathbb{C}[\![T]\!], W)$, which is expected to be the mirror of $(\check{\pi} : \check{\mathfrak{X}} \to \operatorname{Spf} \mathbb{C}[\![\check{T}]\!], \check{\mathfrak{D}})$, and the base $\operatorname{Spf} \mathbb{C}[\![T]\!]$ is a formal neighborhood of the Hodge-theoretic maximally unipotent point in the moduli of complex structures of the Landau-Ginzburg model.

In the case of mirror symmetry of $(\mathbb{P}^2, E)$, we can indeed find a complex analytification of the formal degeneration of the Landau-Ginzburg model, i.e. an analytic family $\pi^{an} : \mathfrak{X}^{an} \to \mathbb{D}_T$ over a disc $\mathbb{D}_T$ with variable T together with an analytic function $W^{an} : \mathfrak{X}^{an} \to \mathbb{C}$ on the total space such that the formal completion of the analytic degeneration of the Landau-Ginzburg model $(\mathfrak{X}^{an}, W^{an})$ along the central fiber $T = 0$ is exactly the formal toric degeneration $(\mathfrak{X}, W)$. To compute the geometry of the complex analytification $\mathfrak{X}^{an}$, an effective way is to compute the algebra of the theta functions and find the algebraic relations among them. According to [GHS], the theta functions are defined by summing over the last monomials of broken lines, and they give a canonical basis of the homogeneous coordinate ring. If we can analytify the algebraic relations of the theta functions, then we can compute the analytification of the formal degeneration. The theta algebra of the mirror toric degeneration of $(\mathbb{P}^2, E)$ was computed explicitly in [GRS1], where the authors reproved the classical local (or log) mirror symmetry of $K_{\mathbb{P}^2}$ by using the Gross-Siebert mirror construction of $(\mathbb{P}^2, E)$. We will use the results in [GRS1] directly and refer the reader to that paper for more details. For the theta algebra of $\mathfrak{X}$, we can compute in the central triangular chamber σ_0 in Figure 4 and obtain the following clean result despite the complexity of the wall structure $\mathscr{S}$

$$\mathfrak{X}^{an} = \operatorname{Proj} \mathbb{C}[X, Y, Z, U, W, T]/(XYZ - U^3, UW - T(X + Y + Z)), \tag{5.2}$$

where X, Y, Z, U are of degree 1 and W, T are of degree 0. The functions X, Y, Z, U are the analytification of the theta functions on $\mathfrak{X}$ and W is exactly the analytification of the superpotential. To be more specific, X, Y, Z are the theta functions corresponding to the three vertices on the boundary of the central triangle chamber σ_0, U is the theta function corresponding to the interior integral point of σ_0, and W is the theta function corresponding to the asymptotic direction ξ.

The Gross-Siebert Landau-Ginzburg model constructed above is indeed a compactification of the Hori-Vafa mirror Landau-Ginzburg model of $\mathbb{P}^2$ in the sense that the superpotential W is proper. In fact, by restricting to the locus where $U \neq 0$ on the generic fiber $\mathfrak{X}_T^{an}$ of the degeneration, one can recover the Hori-Vafa Landau-Ginzburg model for $\mathbb{P}^2$

$$\left((\mathbb{C}^*)^2_{x,y}, W_{\mathbb{P}^2} = T(x + y + \frac{1}{xy})\right) \subseteq (\mathfrak{X}_T^{an}, W), \tag{5.3}$$

where the coordinates x, y are defined to be $x := X/U,\ y := Y/U$. Under the degeneration, the Hori-Vafa torus degenerates to the torus given by the central chamber σ_0. Hence the central chamber combinatorially represents the geometry of the Hori-Vafa mirror.

The mirror curve $\mathfrak{C}^{an}$ of the Landau-Ginzburg model for $K_{\mathbb{P}^2}$ is defined to be the fiber of the superpotential W at $W = 1$[4]. Note that the mirror curve in the Hori-Vafa torus (5.3) is a genus 1 Riemann surface with three punctures, while the mirror curve in the whole geometry $\mathfrak{X}^{an}$ is a compact genus 1 Riemann surface since W is proper. We point out that, in the case of mirror symmetry of the toric degeneration of $(\mathbb{P}^2, E)$, there are elliptic curves on both sides of the mirror symmetry, which are the anticanonical divisor $E \subseteq \mathbb{P}^2$ and the W-fiber

[4]Note that the relation between the mirror superpotential of $K_{\mathbb{P}^2}$ and the mirror superpotential of $\mathbb{P}^2$ is $W_{K_{\mathbb{P}^2}} = -1 + W_{\mathbb{P}^2}$

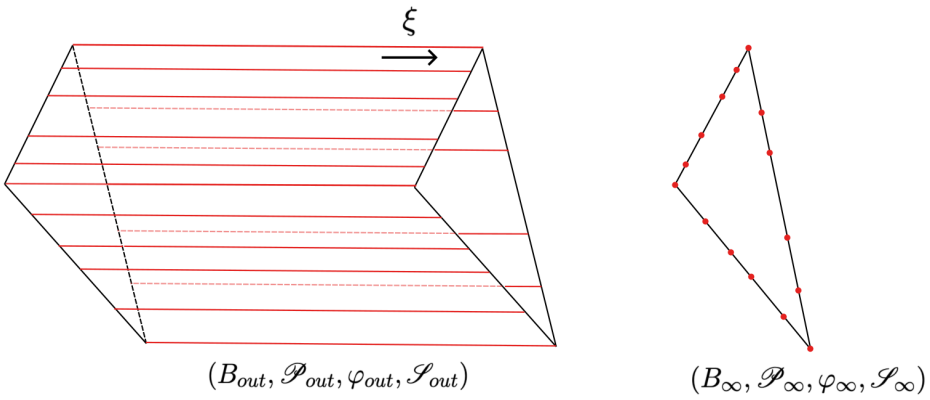


FIGURE 6. Asymptotic Chambers and the Affine Geometry of $(B_{out}, \mathscr{P}_{out}, \varphi_{out}, \mathscr{S}_{out})$ and $(B_\infty, \mathscr{P}_\infty, \varphi_\infty, \mathscr{S}_\infty)$ up to Certain Finite T-order. B_{out} is homeomorphic to an infinite cylinder and the wall structure $\mathscr{S}_{out}$ only consists of the horizontal walls, or equivalently the walls parallel to ξ, in the scattering diagram in Figure 5. The affine geometry near the 1-skeleton of $\mathscr{P}_{out}$ is given by the bottom figure in Figure 4.

$W^{-1}(1)$ respectively. One should not confuse them with each other. In fact, the two elliptic curves are mirror to each other.

5.1.2. *Asymptotic Chambers and Tate Curves.* One of the most intriguing and useful features of the Gross-Siebert point of view of the Landau-Ginzburg model in [CPS] is that it naturally provides a special kind of chamber, which we call the *asymptotic chambers* in this paper, where the enumerative mirror symmetry becomes transparent. The asymptotic chambers are defined to be the chambers at infinity along the asymptotic direction. These asymptotic chambers also give the formal geometry $\mathfrak{X}_{out} \subseteq \mathfrak{X}$ outside the toric surface given by the central chamber σ_0. To be more explicit, let us consider the combinatorial tuple $(B_{out}, \mathscr{P}_{out}, \varphi_{out}, \mathscr{S}_{out})$ given by the asymptotic chambers shown in Figure 6. Note that the asymptotic walls $\mathscr{S}_{out}$ consist only of the horizontal walls, or equivalently the walls parallel to the asymptotic direction ξ. The kinks of φ_{out} are all 3. The toric degeneration $\mathfrak{X}_{out} \to \operatorname{Spf} \mathbb{C}[\![T]\!]$ constructed by the tuple $(B_{out}, \mathscr{P}_{out}, \varphi_{out}, \mathscr{S}_{out})$ is indeed a formal toric degeneration of elliptic surfaces.

One of the most important features of the asymptotic chambers is that the asymptotic wall functions f_{out} and the superpotential W have simple, enumerative expressions in these chambers. Let us denote the toric coordinate defined by the asymptotic direction ξ by w. Then according to [CPS], there is an invertible formal transformation Φ relating w and the superpotential W in the asymptotic chambers

$$W = \Phi(w, T) := w + \sum_{d \geq 1} (3d-1) N^0_{3d-1,1} T^{3d} w^{1-3d}, \tag{5.4}$$

where $N^0_{3d-1,1}$ denotes the genus 0 logarithmic Gromov-Witten invariant with 2 marked points whose contact orders with the anticanonical divisor $E \in \mathbb{P}$ are $3d-1$ and 1, respectively.contact orders with the anticanonical divisor $E \subseteq \mathbb{P}^2$ is $3d-1$ and 1 respectively. Also, the logarithm of the product of all asymptotic wall functions is

$$\log f_{out}(w, T) = \sum_{d=0}^{\infty} 3d N^0_{3d} T^{3d} w^{-3d}, \tag{5.5}$$

where N^0_{3d} denotes the genus 0 logarithmic Gromov-Witten invariant with 1 marked point whose contact ordercontact order with the anticanonical divisor $E \subseteq \mathbb{P}^2$ is 3d. The enumerative meaning of the coefficients $N^0_{1,l}$ and N^0_{3d} comes from the tropical correspondence established in [Gr2; Gr3]. For the general relations between the coefficients and the logarithmic or punctured Gromov-Witten invariants, we refer to the canonical wall structure introduced in [GS4].

In order to see the geometry of $\mathfrak{X}_{out}$, recall that in the paper [CPS], there is a Tate curve description of the mirror curve inside $\mathfrak{X}$. The Tate curve is defined by the 1-dimensional asymptotic polarized tropical manifold $(B_\infty, \mathscr{P}_\infty, \varphi_\infty)$ with asymptotic wall structure $\mathscr{S}_\infty$ shown in Figure 6. For each $b \in B_{out}$ whose projection to B_∞ is $\bar{b}$, there is a canonical identification of $T_{\bar{b}}B_\infty$ with T_bB_{out}/P_b, where P is the subbundle of the tangent bundle TB_{out} defined by the unique outgoing asymptotic direction ξ of order zero in our polarized tropical manifold $(B, \mathscr{P}, \varphi, \mathscr{S})$. If the analytification exists, we obtain a toric degeneration of elliptic curves whose generic fiber is in the following Tate curve form

$$\mathbb{C}^*/ < \exp(2\pi i\tau(T)) >,$$

where $< \exp(2\pi i\tau(T)) >$ denotes the multiplicative cyclic subgroup generated by $\exp(2\pi i\tau(T))$. The modulus $\tau(T)$ is given by

$$\tau = \frac{1}{2\pi i}(\log f_\infty + 9\log T), \tag{5.6}$$

where f_∞ is the product of asymptotic classical wall functions in $\mathscr{S}_\infty$. When T goes to zero, the Tate curve degenerates to a cyclic union of 3 $\mathbb{P}^1$'s as shown in Figure 6.

Based on the above Tate curve description, we can understand the geometry of $\mathfrak{X}_{out}$. Since the polarized tropical manifold $(B_{out}, \mathscr{P}_{out}, \varphi_{out})$ is just the product of $(B_\infty, \mathscr{P}_\infty, \varphi_\infty)$ and the line given by the ξ direction, and the wall functions f_{out} only depend on w, the analytification is indeed a toric degeneration of elliptic surfaces whose generic fiber is in the following quotient form

$$(\mathbb{D}_w \times \mathbb{C}^*_v)/\mathbb{Z} \longrightarrow \mathbb{D}_w, \tag{5.7}$$

where the $\mathbb{Z}$-action is generated by

$$(w, v) \longmapsto (w, vw^{-9}T^9 f_{out}(w, T)).$$

The coordinate v can be identified with the toric coordinate of the asymptotic chamber given by a primitive integral vector ζ_i, $i = 1, 2, 3$ which is linearly independent of ξ, and $\mathbb{D}_w$ is a small disc in the w-variable. The elliptic fibration of the elliptic surface is given by the projection to the w-coordinate, and the fiber of w is equivalent to the fiber of W by the expression (5.4), which defines the mirror curve. It is clear from the $\mathbb{Z}$-action that after compactifying the elliptic fibration at $w = \infty$, the analytic geometry of the generic fiber is an I_9-type elliptic surface whose singular fiber at $w = \infty$ consists of a cyclic union of 9 $\mathbb{P}^1$'s.

To show the existence of the analytification of $\mathfrak{X}_{out}$, the strategy in [GRS1] is to first find the analytification of the whole geometry $\mathfrak{X}$ by computing the theta algebra and then show the analyticity of the formal toric coordinate w and the formal wall function f_{out} by relating them to the period integrals in $\mathfrak{X}^{an}$. The analytification $\mathfrak{X}^{an}$ is given by (5.2), and the period integrals can be computed formally following the methods in [RS; GRS1], which shows the analyticity of the elliptic surface. Also, we can see that enumerative mirror symmetry becomes very transparent in the asymptotic chamber since the coefficients N^0_{3d} of the wall functions f_{out} have enumerative meaning in the geometry $(\mathbb{P}^2, E)$. For example, the complex modulus τ of the mirror curve is clearly given by the logarithmic Gromov-Witten invariants of $(\mathbb{P}^2, E)$, which reveals the mystery of mirror symmetry.

5.2. **Deformation Quantization of Mirror Landau-Ginzburg Model.**

5.2.1. *Construction of the Formal Deformation Quantization $\mathcal{A}_{\mathfrak{X}}$.* We now construct the deformation quantization $\mathcal{A}_{\mathfrak{X}}$ on the classical geometry $\mathfrak{X}$ following the techniques in the first author's papers [Bo3; Bo2; Bo4]. In this paper, the word "formal" always refers to the formal completion with respect to T, and the formal completion with respect to $\hbar$ is implicit in the word "deformation". In fact, according to [Bo4], we can define the quantum wall structure $\mathscr{S}^{\hbar}$ by quantizing the initial wall structure and running the quantum scattering process. To be more explicit, the local model of the formal deformation quantization up to T-order k is given by

$$R^k_{\hbar}[\widehat{P_{\varphi_\rho}}],$$

where P_{φ_ρ} is the monoid consisting of the integral points in the upper graph of φ near the stratum ρ of the polyhedral decomposition $\mathscr{P}$ and R^k is $\mathbb{C}[T]/T^{k+1}$ in the case of mirror symmetry of $(\mathbb{P}^2, E)$. The construction of $R^k_{\hbar}[\widehat{P_{\varphi_\rho}}]$ is as follows. Note that the integral symplectic form ω on the local system of integral tangent vectors on B induces a degenerate skew-symmetric form on the monoid P_{φ_ρ} by setting the T-direction to be the kernel of the skew-symmetric form. We first construct the algebra $R^k[q^{\pm\frac{1}{2}}][\widehat{P_{\varphi_\rho}}]$ whose $R^k[q^{\pm\frac{1}{2}}]$-basis is given by $\hat{z}^m$, $m \in P_{\varphi_\rho}$ and whose multiplication is similar to the quantum torus relation (3.2). After taking the complete tensor product with $R^k[\![\hbar]\!]$, we get the local model of deformation quantization $R^k_{\hbar}[\widehat{P_{\varphi_\rho}}]$.

The gluing procedure of the local model is given by the quantum wall structure constructed in [Bo2; Bo4]. Each quantum wall is assigned a quantum Hamiltonian $\widehat{H}_{\mathfrak{d}}, \mathfrak{d} \in \mathscr{S}^{\hbar}$ by definition and the quantum wall crossing formula for any element $\widehat{A}$ in the local model is given by the conjugation

$$\Psi^{\hbar}_{\mathfrak{d}}(\widehat{A}) = \mathrm{Ad}_{\exp(\frac{\epsilon}{\hbar}\widehat{H}_{\mathfrak{d}})}(\widehat{A}) = \exp(\frac{\epsilon}{\hbar}\widehat{H}_{\mathfrak{d}})\widehat{A}\exp(-\frac{\epsilon}{\hbar}\widehat{H}_{\mathfrak{d}}), \tag{5.8}$$

where ϵ is the sign of the symplectic pairing $\omega(n, m(H_{\mathfrak{d}}))$ and n is the direction of wall crossing. The classical limit of the quantum wall crossing is just the classical Hamiltonian transformation generated by the classical limit of the quantum Hamiltonian $\widehat{H}_{\mathfrak{d}}$, which gives the classical wall crossing. Note that the quantum wall crossing $\Psi^{\hbar}_{\mathfrak{d}}$ has the same form as the time evolution of the operators in the Heisenberg picture of quantum mechanics. To construct the quantum wall structure considered in [Bo4], we need to replace the classical initial wall functions (5.1) by the quantum initial Hamiltonians

$$\widehat{H}_0 = \sum_{k\geq 1} \frac{(-1)^{k-1}\hbar}{k(q^{-k/2} - q^{k/2})}\hat{z}^{km}, \tag{5.9}$$

on each piece of the bounded slabs. From the quantum initial walls (5.9), we can run the Kontsevich-Soibelman quantum scattering process and the consistency condition along a small circle surrounding a joint away from the affine singularities is

$$\mathcal{P}\prod_{\mathfrak{d}} \exp(\frac{\epsilon_{\mathfrak{d}}}{\hbar}\widehat{H}_{\mathfrak{d}}) = 1,$$

where $\mathcal{P}$ is the path-ordering operator and $\mathcal{P}\prod_{\mathfrak{d}}$ means that the product is taken in the order of wall crossings along the path. Just like the classical case, there is a well-defined algorithm to generate the quantum wall structure satisfying the consistency condition order by order and hence we get the quantum wall structure $\mathscr{S}^{\hbar}$ in [Bo4]. From the local model $R^k_{\hbar}[\widehat{P_{\varphi_\rho}}]$ and the quantum wall structure $\mathscr{S}^{\hbar}$, using the quantum construction in [Bo2], we obtain a formal deformation quantization $\mathcal{A}_{\mathfrak{X}}$ on the formal Poisson scheme $\mathfrak{X}$ serving as a one-parameter noncommutative deformation of the structure sheaf $\mathcal{O}_{\mathfrak{X}}$.

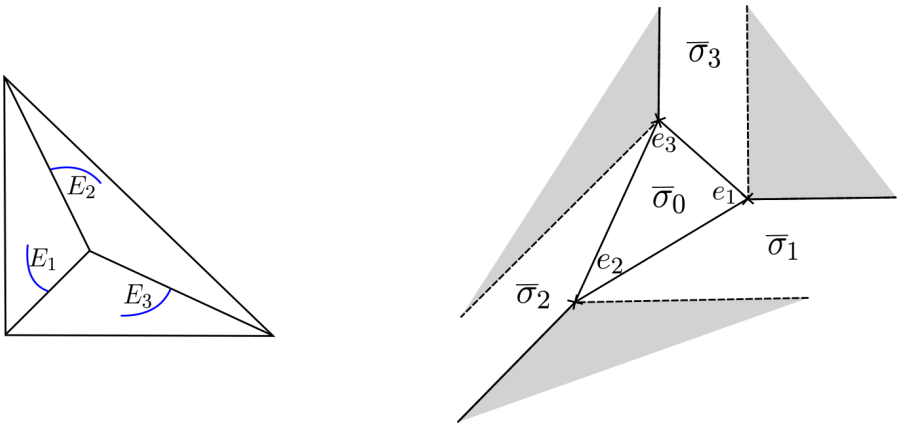


FIGURE 7. $\mathbb{Z}/3$-Symmetric Log Smooth Resolution $\check{\mathfrak{Y}} \longrightarrow \check{\mathfrak{X}}$ and Its Intrinsic Mirror $\mathfrak{Y}$. The left picture shows the central fiber of the log smooth resolution $\check{\mathfrak{Y}}$. Each component of the central fiber is the blow-up of the weighted projective plane $\mathbb{P}(1,1,3)$ at a generic point on a 1-dimensional toric stratum. E_i are the exceptional $\mathbb{P}^1$'s obtained from the blow-ups. The right figure is the generic fiber of the tropicalization of $\check{\mathfrak{Y}}$, or equivalently the intersection complex of the central fiber of the intrinsic mirror $\mathfrak{Y}$. Note that by setting the complex structure parameter T^{E_i} to 1, we recover the original Landau-Ginzburg mirror $(\mathfrak{X}, W)$ shown in Figure 4.

5.2.2. *Local Quantum Theta Functions and Analytification of the Formal Deformation Quantization $\mathcal{A}_{\mathfrak{X}}$.* Following the paper [Bo2], near each vertex in the polyhedral decomposition $\mathscr{P}$, we can define the local quantum theta functions by summing over the last monomials of the quantum broken lines and those local quantum theta functions also form a canonical basis of the global section of the deformation quantization $\mathcal{A}_{\mathfrak{X}}$ on the formal geometry near the vertex. This is a quantum version of the local classical theta functions considered in [GHS]. We will compute the local quantum theta algebra near the three vertices e_1, e_2, e_3 in the polyhedral decomposition $\mathscr{P}$ and patch them together.

To compute the local quantum theta functions, we first take a $\mathbb{Z}/3$-symmetric log smooth resolution $\check{\mathfrak{Y}} \to \check{\mathfrak{X}}$ of the toric degeneration $\check{\mathfrak{X}}$ of $(\mathbb{P}^2, E)$ defined by the upper left tropical manifold in Figure 4 just as the authors did in the classical case in [GRS1]. More explicitly, the resolution is constructed by blowing up the three components of the central fiber of the toric degeneration $\check{\mathfrak{X}}$ respectively and then patching the three different blow-ups together. The resulting resolution $\check{\mathfrak{Y}}$ is a log smooth degeneration. From the intrinsic mirror symmetry construction developed in [GS5; GS4], we can construct the intrinsic mirror degeneration $\mathfrak{Y} \to \operatorname{Spec} \mathbb{C}[\![NE(\check{\mathfrak{Y}})]\!]$ of the log smooth degeneration $\check{\mathfrak{Y}}$. The geometry of the central fibers of $\check{\mathfrak{Y}}$ and $\mathfrak{Y}$ is shown in Figure 7. The relation between the intrinsic mirror $\mathfrak{Y}$ and the toric degeneration mirror $\mathfrak{X}$ is that the mirror toric degeneration $\mathfrak{X}$ is exactly the restriction of the intrinsic mirror family $\mathfrak{Y}$ to $T^{E_i} = 1$, $i = 1, 2, 3$, where E_i are the curve classes of the three exceptional curves. Since we are only interested in the geometry of $\mathfrak{X}$ rather than $\mathfrak{Y}$, we will always restrict to $T^{E_i} = 1$ in the computation of the local quantum theta functions.

Since there is a rotational $\mathbb{Z}/3$-symmetry on the tropical geometry B, it suffices to compute the local quantum theta functions near one vertex. Let us do the computation near the vertex e_1, whose corresponding classical theta function is X. The local tropical geometry and the wall structure are shown in Figure 8. As in the classical case, we can compute the local theta functions in the central chamber σ_0. Since there is no quantum wall in the central chamber

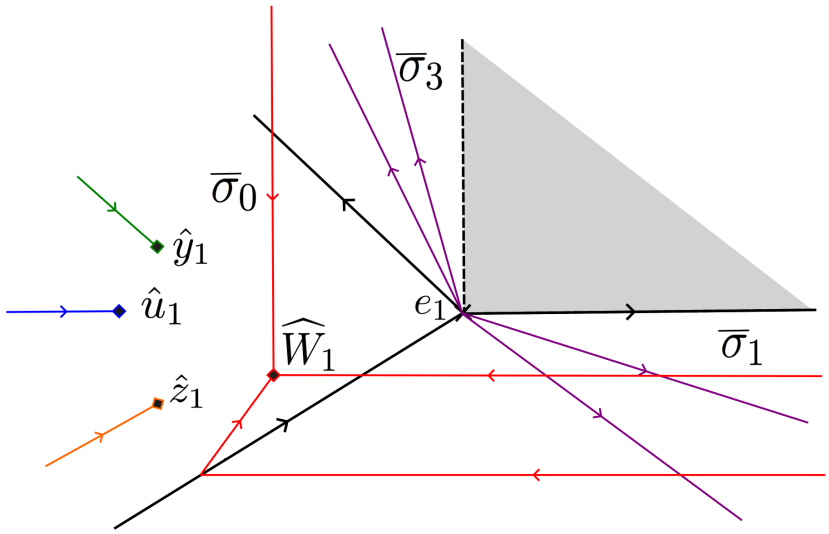


FIGURE 8. Quantum Broken Lines Computation of the Local Quantum Theta Functions Near e_1 in the Chamber $\bar{\sigma}_0$. The quantum walls which are not slabs are shown in purple up to a certain finite T-order. The local quantum theta functions $\hat{y}_1, \hat{u}_1, \hat{z}_1$ are just given by the corresponding quantum toric coordinates. For the local quantum theta function $\widehat{W}_1$, note that the quantum broken lines can only bend at the slabs for combinatorial reasons, and we find that $\widehat{W}_1$ consists of the last monomials of three broken lines shown in red.

σ_0, we have the following computation of the local quantum theta functions ϑ^{e_1} near e_1

$$\begin{aligned}
\hat{u}_1 &:= \hat{\vartheta}^{e_1}_{(-1,0)} = \hat{z}^{(-1,0,0)},\\
\hat{y}_1 &:= \hat{\vartheta}^{e_1}_{(-1,1)} = \hat{z}^{(-1,1,0)},\\
\hat{z}_1 &:= \hat{\vartheta}^{e_1}_{(-2,-1)} = \hat{z}^{(-2,-1,0)},\\
\widehat{W}_1 &:= \hat{\vartheta}^{e_1}_{\xi} = \hat{z}^{(0,1,1)} + \hat{z}^{(1,0,1)} + \hat{z}^{(-1,-1,1)},\\
&= T(\hat{z}^{(0,1,0)} + \hat{z}^{(1,0,0)} + \hat{z}^{(-1,-1,0)}),
\end{aligned}$$

where ξ denotes the asymptotic direction in B. By combining the computation above with the commutation relation of the quantum torus

$$\hat{z}^{(m,n,0)}\hat{z}^{(m',n',0)} = q^{(m,n)\wedge(m',n')}\hat{z}^{(m',n',0)}\hat{z}^{(m,n,0)},$$

where q is defined to be $q := \exp(-\hbar)$ and $\wedge$ means taking the determinant of the 2×2 matrix, we get the following algebraic relations between the local quantum theta functions

$$\begin{aligned}
\hat{u}_1\hat{z}_1 &= q\hat{z}_1\hat{u}_1,\\
\hat{u}_1\hat{y}_1 &= q^{-1}\hat{y}_1\hat{u}_1,\\
\hat{y}_1\hat{z}_1 &= q^{3/2}\hat{u}_1^3,\\
\hat{z}_1\hat{y}_1 &= q^{-3/2}\hat{u}_1^3,\\
\hat{u}_1\widehat{W}_1 &= T(1 + q^{-1/2}\hat{y}_1 + q^{1/2}\hat{z}_1),\\
\widehat{W}_1\hat{u}_1 &= T(1 + q^{1/2}\hat{y}_1 + q^{-1/2}\hat{z}_1).
\end{aligned} \tag{5.10}$$

After taking the complete tensor product with $\mathbb{C}_\hbar$, we get the local quantum theta algebra $A_{\mathfrak{X}_1^{an}}$. Clearly, the local quantum theta algebra $A_{\mathfrak{X}_1^{an}}$ defined by the relations (5.10) above is a deformation quantization of the algebra of regular functions on the affine chart $\mathfrak{X}_1^{an}$ defined by $X \neq 0$. Also, it is indeed an algebraic (and hence analytic) deformation quantization since the relations are algebraic. After Ore localization, we get an analytic deformation quantization on the affine chart $\mathfrak{X}_1^{an}$, which provides the analytification of the formal deformation quantization $\mathcal{A}_{\mathfrak{X}_1}$. Under the classical limit, the quantum coordinates $\hat{u}_1, \hat{y}_1, \hat{z}_1$ become the classical affine coordinates $u_1 := U/X, y_1 := Y/X, z_1 := Z/X$.

To get the global analytic deformation quantization, we need to glue together the local deformation quantizations $A_{\mathfrak{X}_i^{an}}$, $i = 1, 2, 3$, which are computed near the vertices e_i, $i = 1, 2, 3$ respectively. The gluing morphisms can be computed in the central quantum torus represented by σ_0. For example, the identification between the local quantum theta functions in $\mathfrak{X}_1^{an} := \{X \neq 0\}$ and $\mathfrak{X}_3^{an} := \{Y \neq 0\}$ is given by

$$\begin{aligned}
\hat{u}_2 &= q^{-1/2}\hat{u}_1\hat{y}_1^{-1},\\
\hat{x}_2 &= \hat{y}_1^{-1},\\
\hat{z}_2 &= q^{-3/2}\hat{z}_1\hat{y}_1^{-1},\\
\widehat{W}_2 &= \widehat{W}_1,
\end{aligned}$$

where the local quantum theta functions on $\mathfrak{X}_3^{an}$ are defined by $\hat{u}_2 := \hat{\vartheta}^{e_3}_{(0,-1)}$, $\hat{x}_2 := \hat{\vartheta}^{e_3}_{(1,-1)}$, $\hat{z}_2 := \hat{\vartheta}^{e_3}_{(-1,-2)}$, $\widehat{W}_2 := \hat{\vartheta}^{e_3}_{\xi}$. The gluing morphisms from $\mathfrak{X}_1^{an}$ to $\mathfrak{X}_2^{an}$ and from $\mathfrak{X}_2^{an}$ to $\mathfrak{X}_3^{an}$ are similar because of the $\mathbb{Z}/3$-symmetry. It is easy to check the cocycle condition of the gluing morphisms on the triple intersection, which gives the Hori-Vafa quantum torus, and hence we get a global analytic deformation quantization $\mathcal{A}_{\mathfrak{X}^{an}}$ of the analytic toric degeneration $\mathfrak{X}^{an}$.

From the gluing morphisms, we find that the local quantum theta functions $\widehat{W} = \vartheta_\xi$ corresponding to the asymptotic direction ξ glue together to a global section $\widehat{W}$ of the deformation quantization $\mathcal{A}_{\mathfrak{X}^{an}}$. This is indeed an analytification of the formal quantum superpotential defined by summing over the final monomials of the quantum broken lines with asymptotic direction ξ. The restriction of the quantum superpotential $\widehat{W}$ to the Hori-Vafa torus defined by $U \neq 0$ is exactly the quantum Hori-Vafa Landau-Ginzburg superpotential for $\mathbb{P}^2$

$$\widehat{W}_{\mathbb{P}^2} := T(\hat{x} + \hat{y} + q^{-1/2}\hat{x}^{-1}\hat{y}^{-1}). \tag{5.11}$$

In summary, we obtain an analytic deformation quantization $(\mathcal{A}_{\mathfrak{X}^{an}}, \widehat{W})$ of the toric degeneration of Landau-Ginzburg model $(\mathfrak{X}^{an}, W)$. By restricting to $U \neq 0$, one recovers the Hori-Vafa quantum torus and the quantum Landau-Ginzburg superpotential in (5.11).

Remark 5.12. Note that in the first author's paper [Bo2], the definition of deformation quantization is based on the deformation quantization in the algebraic context introduced

in [BK]. To get the deformation quantization theory in the analytic context that we considered in §2, we need an analytification procedure of deformation quantization. To be more explicit, on the generic fiber of each affine chart $\mathfrak{X}_i^{an}$, the noncommutative algebra $A_{\mathfrak{X}_i^{an}}$ is a one-parameter noncommutative deformation of the algebraic global section of $\mathcal{O}_{\mathfrak{X}_i^{an}}$, which is the deformation quantization used in [BK; Bo2]. One-parameter noncommutative deformations of a commutative algebra are controlled by the Hochschild cohomology HH^*, and it is well known that, for a regular commutative algebra in characteristic zero, the Hochschild cochain complex C^* is quasi-isomorphic to the cochain subcomplex C^*_{Diff} consisting of poly-differential operators. In fact, by the famous Hochschild-Kostant-Rosenberg theorem [HKR], we have a sequence of quasi-isomorphisms

$$\wedge^* Der \to C^*_{Diff} \hookrightarrow C^*,$$

where Der is the module of all derivations or equivalently global vector fields and the differential in $\wedge^* Der$ is trivial. The first arrow is given by skew-symmetrization. Therefore, on the generic fiber of the affine chart $\mathfrak{X}_i^{an}$, we can always find algebraic bi-differential operators such that $A_{\mathfrak{X}_1^{an}}$ is isomorphic to the corresponding star algebra (Definition 2.3), and hence we can analytify the algebraic deformation quantization $A_{\mathfrak{X}_i^{an}}$ by analytifying the algebraic bi-differential operators. Then the global analytic DQ algebra $\mathcal{A}_{\mathfrak{X}^{an}}$ on the generic fiber comes from gluing the analytic DQ algebras on the three affine charts and it is exactly the notion of analytic deformation quantization of holomorphic symplectic manifolds in [KS] that we considered in §2.

5.2.3. *Quantum Mirror Curve.* Since the quantum superpotential $\widehat{W}$ is a quantization of the classical superpotential W, we can define a Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ over the DQ algebra $\mathcal{A}_{\mathfrak{X}^{an}}$ by

$$\mathcal{M}_{LG} := \mathcal{A}_{\mathfrak{X}^{an}} / \mathcal{A}_{\mathfrak{X}^{an}}(-1 + \widehat{W}),$$

which serves as a quantization of the mirror curve $\mathfrak{C} := V(-1 + W)$. The restriction of the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ to the Hori-Vafa torus gives the quantization of the mirror curve considered in the physics literature [ACDKV].

5.2.4. *Canonicity of the DQ Algebra $\mathcal{A}_{\mathfrak{X}^{an}}$.* A natural question is whether the DQ algebra $\mathcal{A}_{\mathfrak{X}_T^{an}}$ constructed above is canonical in the sense of Construction 2.10 on the generic fiber. In fact, by evaluating the Deligne-Fedosov class, we can show that it is indeed canonical.

Proposition 5.13. *The DQ algebra $\mathcal{A}_{\mathfrak{X}_T^{an}}$ on any generic fiber is canonical, i.e. the underlying DQ algebroid has trivial Deligne-Fedosov class.*

To prove the canonicity of the DQ algebra $\mathcal{A}_{\mathfrak{X}_T^{an}}$, we need the following proposition on the restriction of the Deligne-Fedosov class to a Lagrangian.

Proposition 5.14. *Let X be a holomorphic symplectic manifold and L be a Lagrangian submanifold. Let $\mathcal{A}_X$ be a DQ algebroid on X. Then there exists a simple holonomic DQ module $\mathcal{M}_L$ over $\mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(L)}$ supported on L if and only if*

$$\mathfrak{D}(\mathcal{A}_X)|_L = 1 \in H^2(L; \mathbb{C}^*_\hbar),$$

where $\mathfrak{D}(\mathcal{A}_X)$ is the Deligne-Fedosov class of $\mathcal{A}_X$.

Proof. Consider the long exact sequence of relative cohomology in Remark 2.35

$$\begin{array}{ccccc} H^2(X, L; \mathbb{C}^*_\hbar) & \longrightarrow & H^2(X; \mathbb{C}^*_\hbar) & \longrightarrow & H^2(L; \mathbb{C}^*_\hbar) \\ \mathfrak{D}(\mathcal{A}_X, \mathcal{M}_L) & \longmapsto & \mathfrak{D}(\mathcal{A}_X) & \longmapsto & \mathfrak{D}(\mathcal{A}_X)|_L \end{array},$$

where $\mathfrak{D}(\mathcal{A}_X, \mathcal{M}_L)$ is the relative Deligne-Fedosov class under a choice of the spin structure $K_L^{1/2}$. From the long exact sequence, we know that if the restriction of the Deligne-Fedosov class $\mathfrak{D}(\mathcal{A}_X)$ to L is 1, then there exists a lift in the relative cohomology $H^2(X, L; \mathbb{C}^*_\hbar)$. By the classification of DQ pairs (Proposition 2.33), there exists a DQ pair whose underlying DQ algebroid is isomorphic to $\mathcal{A}_X$, which implies the existence of a simple holonomic DQ module supported on L. Conversely, if there exists a simple holonomic DQ module $\mathcal{M}_L$, then $\mathfrak{D}(\mathcal{A}_X)|_L$ is the image of $\mathfrak{D}(\mathcal{A}_X, \mathcal{M}_L)$, which is 1 due to the exactness of the sequence. □

Remark 5.15. There is a similar result proven in the algebraic setting of deformation quantization in [BGKP, Theorem 1.4]. Philosophically, the result above means that when a Lagrangian is equipped with a simple holonomic DQ module supported on it, this Lagrangian becomes a quantum Lagrangian in the sense that the restriction of the Deligne-Fedosov class is trivial.

Proof of Proposition 5.13. To show that the DQ algebra $\mathcal{A}_{\mathfrak{X}_T^{an}}$ is canonical, we only need to consider the pairing between the Deligne-Fedosov class and the second homology $H_2(\mathfrak{X}_T^{an}; \mathbb{Z})$. From the Leray spectral sequence computation, we have $H_2(\mathfrak{X}_T^{an}; \mathbb{Z}) \cong \mathbb{Z}^2$, whose basis consists of an SYZ fiber class and a W-fiber class. The SYZ fiber can be chosen to be contained in the Hori-Vafa algebraic torus $(\mathbb{C}^*)^2 \subseteq \mathfrak{X}_T^{an}$. Since the restriction of $\mathcal{A}_{\mathfrak{X}_T^{an}}$ to the torus is the quantum torus, which is canonical, the pairing between the Deligne-Fedosov class and the SYZ fiber must be trivial. For the W-fiber, we can choose the fiber to be $W^{-1}(1)$, which is exactly the quantum curve. Since the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ defined by the quantum superpotential $\widehat{W}$ gives a simple holonomic DQ module supported on the curve, the pairing between the Deligne-Fedosov class and the quantum curve is also trivial by Proposition 5.14. □

Remark 5.16. For the case of local $\mathbb{P}^2$, we have taken the above shortcut to show that the DQ algebra $\mathcal{A}_{\mathfrak{X}}$ is canonical due to the simplicity of the second homology of the generic fiber of $\mathfrak{X}$. For more general local toric del Pezzo surfaces, the DQ algebra $\mathcal{A}_{\mathfrak{X}}$ is still canonical. The idea of the proof is contained in the discussion of the modified quantum wall structure $\widehat{\mathscr{S}^\hbar}$ in §6.1.1. In fact, the construction of the modified quantum wall structure shown in Figure 10 works in great generality and under the modified quantum wall structure, the condition 6.2 is always satisfied, which implies that the canonical natural transformation P^c_{ijk} in the gluing datum of the canonical DQ algebroid (Construction 2.10) is indeed the identity under the gluing datum given by the modified quantum wall structure. Therefore, the gluing datum of the associated DQ algebroid of the DQ algebra $\mathcal{A}_{\mathfrak{X}}$ is equivalent to the gluing datum of the canonical DQ algebroid, which implies the canonicity of the DQ algebra $\mathcal{A}_{\mathfrak{X}}$.

5.3. **Formal Quantum Geometry of the Asymptotic Chambers.**

5.3.1. *Formal Deformation Quantization of the Asymptotic Formal Geometry $\mathfrak{X}_{out}$.* Since the DQ algebra is canonical on the generic fiber, the quantum correction 1-period $\mathrm{QC}(\mathcal{M}_{LG})$ of the Landau-Ginzburg DQ module is well-defined. In order to compute it, we need to analyze the quantum formal geometry of the asymptotic chambers $(B_{out}, \mathscr{P}_{out}, \varphi_{out}, \mathscr{S}^\hbar_{out})$ as we did for the classical geometry. The quantum asymptotic chambers are similar to the classical case and the only difference is that we should replace the classical wall structure $\mathscr{S}_{out}$ by the quantum wall structure $\mathscr{S}^\hbar_{out}$ computed in [Bo4]. The formal deformation quantization constructed from the asymptotic chambers is the restriction of the formal deformation quantization outside the complete toric surface defined by σ_0.

We can describe the formal geometry of the deformation quantization by gluing the local geometry near the three slabs by the product of the quantum walls in the interior of the three chambers $\sigma_1, \sigma_2, \sigma_3$. View i as an element in $\mathbb{Z}/3$. Let us define $U_i, i = 1, 2, 3$ to be the

open subset in $\mathfrak{X}_{out}$ corresponding to the open-star neighborhood of the 1-dimensional cell in $\mathscr{P}_{out}$ in Figure 6 coming from the ray emitted from the vertex e_i. On each cell σ_i, denote the primitive tangent vector parallel to the unique bounded edge in the closure of the cell by ζ_i. We also require that (ζ_i, ξ) forms a symplectic basis of the local system of integral vectors on each cell σ_i. The formal classical geometry of U_i is given by the equation

$$v_{i,+}v_{i,-} = f_{\mathfrak{b}_i}(w,T)T^3w^{-3},$$

where $v_{i,+}$ is the toric coordinate given by ζ_i, $v_{i,-}$ is the toric coordinate given by $-\zeta_{i-1}$, w is the toric coordinate given by the asymptotic direction ξ, and $f_{\mathfrak{b}_i}(w,T)$ is the classical slab function. The formal deformation quantization on U_i is given by

$$\begin{aligned}
\hat{v}_{i,+}\hat{w} &= q\hat{w}\hat{v}_{i,+},\\
\hat{v}_{i,-}\hat{w} &= q^{-1}\hat{w}\hat{v}_{i,-},\\
\hat{v}_{i,+}\hat{v}_{i,-} &= q^{-3/2}\hat{f}_{\mathfrak{b}_i}(\hat{w},T)T^3\hat{w}^{-3},\\
\hat{v}_{i,-}\hat{v}_{i,+} &= q^{3/2}\hat{f}_{\mathfrak{b}_i}(q^{-1}\hat{w},T)T^3\hat{w}^{-3},
\end{aligned}$$

where $\hat{v}_{i,+}$ is the quantum toric coordinate given by ζ_i, $\hat{v}_{i,-}$ is the quantum toric coordinate given by $-\zeta_{i-1}$, $\hat{w}$ is the quantum toric coordinate given by the asymptotic direction ξ, and $\hat{f}_{\mathfrak{b}_i}(w,T)$ is the quantum slab function whose relation with the quantum Hamiltonian $\widehat{H}_{\mathfrak{b}_i}(\hat{w},T)$ is

$$\hat{f}_{\mathfrak{b}_i}(\hat{w},T) := \exp\frac{1}{\hbar}(\widehat{H}_{\mathfrak{b}_i}(q\hat{w},T) - \widehat{H}_{\mathfrak{b}_i}(\hat{w},T)),$$

which recovers the classical slab function $f_{\mathfrak{b}_i}$ by taking the classical limit $\hbar \to 0$. The algebraic relation between $v_{i,+}$ and $v_{i,-}$ can be written formally as

$$\hat{v}_{i,-}^{-1} = \mathrm{Ad}_{\exp\frac{1}{\hbar}(-3\log T\log\hat{w}+\widehat{H}_{\mathfrak{b}_i}(\hat{w},T)+\frac{3}{2}(\log\hat{w})^2)}(\hat{v}_{i,+}),$$

which makes sense on the generic fiber if there exists an analytification. Note that here $\hat{w}$ should be regarded as the exponential of the momentum and $\log\hat{w}$ means the quantum momentum coordinate.

Now we glue the U_i together to obtain the formal asymptotic quantum geometry. On the intersection $U_{i,i+1}, i = 1, 2$, the classical gluing is given by

$$v_{i,+} = (f_{\mathfrak{p}_{i,i+1}}(w,T))^{-1}v_{-,i+1}^{-1},$$

where $f_{\mathfrak{p}_{i,i+1}}(w,T)$ is the product of all the wall functions in the interior of the chamber corresponding to $U_{i,i+1}$, and w is fixed. The quantum gluing is given by

$$\hat{v}_{i,+} = \mathrm{Ad}_{\exp\frac{1}{\hbar}\widehat{H}_{\mathfrak{p}_{i,i+1}}(\hat{w},T)}(\hat{v}_{-,i+1}^{-1}),$$

where $\widehat{H}_{\mathfrak{p}_{i,i+1}}(\hat{w},T)$ is the sum of all the quantum Hamiltonians in the interior of the chamber corresponding to $U_{i,i+1}$, and $\hat{w}$ is also fixed.

The gluing construction above gives the formal degeneration $\mathfrak{X}_{out}$ and the formal deformation quantization $\mathcal{A}_{\mathfrak{X}_{out}}$ on the asymptotic chambers. From this gluing construction, it is clear that if the analytification of the classical formal geometry $\mathfrak{X}_{out}$ exists, it is the same as the elliptic surface mentioned in §5.1.2.

5.3.2. *Formal Quantum Superpotential and Quantum Walls on the Asymptotic Chambers.* As in the classical case discussed in §5.1.2, the quantum superpotential $\widehat{W}$ and the quantum Hamiltonians $\widehat{H}_{out}$ of the walls have expressions with clear enumerative meaning on the asymptotic chambers. The enumerative meaning for $(\mathbb{P}^2, E)$ and more general toric del Pezzo surfaces in the asymptotic chambers was proven in [Gr2; Gr3; Bo4] via the tropical correspondence and the q-refined tropical correspondence. The general theory that explains the

enumerative meaning of the walls and the theta functions was established in the theory of the canonical wall structures in [GS4].

According to [Bo4; Gr2], the sum of the asymptotic quantum Hamiltonians $\widehat{H}_{out}$ takes the following form

$$\widehat{H}_{out}(\hat{w}, T) = \sum_{g\geq 0, d\geq 1} (-1)^g N_{3d}^g T^{3d} \hat{w}^{-3d} \hbar^{2g}, \tag{5.17}$$

where N_{3d}^g is the genus g logarithmic Gromov-Witten invariant with 1 marked point whose contact order with the anticanonical divisor is $3d$ together with the lambda class λ_g insertion. The formal quantum superpotential takes the following form

$$\widehat{W} = \widehat{\Phi}(\hat{w}, T) := \hat{w} + \sum_{g\geq 0, d\geq 1} (-1)^g (3d-1) N_{3d-1,1}^g T^{3d} \hat{w}^{1-3d} \hbar^{2g}, \tag{5.18}$$

where $N_{3d-1,1}^g$ denotes the genus g logarithmic Gromov-Witten invariant with 2 marked points whose contact orders with the anticanonical divisor are $3d-1$ and 1, respectively, together with the lambda class λ_g insertion. The function $\widehat{\Phi}$ is also an invertible formal transformation. Let us denote the inverse formal transformation of $\widehat{\Phi}$ by $\widehat{\Xi}$, whose classical limit is denoted by Ξ. For the computation of the inverse transformation $\widehat{\Xi}$ with respect to $\hat{w}$, see Appendix B in [GRZZ].

We point out that, although the analyticity of the classical Φ and w was proven in [GRS1], at this point we do not know whether the quantum $\widehat{\Phi}$ and $\widehat{H}_{out}$ are analytic; they are only formal. However, in §6.1, we relate these formal functions to well-defined analytic functions and give them an analytic meaning. They are indeed both analytic on certain regions.

6. Quantum Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$

In this section, we give the first proof of quantum closed string mirror symmetry for $K_{\mathbb{P}^2}$. The key idea is that, since we gave an intrinsic definition of the quantum periods (Definition 3.13) and proved its equivalence with the WKB approximation on the Hori-Vafa torus (Theorem 3.15), we can move from the Hori-Vafa torus to the asymptotic chambers to compute the quantum periods instead of struggling with the complicated WKB approximation on the Hori-Vafa torus. The quantum mirror symmetry becomes very transparent in the asymptotic chambers.

6.1. Computation of the Quantum Correction 1-Periods.

From the Leray spectral sequence computation of the SYZ fibration, on the generic fiber of the toric degeneration, the first cohomology $H^1(\mathfrak{X}_T^{an}; \mathbb{C}_\hbar^*)$ is indeed trivial. Therefore, there exists a unique quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}_T^{an}}$ with respect to $(\mathfrak{X}_T^{an}, \mathfrak{C}_T^{an})$. From Theorem 3.15, the quantum correction 1-periods from the WKB approximation are equal to the monodromy of the Hom sheaf

$$\mathcal{H}om_{\mathcal{A}_{\mathfrak{X}_T^{an}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{\mathfrak{C}_T^{an}}),$$

which is the same as the corresponding rank 1 $\mathbb{C}_\hbar^*$-local system of $\mathrm{QC}(M_{LG})$ defined in Definition 3.13. More specifically, we need to evaluate the pairings

$$a^{QC}(T) := \mathcal{N}(< \mathrm{QC}(\mathcal{M}_{LG}), [\alpha] >),\ a_D^{QC}(T) := \mathcal{N}(< \mathrm{QC}(\mathcal{M}_{LG}), [\beta] >),$$

where α is the fiber class of the SYZ fibration on the mirror curve $\mathfrak{C}_T^{an} = V(-1+W)$ and β is a section of the SYZ fibration of the mirror curve in Figure 9. The *normalization operator*

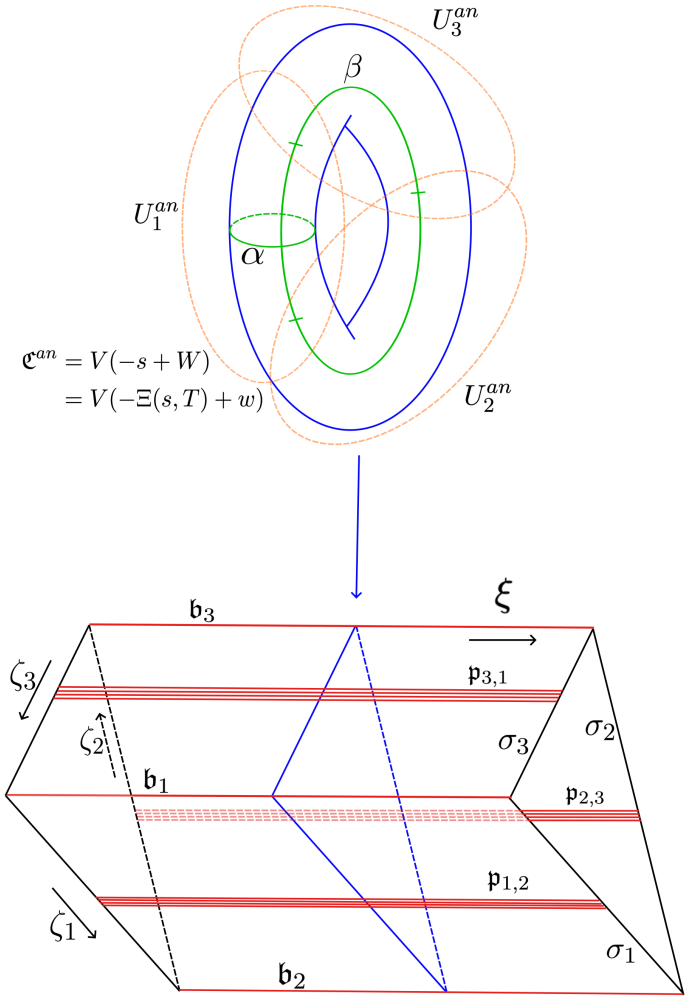


FIGURE 9. Computation of Quantum 1-Periods in the Asymptotic Chambers. Note that in the asymptotic chambers, we can stack all the walls inside the interior of the same 2-cell σ_i together without changing the formal geometry of $\mathfrak{X}_{out}$. For the quantum correction a^{QC}-period, we can compute it in U_1^{an}, where the mirror curve is a cylinder in the quantum torus and the a^{QC}-period is given by the monodromy of the WKB approximation. For the quantum correction a_D^{QC}-period, we need to use the patching computation of the monodromy by the open cover $U_1^{an} \cup U_2^{an} \cup U_3^{an}$.

$\mathcal{N}$ is defined by

$$\begin{aligned} &\mathcal{N} : \mathbb{C}_\hbar^* \longrightarrow \mathbb{C}_\hbar \\ &\sum_{i\geq 0} a_i \hbar^i \mapsto \hbar \log\left(\sum_{i\geq 0}(a_i/a_0)\hbar^i\right) \end{aligned} \tag{6.1}$$

which just means taking the logarithm of the periods. Note that the quantum correction 1-period $\mathrm{QC}(\mathcal{M}_{LG})$ in our case is unitary, and hence normalizing a_0 does not lose any information. These quantum correction 1-periods give the quantum corrections of the classical 2-periods Π_A and Π_B in the classical mirror symmetry where A, B are relative 2-cycles to the mirror curve with boundary α, β respectively. Restricted to the Hori-Vafa torus, these periods are exactly what physicists compute via the WKB approximation in [ACDKV]. Instead of computing the quantum correction 1-periods for one single quantum curve, we are going to compute the quantum correction 1-periods for a family of quantum curves defined by the DQ-modules

$$\mathcal{M}_{LG,s} := \mathcal{A}_{\mathfrak{X}^{an}}/\mathcal{A}_{\mathfrak{X}^{an}}(-s + \widehat{W}),$$

where s is a complex number in a neighborhood $\mathbb{D}_{s,1}$ of 1. Let us denote the corresponding quantum correction 1-periods by

$$a^{QC}(s,T) := \mathcal{N}(< \mathrm{QC}(\mathcal{M}_{LG,s}), [\alpha] >),\ a_D^{QC}(s,T) := \mathcal{N}(< \mathrm{QC}(\mathcal{M}_{LG,s}), [\beta] >).$$

Since the mirror curve is a compact Riemann surface of genus 1 whose canonical bundle is trivial, we will always choose the spin structure $K_{\mathcal{C}^{an}}^{1/2}$ to be the unique spin structure whose underlying line bundle is trivial and omit this subtlety in the computation.

The strategy we use to compute the quantum periods in terms of the formal geometry of the central fiber is similar to the methods in [RS; GRS1], which is to use the analytic approximations of formal coordinates. This means that, for any k and on a Stein neighborhood of the degeneration $\mathfrak{X}^{an}$, there exist analytic sections of $\mathcal{A}_{\mathfrak{X}^{an}}$ such that they are the same as the formal quantum toric coordinates modulo T^{k+1}. In fact, the analytic approximations of formal coordinates can simply be given by only keeping the terms up to T^k and dropping higher terms. We can use the analytic approximations to compute the quantum periods and they will match the actual analytic results modulo T^{k+1}. Up to any finite T-order k, there exist analytic approximations of $\widehat{X}_{b,k}, \widehat{P}_{b,k}$ of the formal quantum toric coordinates $\widehat{X}_b, \widehat{P}_b$ for every chamber b. Since $\widehat{X}_{b,k}, \widehat{P}_{b,k}$ give the quantum torus algebra modulo T^k, the local model of the DQ algebra $\mathcal{A}_{\mathfrak{X}^{an}}$ is just given by the quantum coordinates $\log \widehat{X}_{b,k}, \log \widehat{P}_{b,k}$ on the fundamental domain of the exponential cover $\mathrm{Exp} : \mathbb{C}^2 \longrightarrow (\mathbb{C}^*)^2$. These local models given by the chambers of the quantum wall structure cover the whole degeneration $\mathfrak{X}^{an}$ except the loci

$$V(X, U, W, Y + Z) \cup V(Y, U, W, Z + X) \cup V(Z, U, W, X + Y),$$

whose restriction to each T-fiber is just a disjoint union of 3 points. Since removing the 3 points does not affect the first cohomology of the generic fiber and the 3 points are not relevant in the computation of quantum periods, it is enough to compute the quasi-canonical DQ module and the quantum periods away from those 3 points.

There are two kinds of relevant quantized symplectic transformations between different local models. The first kind corresponds to quantum wall crossing and is given by

$$D \circ \mathrm{Ad}_{\exp \frac{c_{\mathfrak{d}}}{\hbar} \widehat{H}_{\mathfrak{d}}},$$

for some deck transformation of the form

$$D := \mathrm{Ad}_{\exp \frac{1}{\hbar}(-2\pi i m_p \log \widehat{X} + 2\pi i m_x \log \widehat{P})}, (m_x, m_p) \in \mathbb{Z}^2,$$

which corresponds to different choices of branches of the exponential cover Exp. The second kind is induced by the affine monodromy of the focus-focus singularities and takes the form

$$D \circ \mathrm{Ad}_{\exp \frac{1}{2\hbar}(\log \hat{z}^m)^2},$$

where m is the direction along a bounded slab and D is a deck transformation.

6.1.1. *Contributions From the Focus-Focus Singularities and Modification of the Quantum Walls.* Before constructing the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$, we need to make a certain modification of the quantum wall structure $\mathscr{S}^\hbar$ in [Bo4], which is generated from the initial quantum walls (5.9). The reason is condition (1) in the definition of the canonical DQ pair (Definition 2.27). Recall that in order to get the gluing datum of the quasi-canonical DQ module over the canonical DQ algebra $\mathcal{A}_{\mathfrak{X}^{an}}$, we need to choose the quantized contact transformation lifts $\tilde{\Psi}^\hbar_{ij}$ of the quantized symplectic transformations $\Psi^\hbar_{ij}$ above such that

$$\tilde{\Psi}^\hbar_{ij} \circ \tilde{\Psi}^\hbar_{jk} \circ \tilde{\Psi}^\hbar_{ki} = \mathrm{Ad}_{\exp(c_{ijk}\hat{\tau})} \tag{6.2}$$

for some $c_{ijk} \in \mathbb{C}$. Recall that $\hat{\tau} = \hbar^{-1}$ is the operator dual to $\hat{t}$. The natural choices of the quantized contact transformation lifts are

$$\tilde{D} \circ \mathrm{Ad}_{\exp \epsilon_{\mathfrak{d}} \hat{\tau} \widehat{H}_{\mathfrak{d}}}, \ \tilde{D} \circ \mathrm{Ad}_{\exp \frac{\hat{\tau}}{2} (\log \hat{z}^m)^2},$$

where the natural lift $\tilde{D}$ of the deck transformation D is given by

$$\mathrm{Ad}_{\exp \hat{\tau}(-2\pi i m_p \log \widehat{X} + 2\pi i m_x \log \widehat{P})}.$$

Since the quantum Hamiltonians $\widehat{H}_{\mathfrak{d}}$ only depend on $\widehat{X}, \widehat{P}$, the deck transformation $\tilde{D}$ commutes with the quantized contact transformation $\mathrm{Ad}_{\exp \epsilon_{\mathfrak{d}} \hat{\tau} \widehat{H}_{\mathfrak{d}}}$, and hence we can move all the deck transformations $\tilde{D}$ together. From the computations in the proof of Theorem 3.15, we know that the composition of deck transformations $\tilde{D}$ only gives an $\mathrm{Ad}_{\exp(c\hat{\tau})}$-term for some $c \in \mathbb{C}$, which is purely classical and has no quantum correction term like the term P_{ijk} in condition (1) in Definition 2.27. Therefore, condition (6.2) boils down to

$$\mathrm{Ad}_{\exp \hat{\tau} \widehat{H}_{ij}} \circ \mathrm{Ad}_{\exp \hat{\tau} \widehat{H}_{jk}} \circ \mathrm{Ad}_{\exp \hat{\tau} \widehat{H}_{ki}} = \mathrm{Ad}_{\exp(c_{ijk}\hat{\tau})}$$

for some $c_{ijk} \in \mathbb{C}$ as a quantized contact transformation, where $\widehat{H}_{ij}$ is either $\epsilon_{\mathfrak{d}} \widehat{H}_{\mathfrak{d}}$ or $\frac{1}{2}(\log \hat{z}^m)^2$. For joints away from the focus-focus singularities, this is guaranteed by the consistency condition of the quantum wall structure

$$\mathcal{P} \prod_{\mathfrak{d}} \exp(\frac{\epsilon_{\mathfrak{d}}}{\hbar} \widehat{H}_{\mathfrak{d}}) = 1,$$

where $\mathcal{P}$ is the path-ordering operator and $\mathcal{P} \prod_{\mathfrak{d}}$ means the path-ordered product.

However, this consistency condition fails around the focus-focus singularities, which leads to the modification of the quantum wall structure $\mathscr{S}^\hbar$. To understand this, first note that we can analytically continue the initial Hamiltonian (5.9) to the bigger region $\{\hat{z}^m \neq 0, \hat{z}^m \neq -1\}$. In fact, we can express the initial Hamiltonian (5.9) as

$$\begin{aligned}
\widehat{H}_0 &= \sum_{k \geq 1} \frac{(-1)^{k-1}\hbar}{k(q^{-k/2} - q^{k/2})} \hat{z}^{km} = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k^2} \frac{k\hbar/2}{\sinh(k\hbar/2)} \hat{z}^{km} \\
&= \sum_{k \geq 1} \sum_{l \geq 0} \frac{(-1)^{k-1}}{k^2} C_l (\frac{k\hbar}{2})^{2l} \hat{z}^{km} = \sum_{l \geq 0} C_l (\frac{\hbar}{2})^{2l} \sum_{k \geq 1} (-1)^{k-1} k^{2l-2} \hat{z}^{km} \\
&= \sum_{l \geq 0} C_l \frac{\hbar^{2l}}{4^l} F_l(\hat{z}^m),
\end{aligned}$$

where

$$C_l := \frac{2(1 - 2^{2l-1})B_{2l}}{(2l)!}, \ B_{2l} : \ \textit{Bernoulli numbers}$$

are the coefficients of the Taylor expansion of $\frac{z}{\sinh(z)}$ and $F_l(z)$ are (multi-valued) holomorphic functions on $\{z \neq 0, z \neq -1\}$ given by

$$F_0(z) = \int^z \frac{\log(1+u)}{u} du, \;\; F_l(z) = \frac{\partial^{2l} F_0(z)}{(\partial \log z)^{2l}}.$$

From the analytic continuation above, we can compute the sum of two initial Hamiltonians $\widehat{H}_{0,+}, \widehat{H}_{0,-}$ around a focus-focus singularity

$$\widehat{H}_{0,+} + \widehat{H}_{0,-} = \sum_{l \geq 0} C_l \frac{\hbar^{2l}}{4^l} (F_l(\hat{z}^m) + F_l(\hat{z}^{-m})).$$

Since condition (6.2) only concerns $O(\hbar)$-terms, we only focus on the $O(\hbar)$-terms of the sum above, which are given by

$$\begin{aligned}
&\sum_{l \geq 1} C_l \frac{\hbar^{2l}}{4^l} (F_l(\hat{z}^m) + F_l(\hat{z}^{-m})) = \sum_{l \geq 1} C_l \frac{\hbar^{2l}}{4^l} \frac{\partial^{2l} (F_0(\hat{z}^m) + F_0(\hat{z}^{-m}))}{(\partial \log \hat{z}^m)^{2l}} \\
&= \sum_{l \geq 1} C_l \frac{\hbar^{2l}}{4^l} \frac{\partial^{2l}}{(\partial \log \hat{z}^m)^{2l}} (\int^{\hat{z}^m} \frac{\log(1+u)}{u} du + \int^{\hat{z}^{-m}} \frac{\log(1+u)}{u} du) \\
&= \sum_{l \geq 1} C_l \frac{\hbar^{2l}}{4^l} \frac{\partial^{2l}}{(\partial \log \hat{z}^m)^{2l}} (\int^{\hat{z}^m} \frac{\log(1+u)}{u} du - \int^{\hat{z}^m} \frac{\log(1+u^{-1})}{u} du) \\
&= \sum_{l \geq 1} C_l \frac{\hbar^{2l}}{4^l} \frac{\partial^{2l}}{(\partial \log \hat{z}^m)^{2l}} (\frac{1}{2} (\log \hat{z}^m)^2) \\
&= C_1 \frac{\hbar^2}{4} = -\frac{\hbar^2}{24}.
\end{aligned}$$

Therefore, each focus-focus singularity contributes a nontrivial $\frac{\hbar^2}{24}$-term and we have the following equality

$$\mathrm{Ad}_{\exp \frac{-\hat{\tau}}{2} (\log \hat{z}^m)^2} \circ \mathrm{Ad}_{\exp \hat{\tau} \widehat{H}_{0,+}} \circ \mathrm{Ad}_{\exp \hat{\tau} \widehat{H}_{0,-}} = \mathrm{Ad}_{\exp c\hat{\tau}} \circ \mathrm{Ad}_{\exp(\frac{-1}{24\hat{\tau}})},$$

for some constant $c \in \mathbb{C}$ as a quantized contact transformation.

As a result, in order to satisfy condition (6.2), we need to modify the quantum wall structure by this $\frac{\hbar^2}{24}$-term in the way shown in Figure 10. The *modified wall structure*, which will be denoted by $\widetilde{\mathscr{S}^\hbar}$, is obtained by adding the term $\frac{\hbar^2}{24}$ to the quantum Hamiltonians shown in red in Figure 10. In the modified wall structure $\widetilde{\mathscr{S}^\hbar}$, the consistency condition (6.2) is satisfied around every joint including the focus-focus singularities.

Remark 6.3. This mysterious $\frac{\hbar^2}{24}$-term, which contributes to the quantum a_D-period from the computation 6.10, is indeed reasonable from the perspective of physics. A similar term was first computed in Bershadsky-Cecotti-Ooguri-Vafa's computation of the stringy genus 1 partition function of $\mathcal{N}$=2 SCFT in [BCOV1]. It plays a role in the holomorphic anomaly equations [BCOV1; BCOV2] and the quasi-modularity of the topological string partition function [ABK]. For the mathematical treatment of the $(\mathbb{P}^2, E)$ case, please refer to the first author's paper [BFGW] for more details. Also, in the original computation of quantum periods in [ACDKV], this term also appears in the genus-1 part of the partition function. From the point of view of the Gross-Siebert program, this term comes from the focus-focus singularities.

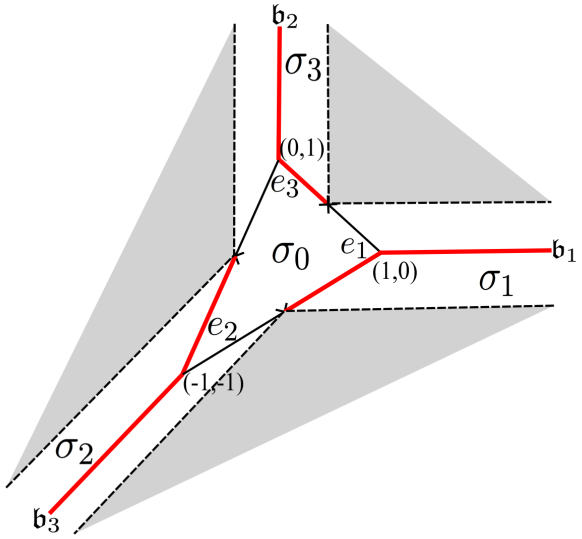


FIGURE 10. Modification of the Quantum Wall Structure. The modified quantum wall structure $\widetilde{\mathscr{S}^\hbar}$ is given by adding the term $\frac{\hbar^2}{24}$ to the quantum Hamiltonians of the slabs in red and keeping other walls unchanged. In the modified quantum wall structure, condition (6.2) is satisfied at every joint including the focus-focus singularities.

6.1.2. *Gluing Datum of the Quasi-canonical DQ Module $\mathcal{M}^c_{\mathfrak{C}^{an}}$.* Now we compute the gluing morphisms of the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$ defined in Definition 2.31. Shrink the analytification U_i^{an} of the neighborhood U_i such that it is a Stein neighborhood in the total space of the degeneration and on the generic fiber, the restriction of the mirror curve in the neighborhood is a cylinder. Let $\hat{w}_k$ and $\hat{v}_{i,\pm,k}$ be the analytic approximations of the formal quantum toric coordinates $\hat{w}, \hat{v}_{i,\pm}$ described in §5.3.1 modulo T^{k+1}. We can refine the cover $U_1^{an} \cup U_2^{an} \cup U_3^{an}$ of the mirror curve $\mathfrak{C}^{an}$ by removing the branch cut of $\log v_{i,\pm,k}$ and then make the classical functions $x_{i,\pm,k} := \log v_{i,\pm,k}$ single-valued, which are compatible with the formal classical wall crossing modulo T^{k+1} in §5.3.1 after fixing a branch of $\log T$.

In order to simplify the notation, from now on, let us just denote $x_{i,+,k}$ by x_i and omit the order k symbol. One should always keep in mind that every computation is under an analytic approximation of order k and is correct modulo T^{k+1}. Also, let us denote $\log \hat{w}$ by $\hat{p}$ since $\log \hat{w}$ will be viewed as the momentum operator in the computation.

For simplicity of computation, we will take the local model of the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$ on $\mathfrak{C}_i := \mathfrak{C}^{an} \cap U_i^{an}$ to be

$$\mathcal{M}^c_{\mathfrak{C}_i} := \mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!] e^{\tau(\log \Xi) x_i},$$

where we recall that we identify τ^{-1} with $\hbar$. Note that $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!] e^{\tau(\log \Xi) x_i}$ has the structure of a microdifferential module over $\widehat{\mathcal{E}}_{\mathfrak{C}_i \times \mathbb{C}}(0)$ supported on the image of the following Legendrian embedding

$$\begin{aligned} \mathfrak{C}_i := \mathfrak{C}^{an} \cap U_i^{an} &\longrightarrow U_i^{an} \times \mathbb{C}_{t_i} \\ x_i &\mapsto (x_i, \log \Xi(s,T), -\log \Xi(s,T) x_i), \end{aligned} \tag{6.4}$$

where $U_i^{an} \times \mathbb{C}_{t_i}$ is regarded as a contact manifold with the contact 1-form $dt_i + pdx_i$. In fact, the action of $\widehat{\mathcal{E}}_{\mathfrak{C}_i \times \mathbb{C}}(0)$ on $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log \Xi)x_i}$ is given by

$$\begin{aligned}
&\hat{t}_i(f(x_i,\tau^{-1})e^{\tau(\log\Xi)x_i}) := -\partial_\tau(f(x_i,\tau^{-1})e^{\tau(\log\Xi)x_i})\\
&= -(\partial_\tau(f(x_i,\tau^{-1}))e^{\tau(\log\Xi)x_i} - f(x_i,\tau^{-1})(\log\Xi)x_i e^{\tau(\log\Xi)x_i} \in \mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i},\\
&\hat{p}(f(x_i,\tau^{-1})e^{\tau(\log\Xi)x_i}) := \tau^{-1}\partial_{x_i}(f(x_i,\tau^{-1})e^{\tau(\log\Xi)x_i})\\
&= \tau^{-1}(\partial_{x_i}(f(x_i,\tau^{-1}))e^{\tau(\log\Xi)x_i} + f(x_i,\tau^{-1})(\log\Xi)e^{\tau(\log\Xi)x_i} \in \mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i},
\end{aligned}$$

and the actions of $\hat{x}_i$ and $\hat{\tau}^{-1}$ are just given by multiplication. The support of $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i}$ is exactly the image of the Legendrian embedding 6.4 and is isomorphic to the quotient module

$$\widehat{\mathcal{E}}_{\mathfrak{C}_i\times\mathbb{C}}(0)/\widehat{\mathcal{E}}_{\mathfrak{C}_i\times\mathbb{C}}(0)(\hat{p} - \log\Xi, \hat{t}_i + (\log\Xi)\hat{x}_i) \tag{6.5}$$

as a microdifferential module. The microdifferential module structure will induce a DQ module structure on $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i}$ over the DQ algebra $\mathcal{A}_{U_i^{an}}$, which is isomorphic to the quotient module

$$\mathcal{A}_{U_i^{an}}/\mathcal{A}_{U_i^{an}}(\hat{p} - \log\Xi).$$

Note that the local model $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i}$ is isomorphic to the local model $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]$ considered in the definition of the canonical DQ pair (Definition 2.27) by the quantized contact transformation $\mathrm{Ad}_{\exp(\hat{\tau}(\log\Xi)\hat{x}_i)}$. It is easy to show that, in the construction of the canonical DQ pair and quasi-canonical DQ module, we can replace the local model of the DQ module by $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i}$ and require that the classical contact transformation lifts of the symplectic transformations preserve the supports of $\mathcal{O}_{\mathfrak{C}_i}[\![\tau^{-1}]\!]e^{\tau(\log\Xi)x_i}$, i.e. the image of the Legendrian embedding 6.4.

Regard the index i as an element in $\mathbb{Z}/3$. Now we evaluate the gluing morphism from U_i^{an} to U_{i+1}^{an}. The quantized symplectic transformation $\Psi^{\hbar}_{i+1,i}$ relating the quantum torus algebra $\mathcal{A}_{U_i^{an}}$ is given by the quantum wall crossing of the modified quantum wall structure $\widetilde{\mathscr{S}^{\hbar}}$ in §6.1.1.

$$\Psi^{\hbar}_{i+1,i} := \mathrm{Ad}_{\exp\frac{1}{\hbar}\widehat{H}_{i,i+1}} : \mathcal{A}_{U_i^{an}} \longrightarrow (\Psi_{i+1,i})_*\mathcal{A}_{U_{i+1}^{an}},$$

where the total quantum Hamiltonian $\widehat{H}_{i,i+1}$ is defined to be

$$\widehat{H}_{i,i+1} := -3(\log T)\hat{p} + \widehat{H}_{\mathfrak{b}_i}(\hat{w}, T) + \widehat{H}_{\mathfrak{p}_{i,i+1}}(\hat{w}, T) + \frac{3}{2}\hat{p}^2 + \frac{\hbar^2}{24},\ i = 1,2,3$$

and $\Psi_{i+1,i}$ is the classical symplectic transformation given by the classical wall crossing. Recall that the last $\frac{\hbar^2}{24}$-term comes from the focus-focus singularity according to §6.1.1.

Next, we lift the quantized symplectic transformation $\Psi^{\hbar}_{i+1,i}$ to a quantized contact transformation $\tilde{\Psi}^{\hbar}_{i+1,i}$ between the contactifications whose classical limit should preserve the Legendrian embedding 6.4 and satisfy conditions (2) and (3) in Definition 2.27. In order to do so, let us separate the classical part from the quantum Hamiltonian. Define the total classical Hamiltonian by

$$\widehat{H}^{Cl}_{i,i+1} := -3(\log T)\hat{p} + \widehat{H}^{Cl}_{\mathfrak{b}_i}(\hat{w}, T) + \widehat{H}^{Cl}_{\mathfrak{p}_{i,i+1}}(\hat{w}, T) + \frac{3}{2}\hat{p}^2,\ i = 1,2,3,$$

where $\widehat{H}^{Cl}_{\mathfrak{b}_i}$ and $\widehat{H}^{Cl}_{\mathfrak{p}_{i,i+1}}(\hat{w}, T)$ are the genus zero parts of the quantum Hamiltonians in (5.17), which take the same form as the classical Hamiltonians of the classical wall crossing. Define the quantized contact transformation $\tilde{\Psi}^{\hbar}_{i+1,i}$ by

$$\tilde{\Psi}^{\hbar}_{i+1,i} := \mathrm{Ad}_{\exp\hat{\tau}(\widehat{H}_{i,i+1}(\hat{w},T) - H^{Cl}_{i,i+1}(\Xi(s,T),T))} : \widehat{\mathcal{E}}_{\mathfrak{C}_i\times\mathbb{C}}(0) \longrightarrow (\tilde{\Psi}_{i+1,i})_*\widehat{\mathcal{E}}_{\mathfrak{C}_{i+1}\times\mathbb{C}}(0),$$

where $\hbar$ should be regarded as the $\hat{\tau}^{-1}$ operator and $H^{Cl}_{i,i+1}(\Xi(s,T),T)$ is just a constant for any given s,T. The constant $H^{Cl}_{i,i+1}(\Xi(s,T),T)$ comes from the requirement that the classical limit of $\tilde{\Psi}^{\hbar}_{i+1,i}$ should preserve the Legendrian embedding 6.4. To see this, we can use the total classical Hamiltonian $\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)$ to do the computation

$$\begin{aligned}
&\mathrm{Ad}_{\exp\hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T))}(\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1})\\
=&\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1}+[\hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T)),\hat{t}_{i+1}]+[\hat{\tau}(H^{Cl}_{i,i+1}(\hat{w},T),(\log\Xi)\hat{x}_{i+1}]\\
=&\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1}+\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T)+\hat{\tau}\partial_{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T))\\
&+(\log\Xi)\hat{w}\partial_w\widehat{H}^{Cl}_{i,i+1}(\widehat{w},T).
\end{aligned}$$

Since the dependence of $\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)$ on τ is through $\hat{w}=\exp(\tau^{-1}\partial_x)$, we have

$$\hat{\tau}\partial_{\tau}\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)=\hat{\tau}\left(\partial_w\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)\right)\hat{w}(-\hat{\tau}^{-2})\partial_x=-\hat{p}\hat{w}\partial_w\widehat{H}^{Cl}_{i,i+1}(\hat{w},T).$$

Finally, we get

$$\begin{aligned}
&\mathrm{Ad}_{\exp\hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T))}(\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1})\\
=&\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1}+\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T)+(\log\Xi-\hat{p})\hat{w}\partial_w\widehat{H}^{Cl}_{i,i+1}(\widehat{w},T)\\
=&\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1}+\frac{\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T)}{\hat{p}-\log\Xi}(\hat{p}-\log\Xi)\\
&-(\hat{p}-\log\Xi)\hat{w}\partial_w\widehat{H}^{Cl}_{i,i+1}(\widehat{w},T).
\end{aligned}\tag{6.6}$$

From this computation, it is clear that $\mathrm{Ad}_{\exp\hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T)-H^{Cl}_{i,i+1}(\Xi,T))}$ maps the left ideal

$$\widehat{\mathcal{E}}_{\mathfrak{C}_i\times\mathbb{C}}(0)(\hat{p}-\log\Xi,\hat{t}_i+(\log\Xi)\hat{x}_i)$$

to the left ideal

$$\widehat{\mathcal{E}}_{\mathfrak{C}_i\times\mathbb{C}}(0)(\hat{p}-\log\Xi,\hat{t}_{i+1}+(\log\Xi)\hat{x}_{i+1})$$

and hence the classical limit preserves the Legendrian embedding 6.4. Also, note that by the argument in §6.1.1, the quantized contact transformation lifts $\tilde{\Psi}^{\hbar}_{i+1,i}$ can be extended to an open cover of the ambient space $\mathfrak{X}^{an}$ minus 3 points so that they satisfy condition (6.2), since we are using the modified quantum wall structure $\widetilde{\mathscr{S}^{\hbar}}$. Therefore, the choices of the quantized contact transformation lifts $\tilde{\Psi}^{\hbar}_{i+1,i}$ satisfy condition (1) in Definition 2.27.

Now we find the transition morphisms $\xi^c_{i+1,i}$ between the local models $\mathcal{M}^c_{\mathfrak{C}_i}$ of the quasi-canonical DQ module satisfying the conditions (2) and (3) in Definition 2.27. Recall that the transition morphism $\xi^c_{i+1,i}$ of the quasi-canonical DQ module is characterized by the following properties:

I. $\xi^c_{i+1,i}$ comes from a microdifferential module isomorphism under the quantized contact transformation $\tilde{\Psi}^{\hbar}_{i+1,i}$ (condition (2) in Definition 2.27),
II. The classical limit of $\xi^c_{i+1,i}$ should be the classical gluing morphism of the trivial spin structure $\mathcal{O}_{\mathfrak{C}^{an}}$ that we have chosen before (condition (3) in Definition 2.27).

In order to find the transition morphism $\xi^c_{i+1,i}$ satisfying the properties above, let us split the quantized contact transformation $\tilde{\Psi}^{\hbar}_{i+1,i}$ into the classical part $\tilde{\Psi}^{Cl}_{i+1,i}$ and the quantum correction part $\tilde{\Psi}^{QC}_{i+1,i}$

$$\tilde{\Psi}^{\hbar}_{i+1,i}:\widehat{\mathcal{E}}_{\mathfrak{C}_i\times\mathbb{C}(0)}\xrightarrow{\tilde{\Psi}^{Cl}_{i+1,i}}(\tilde{\Psi}_{i+1,i})_*\widehat{\mathcal{E}}_{\mathfrak{C}_{i+1}\times\mathbb{C}}(0)\xrightarrow{\tilde{\Psi}^{QC}_{i+1,i}}(\tilde{\Psi}_{i+1,i})_*\widehat{\mathcal{E}}_{\mathfrak{C}_{i+1}\times\mathbb{C}}(0)\ ,$$

where $\tilde{\Psi}^{Cl}_{i+1,i}$ and $\tilde{\Psi}^{QC}_{i+1,i}$ are defined by

$$\tilde{\Psi}^{Cl}_{i+1,i} := \mathrm{Ad}_{\exp \hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T) - H^{Cl}_{i,i+1}(\Xi,T))},$$
$$\tilde{\Psi}^{QC}_{i+1,i} := \mathrm{Ad}_{\exp \hat{\tau}(\widehat{H}_{i,i+1}(\hat{w},T) - \widehat{H}^{Cl}_{i,i+1}(\hat{w},T))}.$$

Note that the classical limit of the quantum correction part $\tilde{\Psi}^{QC}_{i+1,i}$ is the identity. From this splitting, we can construct the transition morphism $\xi^c_{i+1,i}$ as the composition

$$\xi^c_{i+1,i} : \mathcal{M}^c_{\mathfrak{C}_i} \xrightarrow{\xi^{Cl}_{i+1,i}} (\tilde{\Psi}_{i+1,i})_* \mathcal{M}^c_{\mathfrak{C}_{i+1}} \xrightarrow{\xi^{QC}_{i+1,i}} (\tilde{\Psi}_{i+1,i})_* \mathcal{M}^c_{\mathfrak{C}_{i+1}},$$

where $\xi^{Cl}_{i+1,i}$ and $\xi^{QC}_{i+1,i}$ are defined by

$$\xi^{Cl}_{i+1,i} := \mathrm{Ad}_{\exp \hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T) - H^{Cl}_{i,i+1}(\Xi,T))},$$
$$\xi^{QC}_{i+1,i} := \exp \hat{\tau}(\widehat{H}_{i,i+1}(\hat{w},T) - \widehat{H}^{Cl}_{i,i+1}(\hat{w},T))$$

which are microdifferential module isomorphisms under $\tilde{\Psi}^{Cl}_{i+1,i}$ and $\tilde{\Psi}^{QC}_{i+1,i}$ respectively. Note that $\xi^{Cl}_{i+1,i}$ is well-defined since $\mathrm{Ad}_{\exp \hat{\tau}(\widehat{H}^{Cl}_{i,i+1}(\hat{w},T) - H^{Cl}_{i,i+1}(\Xi,T))}$ maps the left ideal

$$\widehat{\mathcal{E}}_{\mathfrak{C}_i \times \mathbb{C}}(0)(\hat{p} - \log \Xi, \hat{t}_i + (\log \Xi)\hat{x}_i)$$

to the left ideal

$$\widehat{\mathcal{E}}_{\mathfrak{C}_i \times \mathbb{C}}(0)(\hat{p} - \log \Xi, \hat{t}_{i+1} + (\log \Xi)\hat{x}_{i+1})$$

by the earlier computation (6.6). It is also easy to verify that the classical limit of $\xi^c_{i+1,i}$ coincides with the transition morphism of $\mathcal{O}_{\mathfrak{C}^{an}}$ and hence $\xi^c_{i+1,i}$ is the required transition morphism of the quasi-canonical DQ module. After identifying $\mathcal{M}^c_{\mathfrak{C}_i}$ with $\mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!] \exp(\frac{1}{\hbar}(\log \Xi)x_i)$, $\xi^c_{i+1,i}$ maps the generator $\exp(\frac{1}{\hbar}(\log \Xi)x_i)$ to

$$\exp \frac{1}{\hbar}(\widehat{H}_{i,i+1}(\Xi, T) - H^{Cl}_{i,i+1}(\Xi, T)) \exp(\frac{1}{\hbar}(\log \Xi)x_{i+1}) \tag{6.7}$$

since $\exp(\frac{1}{\hbar}(\log \Xi)x_{i+1})$ is an eigenfunction of $\hat{w}$ with the eigenvalue Ξ.

6.1.3. *Computation of a^{QC} and Analytification of $\hat{w}$.* We will first compute the period a^{QC}. Since the 1-cycle α does not have monodromy with respect to T, the period $a^{QC}(s,T)$ is indeed a single-valued analytic function on $\mathbb{D}_{s,1} \times \mathbb{D}^*_T := \mathbb{D}_{s,1} \times (\mathbb{D}_T \setminus \{0\})$. We will use the analytic approximation of formal coordinates to find the Laurent series expansion of a^{QC} in terms of the information in the formal geometry $\mathcal{A}_{\mathfrak{X}}$. Consider any one of the neighborhoods, for example U^{an}_1. Since the 1-cycle α can be chosen to be contained in $\mathfrak{C}^{an} \cap U^{an}_1$, we can just compute a^{QC} in U^{an}_1. On a Stein neighborhood, for each k, there always exists an analytic section $\hat{w}_k \in \Gamma(U^{an}_1, \mathcal{A}_{\mathfrak{X}^{an}})$ such that $\hat{w}_k$ is the same as the formal quantum coordinate $\hat{w}$ modulo T^{k+1}. Hence we also know

$$\widehat{W} = \widehat{\Phi}(\hat{w}_k, T) := \hat{w}_k + \sum_{g \geq 0, d \geq 1} (-1)^g (3d-1) N^g_{3d-1,1} T^{3d} \hat{w}_k^{1-3d} \hbar^{2g} \mod T^{k+1},$$

where the quantum superpotential $\widehat{W}$ is a global analytic section of $\mathcal{A}_{\mathfrak{X}^{an}}$ computed in §5.2.2 as a global quantum theta function, and $\widehat{\Phi}$ is so far only a formal function.

Modulo T^{k+1}, the DQ algebra $\mathcal{A}_{U^{an}_1}$ on U^{an}_1 is isomorphic to a restriction of the quantum torus DQ algebra. The Landau-Ginzburg DQ module $\mathcal{M}_{LG,s}$ is isomorphic to

$$\mathcal{M}^k_{LG,s} = \mathcal{A}_{U^{an}_1} / \mathcal{A}_{U^{an}_1}(-\hat{s}'_k + \hat{w}_k) \mod T^{k+1},$$

where $\hat{s}'_k(s,T)$ is an analytic function on $\mathbb{D}_{s,1} \times \mathbb{D}_T$ such that

$$\hat{s}'_k(s,T) = \widehat{\Xi}(s,T) \mod T^{k+1},$$

where we recall that $\widehat{\Xi}$ is the inverse formal transformation of $\widehat{\Phi}$ with respect to $\hat{w}$. By considering the quantum correction 1-period

$$a_k^{QC}(s,T) := \mathcal{N}(< (\mathrm{QC}(\mathcal{M}^k_{LG,s})), [\alpha] >),$$

we get an analytic approximation of the quantum 1-period $a^{QC}(s,T)$ and their Laurent expansions in $\mathbb{D}_{s,1} \times \mathbb{D}^*_T$ coincide up to order k.

In order to compute $a_k^{QC}(s,T)$, note that the local model of the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}_1}$ is given by $\mathcal{O}_{\mathfrak{C}_1}[\![\hbar]\!] \exp(\frac{1}{\hbar}(\log \Xi)x)$, where $x := x_{1,k}$ is regarded as the analytic approximation of $\log v_{1,+}$ modulo T^k. Therefore, the period a_k^{QC} is given by the monodromy of the Hom sheaf

$$\mathcal{H}om_{\mathcal{A}_{U_1^{an}}}(\mathcal{A}_{U_1^{an}}/\mathcal{A}_{U_1^{an}}(-\hat{s}'_k + \hat{w}_k), \mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!] \exp(\frac{1}{\hbar}(\log \Xi)x))$$

along the α-cycle, i.e. the cycle given by moving x to $x + 2\pi i$. The monodromy of the Hom sheaf above is the same as the monodromy of the solution of the difference equation

$$(-\hat{s}'_k(s,T) + \exp(\hbar\partial_x))\psi_k(x) = 0,\ \psi_k \in \mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!] \exp(\frac{1}{\hbar}(\log \Xi)x)$$

whose solution can be easily computed as

$$\psi_k(x) = c \cdot \exp(\log \hat{s}'_k(s,T)x/\hbar),\ c \in \mathbb{C}^*_\hbar,$$

which is the same as the WKB approximation of the difference equation. The classical part of the wavefunction is given by

$$\psi_k^{Cl}(x) = c \cdot \exp(\log \Xi(s,T)x/\hbar),$$

where we recall that Ξ is the inverse transformation of the classical superpotential Φ, which is known to be analytic by [GRS1]. Since the α-cycle is given by moving x to $x + 2\pi i$, from the solution above, the quantum correction 1-period $a_k^{QC}(s,T)$ is given by

$$a_k^{QC}(s,T) = 2\pi i \log \frac{\hat{s}'_k(s,T)}{\Xi(s,T)},$$

Hence the Laurent expansion of $a^{QC}(s,T)$ coincides with

$$a^{QC}(s,T) = 2\pi i \log \frac{\widehat{\Xi}(s,T)}{\Xi(s,T)}. \tag{6.8}$$

Since $a^{QC}(s,T)$ and $\Xi(s,T)$ are both analytic in $\mathbb{D}_{s,1} \times \mathbb{D}_T$, the formal functions $\widehat{\Phi}, \widehat{\Xi}$ are indeed analytic and they can be extended analytically over $T = 0$ since there is no negative T-power term. As a result, there exists an analytification of the formal quantum coordinate $\hat{w}$ given by

$$\hat{w} = \widehat{\Xi}(\widehat{W}, T).$$

Finally, the quantum correction 1-period $a^{QC}(s,T)$ is given by the analytic expression (6.8), whose Taylor expansion is determined by Ξ and $\widehat{\Xi}$, the inverse transformations of the superpotential Φ and $\widehat{\Phi}$ respectively.

Remark 6.9. In the paper [GRZZ], the authors also discovered the relation between the quantum a-period and the inverse of $\widehat{\Phi}$, which is known as the quantum open mirror map (see also our explanation in §7.5.2). We have recovered their result by our study of the quantum geometry. We point out that, with a deeper understanding of the quantum geometry, we give a geometric and intrinsic definition of the quantum periods (Definition 3.13), which

is equivalent to the computational definition in [GRZZ] for the quantum a-period. More importantly, with the help of the intrinsic definition of the quantum periods (Definition 3.13), we can also give the first exact computation of the quantum a_D-period, which is crucial in establishing the quantum closed string mirror symmetry.

6.1.4. *Computation of a_D^{QC} and Analytification of $\widehat{H}_{out}$.* To compute the quantum correction 1-period a_D^{QC}, since the cycle β is only contained in the union $U_1^{an} \cup U_2^{an} \cup U_3^{an}$, we need to use the patching method to compute the β-monodromy of the Hom sheaf.

Let us first evaluate the transition function of the Hom sheaf $\mathcal{H}om(\mathcal{M}_{LG,s}, \mathcal{M}^c_{\mathfrak{C}^{an}})$ from U_1^{an} to U_2^{an}. As in the computation of the a-period, up to any order k, we take the analytic approximations of the formal quantum toric coordinates. We will omit the index k for simplicity of notation in this computation. The restrictions of the DQ algebra to U_1^{an} and U_2^{an} are just quantum torus DQ algebras and they are denoted by $\mathcal{A}_{U_1^{an}}$ and $\mathcal{A}_{U_2^{an}}$. The Landau-Ginzburg DQ modules $\mathcal{M}_{LG,s}$ are just $\mathcal{A}_{U_1^{an}}/\mathcal{A}_{U_1^{an}}(-\widehat{\Xi}(s,T)+\hat{w})$ and $\mathcal{A}_{U_2^{an}}/\mathcal{A}_{U_2^{an}}(-\widehat{\Xi}(s,T)+\hat{w})$, where $\widehat{\Xi}(s,T)$ and $\hat{w}$ are already known to be analytic in the computation of the quantum a-period (6.8). The gluing morphism of the Landau-Ginzburg DQ module is still given by the total quantum Hamiltonian $\mathrm{Ad}_{\exp\frac{1}{\hbar}\widehat{H}_{i,i+1}}$ since $\hat{w}$ commutes with $\widehat{H}_{i,i+1}$.

Now we compute the transition functions $c_{i,i+1}$ of the Hom sheaf using the transition morphisms of the DQ algebra, the Landau-Ginzburg DQ module and the quasi-canonical DQ module computed above. The local trivialization of the Hom sheaf is given by

$$\begin{aligned} &Hom_{\mathcal{A}_{U_i^{an}}}(\mathcal{M}_{LG,s}, \mathcal{M}^c_{\mathfrak{C}_i}) \longrightarrow Hom_{\mathcal{A}_{U_i^{an}}}(\mathcal{A}_{U_i^{an}}/\mathcal{A}_{U_i^{an}}(-\widehat{\Xi}+\hat{w}), \mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!]\exp(\tfrac{1}{\hbar}(\log\Xi)x_i)) \\ &\xrightarrow{Sol_i} \{\psi \in \mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!]\exp(\tfrac{1}{\hbar}(\log\Xi)x_i) | (-\widehat{\Xi}+\exp(\hbar\partial_{x_i}))\psi = 0\} \\ &\xrightarrow{\lambda\exp(\frac{1}{\hbar}(\log\widehat{\Xi})x_i)\longrightarrow\lambda, \lambda\in\mathbb{C}_\hbar} \mathbb{C}_\hbar \end{aligned},$$

where $\exp(\frac{1}{\hbar}(\log\widehat{\Xi})x_i)$ should be regarded as an element in $\mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!]\exp(\frac{1}{\hbar}(\log\Xi)x_i)$ as

$$\exp(\frac{1}{\hbar}(\log\widehat{\Xi}-\log\Xi)x_i)\exp(\frac{1}{\hbar}(\log\Xi)x_i)$$

since $\log\widehat{\Xi}-\log\Xi = O(\hbar)$.

From the local trivialization above, let us compute the transition function from the following diagram

$$\begin{array}{ccc} \mathcal{A}_{U_i^{an}}/\mathcal{A}_{U_i^{an}}(-\widehat{\Xi}+\hat{w}) & \xrightarrow{Sol_i^{-1}(\exp(\frac{1}{\hbar}(\log\widehat{\Xi})x_i))} & \mathcal{O}_{\mathfrak{C}_i}[\![\hbar]\!]\exp(\frac{1}{\hbar}(\log\Xi)x_i) \\ \downarrow{\scriptstyle \Psi^{\hbar}_{i+1,i}} & \overset{c_{i,i+1}}{\swarrow} & \downarrow{\scriptstyle \xi^c_{i+1,i}} \\ (\Psi_{i+1,i})_*\mathcal{A}_{U_{i+1}^{an}}/\mathcal{A}_{U_{i+1}^{an}}(-\widehat{\Xi}+\hat{w}) & \xrightarrow[Sol_{i+1}^{-1}(\exp(\frac{1}{\hbar}(\log\widehat{\Xi})x_{i+1}))]{} & (\Psi_{i+1,i})_*\mathcal{O}_{\mathfrak{C}_{i+1}}[\![\hbar]\!]\exp(\frac{1}{\hbar}(\log\Xi)x_{i+1}) \end{array}$$

The transition function $c_{i,i+1}$ satisfies

$$c_{i,i+1}\xi^c_{i+1,i}(\exp(\frac{1}{\hbar}(\log\widehat{\Xi})x_i)) = \exp(\frac{1}{\hbar}(\log\widehat{\Xi})x_{i+1}).$$

From the definition of $\xi^c_{i+1,i}$, we can compute the transition function

$$\begin{aligned} c_{i,i+1} &= \exp\frac{1}{\hbar}(\widehat{H}_{i,i+1}(\Xi,T) - \widehat{H}_{i,i+1}(\widehat{\Xi},T))\exp\frac{1}{\hbar}(\widehat{H}^{Cl}_{i,i+1}(\Xi,T) - \widehat{H}_{i,i+1}(\Xi,T)) \\ &= \exp\frac{1}{\hbar}(\widehat{H}^{Cl}_{i,i+1}(\Xi,T) - \widehat{H}_{i,i+1}(\widehat{\Xi},T)), \end{aligned}$$

where the second term on the first row comes from the constant term in (6.7).

In the exponent of $c_{i,i+1}$, there is a $3\log T\log(\widehat{\Xi}/\Xi) = \frac{3}{2\pi i}a^{QC}\log T$ term, which gives the monodromy of the transition function $c_{i,i+1}$ when T goes around the origin. This is expected since when T goes around, locally on the intersection $U^{an}_{i,i+1}$, the 1-cycle β will be Dehn twisted 3 times by the 1-cycle α and the intersection number of the 1-cycle β and the logarithmic branch cut will change by 3. Hence the transition function $c_{i,i+1}$ has monodromy given by the a^{QC}-period.

Finally, the monodromy of the Hom sheaf $\mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{M}_{LG,s},\mathcal{M}^c_{\mathfrak{C}^{an}})$ along β is given by

$$\begin{aligned} <\mathrm{QC}(\mathcal{M}_{LG,s}),[\beta]> &= c_{1,2}c_{2,3}c_{3,1} \\ &= \prod_{i=1}^{3}\exp\frac{1}{\hbar}(\sum_i \widehat{H}^{Cl}_{i,i+1}(\Xi,T) - \sum_i \widehat{H}_{i,i+1}(\widehat{\Xi},T)) \\ &= \exp\frac{1}{\hbar}(9(\log\widehat{\Xi}(s,T)/\Xi(s,T))\log T - \widehat{H}_{out}(\widehat{\Xi}(s,T),T) \\ &\quad + \widehat{H}^{Cl}_{out}(\Xi(s,T),T)) - \frac{9}{2}(\log\widehat{\Xi}(s,T))^2 + \frac{9}{2}(\log\Xi(s,T))^2 - \frac{\hbar^2}{8}), \end{aligned}$$

where the $\log T$-term comes from the T-monodromy of β, and the two quadratic log-terms come from the affine monodromy, and $\widehat{H}^{Cl}_{out}$ is the genus zero part of the asymptotic quantum Hamiltonian $\widehat{H}_{out}$. The quantum correction a^{QC}_D-period is given by

$$\begin{aligned} a^{QC}_D(s,T) =& 9(\log\widehat{\Xi}(s,T)/\Xi(s,T))\log T - \widehat{H}_{out}(\widehat{\Xi}(s,T),T) \\ &+ \widehat{H}^{Cl}_{out}(\Xi(s,T),T)) - \frac{9}{2}(\log\widehat{\Xi}(s,T))^2 + \frac{9}{2}(\log\Xi(s,T))^2 - \frac{\hbar^2}{8}. \end{aligned} \tag{6.10}$$

In short, the quantum correction a^{QC}_D is just given by the difference of the asymptotic quantum Hamiltonian and the asymptotic classical Hamiltonian.

Since the expression (6.10) above holds up to any T-order k and the quantum period $a^{QC}_D(s,T)$ is analytic on $\mathbb{D}_{s,1}\times\mathbb{D}^*_T$, we get that the asymptotic quantum Hamiltonian

$$\widehat{H}_{out}(\widehat{\Xi}(s,T),T)$$

is also analytic on $\mathbb{D}_{s,1}\times\mathbb{D}^*_T$.

Remark 6.11. From the point of view of the integrable systems, the superpotential W and the quantum superpotential $\widehat{W}$ give $\mathfrak{X}^{an}$ the geometry of classical and quantum integrable systems, respectively. The toric coordinates $w, v_{i,\pm}$ and the quantum toric coordinates $\hat{w}, \hat{v}_{i,\pm}$ in the asymptotic chambers indeed give the classical and quantum action-angle coordinates of the integrable system and this philosophically explains the reason why the computations of the quantum periods can be simplified in the asymptotic chambers.

6.2. **Review of Classical Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$.** In this section, we review the results on classical closed string local mirror symmetry of $K_{\mathbb{P}^2}$, or equivalently Takahashi's log mirror symmetry of $(\mathbb{P}^2,E)$, obtained via the Gross-Siebert program in [GRS1]. The classical mirror symmetry of toric Calabi-Yau threefolds was proven more than a decade ago by comparing the equivariant localization computations in the A-model with the solutions of the Picard-Fuchs equations in the B-model (see, for example, [FL] for

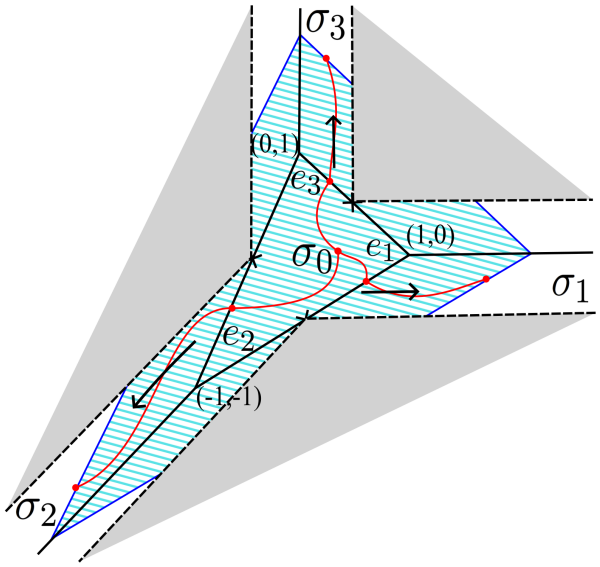


FIGURE 11. Relative Tropical 1-Cycle γ_A^{trop} and Relative Tropical 2-Cycle γ_B^{trop}. The dark blue lines show the image of the mirror curve under the generalized moment map or the SYZ fibration. The relative tropical 1-cycle γ_A^{trop}, which is drawn in red, is a 1-cycle on B relative to the image of the mirror curve with coefficients in the local system of integral tangent vectors. The tangent vector coefficients, which are shown by the black arrows, are (1,0),(0,1),(-1,-1) on the three pieces of γ_A^{trop} respectively. The relative tropical 1-cycle γ_A^{trop} defines a relative 2-cycle Γ_A on $\mathfrak{X}_T^{an}$ which is generically an S^1-fibration over γ_A^{trop} by the conormal construction in [RS]. The tropical 2-cycle γ_B^{trop} is just the relative 2-cycle on B shown as the light blue region. The tropical 2-cycle γ_B^{trop} defines a relative 2-cycle Γ_B on $\mathfrak{X}_T^{an}$ which is a positive real section of γ_B^{trop} under the SYZ fibration.

a general proof). However, the quantum closed string mirror symmetry of $K_{\mathbb{P}^2}$ remained an open question before our paper due to the computational complexity of the WKB approximation. Our paper can be regarded as a quantum enhancement of the classical treatment in [GRS1].

In the classical closed mirror symmetry, we need to compute the period integrals of the holomorphic symplectic form, or the Calabi-Yau 2-form, ω along the relative 2-cycles on the generic fiber $\mathfrak{X}_T^{an}$ whose boundaries lie on the mirror curve $\mathfrak{C}_T^{an}$. For the case of local mirror symmetry of $K_{\mathbb{P}^2}$, the relevant period integrals are

$$\Pi_A(T) = \frac{1}{2\pi i}\int_{\Gamma_A}\omega,\ \Pi_B(T) = \int_{\Gamma_B}\omega,\ A, B \in H_2(\mathfrak{X}_T^{an}, \mathfrak{C}_T^{an}; \mathbb{Z}), \tag{6.12}$$

where Γ_A is a relative 2-cycle defined by a relative tropical 1-cycle γ_A^{trop} with coefficients in the local system of integral tangent vectors on B, and Γ_B is a relative 2-cycle defined by the positive real locus of a relative tropical 2-cycle γ_B^{trop} on B. The picture of the tropical cycles γ_A^{trop} and γ_B^{trop} is depicted in Figure 11.

The results of the classical period integral computations in [GRS1] are

$$\Pi_A(T) = \frac{1}{2\pi i}\int_{\Gamma_A}\omega = \pi i + 3\log T - 3\log \Xi(1,T),$$
$$\Pi_B(T) = \int_{\Gamma_B}\omega = \frac{9}{2}(\log T - \log\Xi(1,T))^2 + \sum_{d\geq 1} N_d^0 (T/\Xi(1,T))^{3d} - 3\zeta(2),$$

where N_d^0 is the genus zero logarithmic Gromov-Witten invariant in $(\mathbb{P}^2, E)$ with one marked point such that the only intersection point with E is of contact order $3d$. From the results above we know that the closed flat coordinate Q is given by

$$Q = \exp(\Pi_A(T)) = -\frac{T^3}{\Xi(1,T)^3}$$

and the dependence of the Π_B-period on Q is

$$\Pi_B = \frac{1}{2}(\log(-Q))^2 + \sum_{d\geq 1} N_d^0 (-Q)^d - 3\zeta(2),$$

which proves the special geometry relation in the classical closed string mirror symmetry

$$\frac{\partial F_0^{K_{\mathbb{P}^2}}}{\partial \log Q} = \Pi_B,$$

where $F_0^{K_{\mathbb{P}^2}}$ is the partition function of genus zero local Gromov-Witten invariants of $K_{\mathbb{P}^2}$, which is the same as the partition function of genus zero logarithmic Gromov-Witten invariants in $(\mathbb{P}^2, E)$ by the log-local correspondence.

6.3. **Proof of Quantum Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$.** Now we give the quantum corrections of the classical period integrals in (6.12). According to Theorem 3.15, the quantum 2-periods are given by the integral of the following quantum 2-class $[\omega_\hbar]$

$$[\omega_\hbar] := [\omega] + \mathcal{N}(\mathfrak{D}(\mathcal{A}_{\mathfrak{X}^{an}}, \mathcal{M}_{LG})) = [\omega] - \delta\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG})),$$

where we recall that $\mathfrak{D}(\mathcal{A}_{\mathfrak{X}^{an}}, \mathcal{M}_{LG})$ is the relative Deligne-Fedosov class defined in Definition 2.34, $\mathrm{QC}(\mathcal{M}_{LG})$ is the quantum correction 1-period defined in Definition 3.13, and $\mathcal{N}$ is the normalization operator (6.1). Based on the computations of a^{QC} and a_D^{QC} in (6.8) and (6.10), the quantum 2-periods are given by

$$\begin{aligned}
\Pi_A^\hbar(T) &:= \frac{1}{2\pi i}\int_{\Gamma_A}[\omega_\hbar] = \frac{1}{2\pi i}\int_{\Gamma_A}[\omega] - \frac{1}{2\pi i}\int_{\Gamma_A}\delta(\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG}))) \\
&= \Pi_A(T) - \frac{1}{2\pi i}\int_{\partial\Gamma_A}\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG})) \\
&= \Pi_A(T) - \frac{1}{2\pi i}\int_{3\alpha}\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG})) \\
&= \Pi_A(T) - \frac{3}{2\pi i}a^{QC}(1,T) \\
&= \pi i + 3\log T - 3\log\widehat{\Xi}(1,T)
\end{aligned}$$

and

$$\begin{aligned}\Pi_B^\hbar(T) &:= \int_{\Gamma_B}[\omega_\hbar] = \int_{\Gamma_B}[\omega] - \int_{\Gamma_B}\delta(\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG})))\\ &= \Pi_B(T) - \int_{\partial\Gamma_B}\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG}))\\ &= \Pi_B(T) - \int_{\beta}\mathcal{N}(\mathrm{QC}(\mathcal{M}_{LG}))\\ &= \Pi_B(T) - a_D^{QC}(1,T)\\ &= \frac{9}{2}(\log T - \log\widehat{\Xi}(1,T))^2 + \sum_{d\geq 1, g\geq 0}(-1)^g N_d^g (T/\widehat{\Xi}(1,T))^{3d}\hbar^{2g} - 3\zeta(2) + \frac{\hbar^2}{8},\end{aligned}$$

where N_d^g is the genus g logarithmic Gromov-Witten invariant in $(\mathbb{P}^2, E)$ with one marked point such that the only intersection point with E is of contact order $3d$ and with the λ_g-class insertion. By our intrinsic definition of the quantum periods (Definition 3.13), the quantum corrections in the periods $\Pi_A^\hbar, \Pi_B^\hbar$ are exactly the same as those one would obtain from the WKB approximation (see Theorem 3.15). The quantum closed flat coordinate $Q^\hbar$ is given by

$$Q^\hbar = \exp(\Pi_A^\hbar(T)) = -\frac{T^3}{\widehat{\Xi}(1,T)^3}$$

and the dependence of $\Pi_B^\hbar$ on $Q^\hbar$ is[5]

$$\Pi_B^\hbar = \frac{1}{2}(\log(-Q^\hbar))^2 + \sum_{d\geq 1, g\geq 0}(-1)^g N_d^g (-Q^\hbar)^d \hbar^{2g} - 3\zeta(2) + \frac{\hbar^2}{8}.$$

Finally, we get the quantum special geometry relation

$$\frac{\partial F^{(\mathbb{P}^2,E)}}{\partial \log Q^\hbar} = \Pi_B^\hbar,$$

where $F^{(\mathbb{P}^2,E)}$ is the partition function of all-genus logarithmic Gromov-Witten invariants of $(\mathbb{P}^2, E)$. From the proof of the higher-genus Takahashi's conjecture by the first author in [Bo4], we know that

$$F^{(\mathbb{P}^2,E)} = F_{NS}^{K_{\mathbb{P}^2}},$$

where $F_{NS}^{K_{\mathbb{P}^2}}$ is the all-genus partition function of the NS-limit of the refined topological string A-model on $K_{\mathbb{P}^2}$. Hence we obtain the main result of this paper.

Theorem 6.13 (Quantum Closed String Mirror Symmetry of $K_{\mathbb{P}^2}$)**.** *The quantum 2-periods $\Pi_A^\hbar, \Pi_B^\hbar$ computed from the WKB approximation of the quantum curve equation on the mirror Landau-Ginzburg model satisfy the quantum special geometry relation*

$$\frac{\partial F_{NS}^{K_{\mathbb{P}^2}}}{\partial \log Q^\hbar} = \Pi_B^\hbar,$$

where $Q^\hbar := \exp(\Pi_A^\hbar)$ is the quantum closed flat coordinate and $F_{NS}^{K_{\mathbb{P}^2}}$ is the all-genus partition function of the NS-limit of the refined topological string A-model on $K_{\mathbb{P}^2}$.

Remark 6.14. Owing to the generality of the intrinsic definition of quantum periods (Definition 3.13) and of the Gross-Siebert construction of the mirror Landau-Ginzburg model in [CPS], our method can be directly generalized to all local toric del Pezzo surfaces.

[5]Please notice our convention of the reduced Planck constant $\hbar = -i\hbar_{phy}$ when one tries to compare with the physics literature like [ACDKV; HW].

7. Remarks on Quantum Open String Mirror Symmetry of $K_{\mathbb{P}^2}$

7.1. **Sheaf of WKB Wavefunctions and the Local Representations.** In this section, we make some remarks on how to use the Landau-Ginzburg model construction from the Gross-Siebert program to study the quantum open string mirror symmetry. Recall that since the first cohomology of $\mathfrak{X}_T^{an}$ on the generic fiber is trivial, there exists a unique quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$. From our study of the quantum closed string mirror symmetry, the central intrinsic object of interest is the Hom sheaf $\mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{\mathfrak{C}^{an}})$, whose monodromy gives the quantum correction 1-period in Definition 3.13. From our computation above, it is transparent that the local sections of the Hom sheaf $\mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{\mathfrak{C}^{an}})$ have the physics meaning of wavefunctions. Therefore, we also call the Hom sheaf $\mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{\mathfrak{C}^{an}})$ the *sheaf of WKB wavefunctions* $\mathcal{WKB}$ on the mirror curve $\mathfrak{C}^{an}$

$$\mathcal{WKB} := \mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{\mathfrak{C}^{an}}).$$

There is a natural embedding of the sheaf of WKB wavefunctions $\mathcal{WKB}$ into the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$

$$\mathcal{WKB} := \mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{M}_{LG}, \mathcal{M}^c_{\mathfrak{C}^{an}}) \hookrightarrow \mathcal{H}om_{\mathcal{A}_{\mathfrak{X}^{an}}}(\mathcal{A}_{\mathfrak{X}^{an}}, \mathcal{M}^c_{\mathfrak{C}^{an}}) = \mathcal{M}^c_{\mathfrak{C}^{an}}.$$

Hence, the sheaf of WKB wavefunctions $\mathcal{WKB}$ is a rank 1 sub $\mathbb{C}_\hbar$-local system of the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$. This sub local system is intrinsic to the deformation quantization and is an abstract object without any explicit expression. Wavefunctions can be regarded as local sections of $\mathcal{WKB}$.

In order to get the information of the open string theory, we need more data than the deformation quantization. The data we need is exactly a local trivialization of $(\mathcal{A}_{\mathfrak{X}^{an}}, \mathcal{M}^c_{\mathfrak{C}^{an}})$, which can be induced by a choice of the quantum Darboux coordinates $\hat{x}, \hat{p}$ locally on the mirror curve $\mathfrak{C}^{an}$ given by each chamber in the quantum wall structure. To be more explicit, the local trivialization gives an identification of the quasi-canonical DQ module $\mathcal{M}^c_{\mathfrak{C}^{an}}$ with

$$\mathcal{O}_{\mathfrak{C}^{an}}[\![\hbar]\!] \exp(\frac{1}{\hbar}\int^x f(x)dx),$$

where $p = f(x)$ is the local defining equation of the mirror curve $\mathfrak{C}^{an}$. As a result, locally the sheaf of WKB wavefunctions $\mathcal{WKB}$ is a rank 1 sub $\mathbb{C}_\hbar$-constant sheaf of $\mathcal{O}_{\mathfrak{C}^{an}}[\![\hbar]\!] \exp(\frac{1}{\hbar}\int^x f(x)dx)$ and the embedding

$$\mathcal{WKB} \hookrightarrow \mathcal{O}_{\mathfrak{C}^{an}}[\![\hbar]\!] \exp(\frac{1}{\hbar}\int^x f(x)dx)$$

represents the sheaf $\mathcal{WKB}$ as an explicit expression

$$\mathbb{C}_\hbar\psi \hookrightarrow \mathcal{O}_{\mathfrak{C}^{an}}[\![\hbar]\!] \exp(\frac{1}{\hbar}\int^x f(x)dx).$$

Hence the local trivialization of $(\mathcal{A}_{\mathfrak{X}^{an}}, \mathcal{M}^c_{\mathfrak{C}^{an}})$ gives a local representation of the sheaf of wavefunctions $\mathcal{WKB}$ as an element in $\mathcal{O}_{\mathfrak{C}^{an}}[\![\hbar]\!] \exp(\frac{1}{\hbar}\int^x f(x)dx)$, which is exactly the wavefunction we want on the quantum open string B-model up to the subtlety of quantum open mirror maps.

7.2. **Additional Data in the Open String Theory and Quantum Open String Mirror Symmetry.** The data of local trivialization mentioned in the quantum B-model, or the choices of the quantum Darboux coordinates $\hat{x}, \hat{p}$, also have an open string meaning in the A-model. To be more explicit, according to [ACDKV], the choice of the $\hat{p}$ coordinate corresponds to the choice of the phase of the Lagrangian A-branes, while the choice of the position coordinate $\hat{x}$ corresponds to the framing of the boundary knot or link of the open Riemann

surfaces. In physics, this degree of freedom of framing comes from the 3d Chern-Simons theory on the Lagrangian A-brane, which defines the so-called quantum knot or link invariants. The additional degrees of freedom coming from open string theory are shown in Table 1.

Open String A-Model	Open String B-Model
Phase of Lagrangian A-Brane	Choice of Quantum Momentum Coordinate $\hat{p}$
Framing of Boundary Links	Choice of Quantum Position Coordinate $\hat{x}$

TABLE 1. Correspondence of the Additional Data in the Open String Theory Between A-model and B-model. These data do not arise in closed string theory.

In the context of open mirror symmetry of toric Calabi-Yau threefolds in [AV; AKV; ACDKV], the mirror curve is also regarded as the moduli space of D-branes. In the A-model, the natural way to parametrize A-branes is to use the area of the minimal disc whose boundary is contained in the given A-brane. In the B-model, the natural way to parametrize B-branes is by the position coordinate x. Apart from the closed mirror maps given by period integrals, there are open mirror maps relating the different open string parameters of the moduli space of D-branes. Unlike closed mirror symmetry, open mirror symmetry must also take the open string parameters and open mirror maps into account, which is crucial in the context of open mirror symmetry.

Conjecture 7.1 (Quantum Open String Mirror Symmetry [ACDKV])**.** *The A-model partition function of the NS-limit of refined open topological string theory on a toric Calabi-Yau threefold is equal to the WKB approximation of the quantum curve equation on its mirror Landau-Ginzburg model after imposing the quantum open and closed mirror maps.*

What we want to emphasize about the open mirror symmetry conjecture (Conjecture 7.1) is that, in order to give the WKB approximation an enumerative meaning, the corresponding trivialization of the deformation quantization, or the choice of quantum Darboux coordinates, should not be arbitrary. They should be the quantum open flat coordinates obtained by applying the quantum open-closed mirror maps to the originally chosen coordinates. This subtlety is important in the statement of open mirror symmetry and leads us to the question of how to find the quantum flat open coordinates.

7.3. **Quantum Flat Coordinates and Quantum Mirror Curve Under the Quantum Flat Coordinates.** In the physics literature [AV; AKV; ACDKV], the open mirror map comes from the small periods, i.e. periods along the 1-cycles which are not contractible in the ambient Hori-Vafa torus. To be more specific, the classical open mirror map comes from the period integral of the 1-form $\log P d\log X$ along a cycle which does not cross the branch cut of $\log P$ but crosses the branch cut of $\log X$. The quantum open mirror map comes from the integral of the quantum 1-form $\lambda_\hbar = \log P d\log X + \lambda_{QC}$ along the same cycle. In the context of local mirror symmetry of $K_{\mathbb{P}^2}$, this is just equal to the period integral of $\log wd\log v_{1,+}$ and its quantum analog along the α cycle, which is equal to the inverse of Φ and $\widehat{\Phi}$ (see [GRZ; GRZZ] or our explanation in §7.5.2).

On the tropical manifold $(B, \mathscr{P}, \varphi, \mathscr{S}^h)$, for each choice of the symplectic basis of the local system of integral vectors and each choice of the chamber, at least formally, we obtain a choice of quantum Darboux coordinates on $\mathfrak{X}$ from the toric coordinates corresponding to the basis. However, these coordinates are not the quantum flat coordinates in the context of local mirror symmetry and there is a nontrivial open mirror map relating them to the quantum flat coordinates. Consider the affine geometry outside the central chamber σ_0. Take the asymptotic direction ξ to be the direction corresponding to the momentum $\widehat{P}$ and

take the primitive integral vector ζ_i which is parallel to one of the bounded edges of $\mathscr{P}$ to be the direction corresponding to the position $\widehat{X}$. Under this choice of symplectic basis (ζ_i, ξ), on each chamber b outside σ_0, the quantum open-closed mirror map and the quantum flat coordinates are given by

$$\widehat{X}'_b = \widehat{X}_b,\ \widehat{P}'_b = \widehat{P}_b \cdot \widehat{\Xi}(1,T)^{-1},\ T' = T \cdot \widehat{\Xi}(1,T)^{-1}, \tag{7.2}$$

where $\widehat{X}'_b, \widehat{P}'_b, T' \in \mathcal{A}_{\mathfrak{X}}$ are the quantum flat coordinates and $\widehat{\Xi}(s,T)$ gives the quantum open mirror map as in [GRZZ] and (6.8). For the central chamber $b = \sigma_0$, the quantum open-closed mirror map preserves all toric coordinates and the transformation on the parameter T is $T' = T \cdot \widehat{\Xi}(1,T)^{-1}$.

Remark 7.3. We point out that the flatness condition of the quantum open coordinates in local mirror symmetry refers to the dependence on T, and on each generic fiber, both $\widehat{X}'_b, \widehat{P}'_b$ and $\widehat{X}_b, \widehat{P}_b$ are quantum Darboux coordinates since the quantum open mirror maps (7.2) are just multiplication by a constant. The reason why the issue of open flat coordinates does not appear in the closed string mirror symmetry is that both the classical and quantum closed string B-models only depend on the intrinsic invariants from the geometry, i.e. period integrals and the relative Deligne-Fedosov class (recall Definitions 2.34 and 3.13).

Using the quantum flat coordinates $\widehat{X}'_b, \widehat{P}'_b, T' \in \mathcal{A}_{\mathfrak{X}}$ for each chamber b, at least formally, we can find the local trivialization mentioned in §7.1 and give the sheaf of WKB wavefunctions $\mathcal{WKB}$ an explicit expression. The quantum open mirror symmetry conjecture (Conjecture 7.1) predicts that this WKB expression is the same as the partition function of the NS-limit of refined open topological string theory on $K_{\mathbb{P}^2}$. If b is an asymptotic chamber $b = \infty$, then $\widehat{P}_\infty, \widehat{X}_\infty$ can be identified with $\hat{w}, \hat{v}_{i,+}$ that we considered in §5.3.1. On $\mathfrak{X}_{out}$, the Hori-Vafa definition of the quantum mirror curve is defined by the Landau-Ginzburg DQ module

$$\mathcal{M}_{LG}^{HV} := \mathcal{A}_{\mathfrak{X}_{out}}/\mathcal{A}_{\mathfrak{X}_{out}}(-1+\widehat{W}) = \mathcal{A}_{\mathfrak{X}_{out}}/\mathcal{A}_{\mathfrak{X}_{out}}(-\widehat{\Xi}(1,T)+\hat{w}) = \mathcal{A}_{\mathfrak{X}_{out}}/\mathcal{A}_{\mathfrak{X}_{out}}(-1+\hat{w}').$$

Therefore, in terms of the quantum flat coordinates, the Landau-Ginzburg DQ module $\mathcal{M}_{LG}$ is exactly the Landau-Ginzburg DQ module corresponding to the superpotential $\hat{w}'$. Under the flat coordinates, the mirror curve is just defined by the equation $-1 + w' = 0$ and the quantum mirror curve takes the simple form $\mathcal{A}_{\mathfrak{X}_{out}}/\mathcal{A}_{\mathfrak{X}_{out}}(-1+\hat{w}')$.

7.4. **Hori-Vafa Picture and Gross-Siebert Picture of the Mirror Curve.** Throughout the whole paper, we have been using the Hori-Vafa definition of the classical and quantum mirror curves, which are defined by the fiber of the superpotential W at 1, in order to compare with the original local mirror symmetry conjecture in the physics literature [HV; AV; AKV; ACDKV]. However, as we have seen in the physics literature, in [GRS1] and in our computation (6.8), a nontrivial open-closed mirror map is needed to obtain the flat coordinates for the local mirror symmetry.

In fact, in the Gross-Siebert construction of the Landau-Ginzburg model in [CPS] that we have been using throughout the paper, the authors proposed a different definition, or picture, of the mirror curve. The Gross-Siebert definition of the quantum mirror curve is given by

$$\mathcal{M}_{LG}^{GS} := \mathcal{A}_{\mathfrak{X}}/\mathcal{A}_{\mathfrak{X}}(-\widehat{\Phi}(1,T)+\widehat{W}),$$

whose restriction to $\mathfrak{X}_{out}$ is given by

$$\mathcal{A}_{\mathfrak{X}_{out}}/\mathcal{A}_{\mathfrak{X}_{out}}(-1+\hat{w}),$$

which takes the same form as the quantum mirror curve under flat coordinates.

In fact, the different definitions of the mirror curve just refer to different parametrizations of the mirror geometry and the quantum geometry of the Hori-Vafa mirror curve $(\mathcal{A}_{\mathfrak{X}_{out}}, \mathcal{M}_{LG}^{HV})$ is isomorphic to the quantum geometry of the Gross-Siebert mirror curve $(\mathcal{A}_{\mathfrak{X}_{out}}, \mathcal{M}_{LG}^{GS})$. We

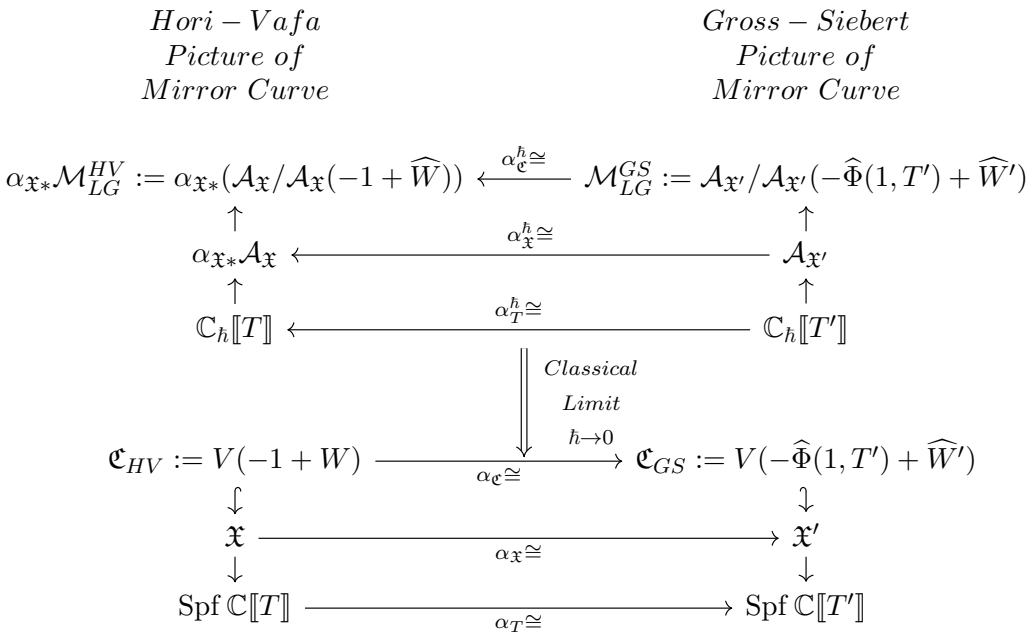


FIGURE 12. Comparison between the Hori-Vafa Picture and the Gross-Siebert Picture of the Mirror Curve. They are related exactly by the classical open-closed mirror maps $\alpha_{\mathfrak{X}}, \alpha_{\mathfrak{C}}, \alpha_T$ and the quantum open-closed mirror maps $\alpha_{\mathfrak{X}}^{\hbar}, \alpha_{\mathfrak{C}}^{\hbar}, \alpha_T^{\hbar}$. The coordinates in the Gross-Siebert picture of the mirror curve are automatically flat coordinates.

point out that, although the two definitions of the quantum mirror curve are different, the definitions of the ambient symplectic surface are the same. The following proposition and Figure 12 compare the Hori-Vafa picture of the mirror curve with the Gross-Siebert picture of the mirror curve. The isomorphism between the Hori-Vafa picture and the Gross-Siebert picture is exactly given by the quantum open-closed mirror map. In order to distinguish the coordinates in these two different pictures, we will denote the coordinates in the Gross-Siebert picture with $'$ and denote the coordinates in the Hori-Vafa picture without $'$.

Proposition 7.4. *For the chamber b outside the central chamber σ_0, denote the quantum toric coordinate given by the asymptotic direction ξ by $\widehat{P}_b, \widehat{P}_b'$. In the closure of the 2-cell σ_i in $\mathscr{P}$ containing b, there is a unique bounded edge, and we denote the quantum toric coordinate given by the direction ζ_i along the bounded edge by $\widehat{X}_b, \widehat{X}_b'$ such that ζ_i, ξ form a symplectic basis of the local system of integral vectors on the chamber b. Define the isomorphism from the quantum Hori-Vafa picture to the quantum Gross-Siebert picture by*

$$\widehat{X}_b' \mapsto \widehat{X}_b,\ \widehat{P}_b' \mapsto \widehat{P}_b \cdot \widehat{\Xi}(1,T)^{-1},\ T' \mapsto T \cdot \widehat{\Xi}(1,T)^{-1}.$$

For the chamber $b = \sigma_0$, the isomorphism is the identity on every toric coordinate and the transformation on T is $T' \mapsto T \cdot \widehat{\Xi}(1,T)^{-1}$. Then the quantum coordinate transformations defined above glue to well-defined isomorphisms on $\mathcal{A}_{\mathfrak{X}}, \mathcal{M}_{LG}, \mathbb{C}_\hbar[\![T]\!]$ shown in Figure 12. The induced isomorphisms are denoted by $\alpha_{\mathfrak{X}}^{\hbar}, \alpha_{\mathfrak{C}}^{\hbar}, \alpha_T^{\hbar}$ and the classical limits are denoted by $\alpha_{\mathfrak{X}},\ \alpha_{\mathfrak{C}},\ \alpha_T$ respectively.

Proof. The key observation for proving this proposition is that for all quantum walls that are not slabs, every monomial involved in the quantum Hamiltonians takes the form

$$cT^m \widehat{X}_b^n \widehat{P}_b^{-m},\ c \in \mathbb{C}_\hbar,\ m, n \in \mathbb{Z},$$

and for all slabs, every monomial involved in the quantum Hamiltonians is either $c\widehat{X}^n$ or $cT^m\widehat{P}^{-m}$, where $\widehat{X}$ or $\widehat{P}$ refer to the quantum toric coordinates along the direction of the corresponding slab. Therefore, all the quantum Hamiltonians are invariant under the quantum coordinate transformations defined above and these quantum coordinate transformations give rise to well-defined isomorphisms $\alpha^\hbar_{\mathfrak{X}}, \alpha^\hbar_T$. To get a well-defined $\alpha^\hbar_{\mathfrak{C}}$, we only need to verify that $\alpha^\hbar_{\mathfrak{X}}$ maps the left ideal $\mathcal{A}_{\mathfrak{X}'}(-\widehat{\Phi}(1,T')+\widehat{W}')$ to the left ideal $\mathcal{A}_{\mathfrak{X}}(-1+\widehat{W})$. Note that the induced transformation on $\widehat{W}$ is $\widehat{W}' \mapsto \widehat{W}\cdot\widehat{\Xi}(1,T)^{-1}$. Then the well-definedness is achieved by the following computation

$$\begin{aligned}-\widehat{\Phi}(1,T')+\widehat{W}' \mapsto &-\widehat{\Phi}(1,T\cdot\widehat{\Xi}(1,T)^{-1})+\widehat{W}\cdot\widehat{\Xi}(1,T)^{-1}\\ &=-\widehat{\Phi}(\widehat{\Xi}(1,T),T)\cdot\widehat{\Xi}(1,T)^{-1}+\widehat{W}\cdot\widehat{\Xi}(1,T)^{-1}\\ &=(-1+\widehat{W})\cdot\widehat{\Xi}(1,T)^{-1},\end{aligned}$$

where the first equality comes from the concrete expression of $\widehat{\Phi}$ in (5.18). □

The isomorphisms $\alpha^\hbar_{\mathfrak{X}}, \alpha^\hbar_{\mathfrak{C}}, \alpha^\hbar_T$ between the Hori-Vafa picture of the mirror curve and the Gross-Siebert picture of the mirror curve are exactly the open-closed mirror maps we mentioned in §7.3. Therefore, the quantum toric coordinates and the coordinate T' on the base in the Gross-Siebert picture are automatically the quantum flat coordinates and the quantum open-closed mirror maps are trivialized in the Gross-Siebert picture. One can also verify this by substituting $s=\widehat{\Phi}(1,T')$ into our computation of the quantum a-period in (6.8), which gives the trivial mirror map. In summary, we conclude that the quantum flat coordinates we were searching for are just the natural coordinates in the Gross-Siebert picture of the mirror curve. This is an example of the following slogan about the canonical or flat coordinates in the Gross-Siebert program.

Slogan 7.5 (Canonical or Flat Coordinates in the Gross-Siebert Program)**.** The coordinates naturally arising in the Gross-Siebert program are automatically canonical or flat coordinates in the context of mirror symmetry.

Remark 7.6. In the case of closed string mirror symmetry, our paper provides an example of the above slogan in the case where the relevant period integrals are along relative cycles. When the period integrals are along usual cycles without boundary, this slogan was proven in the paper [RS]. In this sense, one may think that the Gross-Siebert construction of mirrors is better than the constructions by physicists from the mathematical perspective since every coordinate is automatically canonical or flat. This also philosophically explains the reason why the wall functions in the Gross-Siebert program have a clear enumerative meaning since the wall functions are already written in terms of the canonical or flat coordinates.

7.5. **Quantum Wall Crossing of Wavefunctions and Punctured-Open Correspondence.** Now we have found the quantum flat coordinates we need for the quantum open string mirror symmetry, which is equivalent to working directly with the Gross-Siebert picture of the mirror curve instead of the Hori-Vafa picture. The mechanism of open mirror symmetry is then hidden in the quantum wall structure arising in the Gross-Siebert program.

7.5.1. *Quantum Wall Crossing of Wavefunctions.* For each chamber b and each choice of the symplectic basis ζ,ξ of the local system of integral vectors on b, at least formally or up to a finite T-order, we get a choice of quantum Darboux coordinates $\hat{x}_b, \hat{p}_b$ on the neighborhood of the mirror curve $\mathfrak{C}^{an}$, which are automatically flat coordinates in local mirror symmetry if we use the Gross-Siebert picture of the mirror curve. Under this choice of quantum Darboux coordinates, we can represent the sheaf of WKB wavefunctions $\mathcal{WKB}$ formally as a wavefunction ψ_b as explained in §7.1.

When we move from one chamber b to an adjacent chamber b' through a quantum wall $\mathfrak{d} \in \mathscr{S}^\hbar$, there is a quantum wall crossing of the quantum Darboux coordinates given by the adjoint action

$$\widehat{A}_{b'} = \mathrm{Ad}_{\exp(\frac{\epsilon}{\hbar}\hat{H}_\mathfrak{d})}(\widehat{A}_b) = \exp(\frac{\epsilon}{\hbar}\hat{H}_\mathfrak{d})\widehat{A}_b \exp(-\frac{\epsilon}{\hbar}\hat{H}_\mathfrak{d}),$$

where ϵ is the sign of $\omega(n, m(H_\mathfrak{d}))$ and n is the direction of wall crossing. The wall crossing formula above naturally induces the following quantum wall crossing formula for the wavefunctions

$$\psi_{b'} = \exp(\frac{\epsilon}{\hbar}\hat{H}_\mathfrak{d})\psi_b. \tag{7.7}$$

It is easy to see that the wall crossing of the WKB wavefunctions, or equivalently local solutions of the quantum curve equation, are still the WKB wavefunctions under the corresponding quantum Darboux coordinates since the quantum superpotential $\widehat{W}$ is changed by the adjoint action under the quantum wall crossing.

7.5.2. *Quantum a-period and Quantum Open Mirror Map.* As an application of the quantum wall crossing of wavefunctions, we explain the meaning of the quantum a-period along the α-cycle in (6.8) from the B-model perspective. In fact, from the simple computations below, we find that

$$\frac{\psi^{HV}(x + 2\pi i)}{\psi^{HV}(x)}$$

is invariant under quantum wall crossing, where $\psi^{HV}(x)$ means the WKB wavefunction under the Hori-Vafa picture of the quantum mirror curve.

Proposition 7.8. *For a chamber b, if the WKB wavefunction $\psi_b(x)$ satisfies*

$$\frac{\psi_b(x + 2\pi i)}{\psi_b(x)} = \frac{\psi_b(x + 2\pi i + m\hbar)}{\psi_b(x + m\hbar)}$$

for any $m \in \mathbb{Z}$, then after a quantum wall crossing through a wall $\mathfrak{d}$ in $\mathscr{S}^\hbar$ to an adjacent chamber b', the WKB wavefunction $\psi_{b'}(x)$ satisfies

$$\frac{\psi_{b'}(x + 2\pi i)}{\psi_{b'}(x)} = \frac{\psi_{b'}(x + 2\pi i + m'\hbar)}{\psi_{b'}(x + m'\hbar)} = \frac{\psi_b(x + 2\pi i)}{\psi_b(x)} = \frac{\psi_b(x + 2\pi i + m\hbar)}{\psi_b(x + m\hbar)}$$

for any $m, m' \in \mathbb{Z}$.

Proof. Notice that $\exp(\frac{\epsilon}{\hbar}H_\mathfrak{d})$ in $\mathscr{S}^\hbar$ always takes the form

$$\exp(\frac{\epsilon}{\hbar}H_\mathfrak{d}) = \sum_{a\geq 0} b_a T^{am}\widehat{X}^{an}\widehat{P}^{-am},\ b_a \in \mathbb{C}((\hbar)),\ (m, n) \in \mathbb{Z}^2.$$

Then we have the following computation

$$\begin{aligned}
\frac{\psi_{b'}(x + 2\pi i)}{\psi_{b'}(x)} &= \frac{\sum_{a\geq 0} b_a T^{am}\exp(anx + 2an\pi i)\psi_b(x - am\hbar + 2\pi i)}{\sum_{a\geq 0} b_a T^{am}\exp(anx)\psi_b(x - am\hbar)} \\
&= \frac{\sum_{a\geq 0} b_a T^{am}\exp(anx)\psi_b(x - am\hbar)\frac{\psi_b(x+2\pi i-am\hbar)}{\psi_b(x-am\hbar)}}{\sum_{a\geq 0} b_a T^{am}\exp(anx)\psi_b(x - am\hbar)} \\
&= \frac{\sum_{a\geq 0} b_a T^{am}\exp(anx)\psi_b(x - am\hbar)\frac{\psi_b(x+2\pi i)}{\psi_b(x)}}{\sum_{a\geq 0} b_a T^{am}\exp(anx)\psi_b(x - am\hbar)} \\
&= \frac{\psi_b(x + 2\pi i)}{\psi_b(x)}.
\end{aligned}$$

Similarly, we also have

$$\frac{\psi_{b'}(x+2\pi i+m'\hbar)}{\psi_{b'}(x+m'\hbar)}=\frac{\psi_b(x+2\pi i)}{\psi_b(x)}$$

for any $m'\in\mathbb{Z}$. □

Since in the asymptotic chamber $b=\infty$, the wavefunction in the Hori-Vafa picture of the quantum mirror curve takes the form

$$\psi_\infty^{HV}=\exp(\frac{1}{\hbar}\widehat{\Xi}(1,T)x)$$

as shown in the computation of the quantum a-period in (6.8), and satisfies the condition in Proposition 7.8, we know that $\frac{\psi^{HV}(x+2\pi i)}{\psi^{HV}(x)}$ is the same for every chamber. Therefore, in the Hori-Vafa central chamber σ_0, we have

$$\frac{\psi_{\sigma_0}^{HV}(x+2\pi i)}{\psi_{\sigma_0}^{HV}(x)}=\frac{\psi_\infty^{HV}(x+2\pi i)}{\psi_\infty^{HV}(x)}=\exp(\frac{2\pi i}{\hbar}\widehat{\Xi}(1,T)),$$

which means that the quantum open mirror map defined in [ACDKV] is given by $\widehat{\Xi}(1,T)$. This explains the relation between the quantum a-period along the α-cycle in (6.8) and the quantum open mirror map from the quantum B-model perspective.

7.5.3. *Punctured-Open Correspondence.* Via the quantum wall crossing of the wavefunctions, we can also reformulate the quantum open mirror symmetry conjecture (Conjecture 7.1). Up to any finite T-order, as we have already seen in the proof of quantum closed mirror symmetry in §6, among all the chambers in the quantum wall structure $\mathscr{S}^\hbar$, there are two kinds of chambers that are of great interest. One kind is the asymptotic chambers $b=\infty$, where the computation is trivial; the other is the Hori-Vafa central chamber σ_0, where every computation is the same as the computation in the physics literature [ACDKV].

For the asymptotic chambers $b=\infty$, if we use ζ_i,ξ as chosen in Propositionin Proposition 7.4, then the quantum mirror curve in the Gross-Siebert picture, or equivalently the quantum mirror curve under the quantum flat coordinates, is given by

$$\mathcal{A}_{\mathfrak{x}_{out}}/\mathcal{A}_{\mathfrak{x}_{out}}(-1+\hat{w}).$$

By regarding $\hat{w}$ as the exponential of the quantum momentum coordinate, the corresponding WKB wavefunction takes the trivial form

$$\psi_\infty=1.$$

For the Hori-Vafa central chamber $b=\sigma_0$, the quantum mirror curve in the Gross-Siebert picture is exactly the quantum curve considered in the physics literature [ACDKV] after using the quantum flat coordinates. The quantum open mirror symmetry conjecture (Conjecture 7.1) predicts that the WKB wavefunction ψ_{σ_0} in the Hori-Vafa chamber is the same as the partition function of the NS-limit of refined open topological string theory on $K_{\mathbb{P}^2}$ with an internal Aganagic-Vafa A-brane if the symplectic basis is obtained by parallel transport of ζ_i,ξ in the asymptotic chambers.

However, up to a finite T-order, the asymptotic wavefunction $\psi_\infty=1$ and the Hori-Vafa chamber wavefunction are related exactly by the quantum wall crossing of wavefunctions in (7.7), i.e.

$$\psi_{\sigma_0}=\mathcal{P}\prod_{\mathfrak{d}}\exp(\epsilon_{\mathfrak{d}}\hat{H}_{\mathfrak{d}})\psi_\infty,$$

where $\mathcal{P}$ is the path-ordering operator and $\mathcal{P}\prod_{\mathfrak{d}}$ means that the product is taken along the path of the wall crossing. Therefore, the quantum open mirror symmetry conjecture (Conjecture 7.1) is equivalent to the following Punctured-Open Correspondence in enumerative geometry, which is formulated in terms of the wall crossing of wavefunctions (7.7).

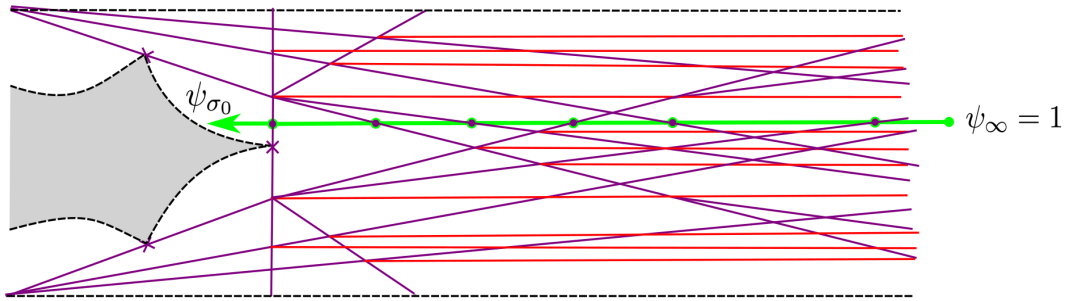


FIGURE 13. Quantum Wall Crossing of Wavefunctions from the Asymptotic Chambers to the Hori-Vafa Central Chamber σ_0. The asymptotic, or horizontal, quantum walls are in red and the non-horizontal quantum walls are in purple. Note that if we perform the quantum wall crossing (7.7) from the asymptotic wavefunction $\psi_\infty = 1$ to get the Hori-Vafa chamber wavefunction ψ_{σ_0} along the green path, the only quantum walls that contribute nontrivially to the wavefunctions are some of the non-horizontal walls, whose enumerative meaning is given by the higher-genus punctured Gromov-Witten invariants defined in [ACGS]. Different wall crossing paths give the same result because of the consistency of the quantum wall structure $\mathscr{S}^\hbar$.

Conjecture 7.9 (Punctured-Open Correspondence)**.** *Up to any finite T-order, the partition function of the NS-limit of refined open topological string theory A-model on $K_{\mathbb{P}^2}$ (more generally, on the canonical bundles of toric del Pezzo surfaces) with an internal Aganagic-Vafa A-brane is the same as*

$$\mathcal{P}\prod_{\mathfrak{d}} \exp(\epsilon_{\mathfrak{d}}\hat{H}_{\mathfrak{d}})\psi_\infty,$$

where the wall crossing path is from the asymptotic chamber $b = \infty$ to the Hori-Vafa central chamber $b = \sigma_0$ in the quantum wall structure of the mirror Landau-Ginzburg model of $(\mathbb{P}^2, E)$ (more generally a toric del Pezzo surface together with a smooth anticanonical divisor) and ψ_∞ is just the constant function 1*.*

The reason why Conjecture 7.9 above is called the Punctured-Open Correspondence is that, if we take the wall crossing path from an asymptotic chamber to the bounded chamber shown in Figure 13, we find that the quantum walls that contribute to the wall crossing are some of the non-horizontal walls. From the canonical wall structure introduced in [GS4], the enumerative meaning of the non-horizontal walls is exactly the punctured Gromov-Witten invariants defined in [ACGS] and Conjecture 7.9 is just an equivalence between the punctured Gromov-Witten invariants and certain open Gromov-Witten invariants. Punctured Gromov-Witten invariants count the so-called punctured curves on the toric degeneration of $(\mathbb{P}^2, E)$ whose intersection numbers with the components of the central fiber are allowed to be negative. Curves with negative intersection number with a component of the central fiber can never be smoothed out to closed curves in the generic fiber. Instead, they can be smoothed out to Riemann surfaces with boundary. Therefore, the punctured Gromov-Witten invariants are the algebro-geometric analogs of the open Gromov-Witten invariants, which is why we expect the Punctured-Open Correspondence (Conjecture 7.9) to hold. On the other hand, there is much work on the symplectic geometric meaning of the wall structure

considered in [CPS] in the genus zero case, for example Auroux's work [Au] and the further work [Li3; CJL; LLL; Li4]. In the genus zero case, the symplectic geometric meaning of the non-horizontal walls is the count of Maslov index zero discs whose boundaries are on the special Lagrangian torus fiber of the SYZ-fibration, which provides some evidence for the Punctured-Open Correspondence (Conjecture 7.9). A rigorous treatment and proof of this Punctured-Open Correspondence (Conjecture 7.9) is the subject of work in progress by the authors.

Remark 7.10. In the papers [TZ; SSZ; HSZ; HS], the authors also found that the moduli space of D-branes is a holomorphic Lagrangian submanifold inside an ambient space with cluster structure and there is a wall crossing structure of the partition functions of skein-valued open Gromov-Witten theory defined by Ekholm-Shende in [ES] in the case of $\mathbb{C}^3$ and resolved conifolds in a different set-up. In their set-up, the wall crossing of the partition function corresponds to certain surgery of the Lagrangian A-branes coming from the Lagrangian filling of certain Legendrian surfaces. It is an interesting question to determine the relation between their wall structures and the quantum wall structures in our study of the quantum mirror symmetry of local toric del Pezzo surfaces.

Appendix A. Categorical Characterization of Canonical Deformation Quantization

This appendix is an extension of Remark 2.32 and is inspired by a discussion with Pavel Safronov. In Construction 2.10 and Definition 2.27, we used the gluing data with certain characteristic properties to define the canonical isomorphism classes of DQ algebroids and DQ pairs on a given classical geometry. However, if we only consider DQ algebroids and DQ pairs without any additional data, as we have done throughout the paper, then there is no way to single out a canonical isomorphism between two deformation quantizations in the canonical isomorphism class, and we always need to take care of the ambiguity from the automorphisms. For the case of DQ algebroids, this ambiguity of automorphisms can be fixed by keeping track of some additional data. In fact, we can construct a contractible 2-groupoid of canonical DQ algebroids by considering enough additional data. By forgetting these data, we recover the canonical isomorphism class of DQ algebroids in Construction 2.10. Also, by considering these additional data, we can give a more categorical definition of canonical DQ pairs, which is equivalent to the gluing definition (Definition 2.27).

A.1. Categorical Characterization of Canonical DQ Algebroids.

This section mainly follows [DK, Appendix A]. Let (X,ω) be a holomorphic symplectic manifold and let us consider the following additional data on a DQ algebroid on X.

Definition A.1 (Strict Anti-linear Involution)**.** Let $\mathcal{A}_X$ be a DQ algebroid on X. A $\mathbb{C}_\hbar$-anti-linear equivalence of algebroids

$$* : \mathcal{A}_X \longrightarrow \mathcal{A}_X^{op}$$

is a *strict anti-linear involution* if it satisfies $*^2 = Id_{\mathcal{A}_X}$. The $\mathbb{C}_\hbar$-anti-linearity means that it is a linear functor under the map

$$\begin{aligned} * : \mathbb{C}_\hbar &\longrightarrow \mathbb{C}_\hbar \\ \sum_{i\geq 0} a_i \hbar^i &\mapsto \sum_{i\geq 0} (-1)^i a_i \hbar^i, \end{aligned}$$

which we will also denote by $*$ if there is no ambiguity.

On the holomorphic symplectic manifold (X, ω), let $(\mathbb{C}[\epsilon]/\epsilon^2)_\omega$ be an invertible $\mathbb{C}[\epsilon]/\epsilon^2$-algebroid defined by the image of the symplectic class $[\omega]$ under the homomorphism

$$H^2(X;\mathbb{C}) \longrightarrow H^2\left(X;(\mathbb{C}[\epsilon]/\epsilon^2)^*\right)$$

induced by the group homomorphism

$$\begin{aligned}\mathbb{C} &\longrightarrow (\mathbb{C}[\epsilon]/\epsilon^2)^* \\ \lambda &\mapsto 1+\epsilon\lambda.\end{aligned}$$

Definition A.2 (Strict $\mathbb{C}$-linear Derivation)**.** A *strict $\mathbb{C}$-linear derivation* on a DQ algebroid $\mathcal{A}_X$ is a $\mathbb{C}$-linear functor

$$\nu : \mathcal{A}_X \longrightarrow \mathcal{A}_X \otimes_{\mathbb{C}} (\mathbb{C}[\epsilon]/\epsilon^2)_\omega$$

such that the composition

$$\mathcal{A}_X \xrightarrow{\ \nu\ } \mathcal{A}_X \otimes_{\mathbb{C}} (\mathbb{C}[\epsilon]/\epsilon^2)_\omega \xrightarrow{\ /\epsilon\ } \mathcal{A}_X$$

is equal to the identity functor $Id_{\mathcal{A}_X}$, where the second arrow means quotienting by ϵ.

Note that for DQ algebras, the $\mathbb{C}$-linear derivation above recovers the usual definition of $\mathbb{C}$-linear derivation defined by the Leibniz rule.

Now we want to consider a DQ algebroid $\mathcal{A}_X$ with the following additional data

(1) a strict anti-linear involution $*$, which defines a formal adjoint operator on $\mathcal{A}_X$,
(2) an equivalence of $\mathbb{C}$-algebroids
$$\mathrm{cl} : \mathcal{A}_X/\hbar\mathcal{A}_X \longrightarrow (\mathcal{O}_X)^+,$$
which stands for the classical limit morphism.
(3) a strict $\mathbb{C}$-linear derivation
$$\nu : \mathcal{A}_X \longrightarrow \mathcal{A}_X \otimes_{\mathbb{C}} (\mathbb{C}[\epsilon]/\epsilon^2)_\omega$$
satisfying $\nu(\hbar) = \hbar + \epsilon\hbar^2, *\nu = \nu*$. Here we define $\epsilon^* = -\epsilon$, where ϵ is the formal variable in $\mathbb{C}[\epsilon]/\epsilon^2$.

Example A.3 (Local Model of Tuple $(\mathcal{A}_X, *, \mathrm{cl}, \nu)$)**.** On the cotangent bundle local model T^*M, there is a natural way to construct the DQ algebra with the data above. The local model of the DQ algebra is given by $\mathcal{A}_{T^*M} = \rho_* \widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C},\partial_t}$ defined in Construction 2.7. The anti-linear involution $*$ comes from the formal adjoint operator on the sheaf of formal microdifferentials. The classical limit comes from the usual classical limit map on $\mathcal{A}_{T^*M}$. The $\mathbb{C}$-linear derivation is given by

$$\begin{aligned}\nu : \mathcal{A}_{T^*M} &\longrightarrow \mathcal{A}_{T^*M} \otimes_{\mathbb{C}} \mathbb{C}[\epsilon]/\epsilon^2 \\ Q &\mapsto Q + \epsilon\hbar ad_{\hat{t}\partial_t}(Q),\end{aligned}$$

where $ad_{\hat{t}\partial_t}(Q)$ means the commutator $[\hat{t}\partial_t, Q]$ in the sheaf of formal microdifferentials $\rho_*\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$. Since $ad_{\hat{t}\partial_t}$ maps an element in $\mathcal{A}_{T^*M}$ to an element in $\mathcal{A}_{T^*M}$, it gives a well-defined $\mathbb{C}$-linear derivation on $\mathcal{A}_{T^*M}$. It is easy to check that $\nu(\hbar) = \hbar+\epsilon\hbar^2, *\circ\nu = \nu\circ*$. From the local model above, one can see that the $\mathbb{C}$-linear derivation ν encodes the information of the t-direction in the contactification of T^*M in (2.8).

Definition A.4 (Strict Canonical DQ Algebroid)**.** A *strict canonical DQ algebroid* on a holomorphic symplectic manifold (X, ω) is defined to be the tuple $(\mathcal{A}_X, *, \mathrm{cl}, \nu)$ such that it satisfies $\nu(\hbar) = \hbar + \epsilon\hbar^2, * \circ \nu = \nu \circ *$ and is locally isomorphic to the local model in Example A.3.

Now we study the 2-groupoid of the strict canonical DQ algebroids defined above. Let us use $\bullet$ to denote the horizontal composition of functors.

Definition A.5 (2-Groupoid $\mathfrak{DQ}^{can}_{X,\omega}$ of Strict Canonical DQ Algebroids)**.** The *2-groupoid $\mathfrak{DQ}^{can}_{X,\omega}$ of strict canonical DQ algebroids* on the holomorphic symplectic manifold (X,ω) is defined as follows:

(1) Objects: strict canonical DQ algebroids $(\mathcal{A}_X, *, \mathrm{cl}, \nu)$,

(2) 1-Morphisms from $(\mathcal{A}_X, *, \mathrm{cl}, \nu)$ to $(\mathcal{A}'_X, *', \mathrm{cl}', \nu')$: pairs (Ψ, η), where

$$\Psi : \mathcal{A}_X \longrightarrow \mathcal{A}'_X$$

is an equivalence of $\mathbb{C}_\hbar$-algebroids satisfying $\mathrm{cl} = \mathrm{cl}' \circ \overline{\Psi}$ and $\Psi^{op} \circ * = *' \circ \Psi$, and

$$\eta : (\Psi \otimes_{\mathbb{C}} Id_{(\mathbb{C}[\epsilon]/\epsilon^2)_\omega}) \circ \nu \Longrightarrow \nu' \circ \Psi$$

is a natural isomorphism such that we have the following equalities of the horizontal compositions of natural transformations:

(a)
$$Id_{/\epsilon} \bullet \eta = Id_\Psi : \Psi \Longrightarrow \Psi,$$

(b)
$$\eta^{op} \bullet Id_* = Id_* \bullet \eta : * \circ (\Psi \otimes_{\mathbb{C}} Id_{(\mathbb{C}[\epsilon]/\epsilon^2)_\omega}) \circ \nu \Longrightarrow * \circ \nu' \circ \Psi,$$

where $\overline{\Psi}$ and Ψ^{op} are the equivalences induced by Ψ and $\bullet$ means the horizontal composition of natural transformations.

(3) 2-Morphisms from (Ψ, η) to (Ψ', η'): pairs (P, Λ), where

$$P : \Psi \Longrightarrow \Psi'$$

is a natural isomorphism between the functors Ψ, Ψ' and Λ is a natural isomorphism

$$\Lambda : Id_{(\mathbb{C}[\epsilon]/\epsilon^2)_\omega} \Longrightarrow Id_{(\mathbb{C}[\epsilon]/\epsilon^2)_\omega}$$

of the identity functor $Id_{(\mathbb{C}[\epsilon]/\epsilon^2)_\omega}$ of $(\mathbb{C}[\epsilon]/\epsilon^2)_\omega$. Also, (P, Λ) should satisfy the following equalities of the horizontal compositions of natural transformations:

(a)
$$Id_{/\epsilon} \bullet \Lambda = Id_{/\epsilon} : /\epsilon \Longrightarrow /\epsilon$$

(b)
$$Id_{*'} \bullet P = P^{op} \bullet Id_* : *' \circ \Psi \Longrightarrow *' \circ \Psi'$$

(c)
$$Id_{\mathrm{cl}'} \bullet P = Id_{\mathrm{cl}} : \mathrm{cl}' \circ \Psi \Longrightarrow \mathrm{cl}' \circ \Psi'$$

(d)
$$\eta' \circ (P \otimes_{\mathbb{C}} \Lambda \bullet Id_\nu) = (Id_{\nu'} \bullet P) \circ \eta : \Psi \otimes_{\mathbb{C}} Id_{(\mathbb{C}[\epsilon]/\epsilon)_\omega} \circ \nu \Longrightarrow \nu' \circ \Psi'$$

The 2-groupoid of strict canonical DQ algebroids $\mathfrak{DQ}^{can}_{X,\omega}$ defined above is in fact contractible, i.e. it is equivalent to the trivial 2-groupoid with a single object. In particular, the 2-groupoid $\mathfrak{DQ}^{can}_{X,\omega}$ defines a canonical isomorphism class of DQ algebroids, which justifies our terminology in Definition A.4.

Proposition A.6. *The 2-groupoid of strict canonical DQ algebroids $\mathfrak{DQ}^{can}_{X,\omega}$ is contractible.*

Proof. Existence of Strict Canonical DQ Algebroids: This is contained in [DK, Appendix A]. In fact, we can enhance the gluing datum of the canonical DQ algebroid in Construction 2.10 to a gluing datum of a strict canonical DQ algebroid defined in Definition A.4. First, we can endow the local model $\mathcal{A}_{T^*M_i}$ with the additional data $(*, \mathrm{cl}, \nu)$ as in Example A.3. Also, the quantized contact transformation lifts $\tilde{\Psi}^\hbar_{ij}$ give the 1-morphisms in $\mathfrak{DQ}^{can}_{X,\omega}$ as

$$(\Psi^\hbar_{ij}, \eta_{ij} := 1 + \epsilon(\hat{t}_i - \tilde{\Psi}^\hbar_{ij}(\hat{t}_j))).$$

The 2-morphisms on the triple intersection in the gluing datum can be written as

$$(P_{ijk}, \Lambda_{ij} := 1 + \epsilon c_{ijk}),$$

where P_{ijk} is the same as the natural transformation in Construction 2.10 and c_{ijk} is the *Čech* 2-cocycle of the symplectic class $[\omega]$. Due to the fact that the gluing datum in Construction 2.10 satisfies the conditions in Proposition 2.11, the gluing data specified above are compatible with each other and glue to a global strict canonical DQ algebroid defined in Definition A.4, which implies the existence of strict canonical DQ algebroids.

Contractibility of the 2-Groupoid $\mathfrak{DQ}^{can}_{X,\omega}$: This follows from the following lemma, proven in [DK, Lemma A.1.5].

Lemma A.7 ([DK, Lemma A.1.5])**.** *On the local model $(\mathcal{A}_{T^*M}, *, \mathrm{cl}, \nu)$ in Example A.3, we define $\mathcal{A}^{*,1,u}_{T^*M}$ to be the subsheaf of groups of the DQ algebra $\mathcal{A}_{T^*M}$*

$$\mathcal{A}^{*,1,u}_{T^*M} := \{P \in \mathcal{A}_{T^*M} | \sigma_0(P) = 1, PP^* = 1\}$$

and $\mathcal{A}ut(\mathcal{A}_{T^*M}, *, \nu)$ to be the sheaf of automorphisms of the tuple $(\mathcal{A}_{T^*M}, *, \nu)$

$$\mathcal{A}ut(\mathcal{A}_{T^*M}, *, \nu) :=$$

$$\{(\Psi, Q) \in \mathcal{A}ut(\mathcal{A}_{T^*M}) \ltimes \mathcal{A}_{T^*M} | \Psi^{op} \circ * = * \circ \Psi, ad_{\hat{t}\partial_t} \circ \Psi - \Psi \circ ad_{\hat{t}\partial_t} = \frac{1}{\hbar} ad_Q \circ \Psi, Q = Q^*\}$$

with multiplication given by the semi-direct product

$$(\Psi, Q) \cdot (\Psi', Q') := (\Psi \circ \Psi', Q + \Psi(Q')).$$

Note that every pair $(\Psi, Q) \in \mathcal{A}ut(\mathcal{A}_{T^*M}, *, \nu)$ defines a 1-morphism in $\mathfrak{DQ}^{can}_{X,\omega}$ by $(\Psi, 1+\epsilon Q)$. Then there is an isomorphism of sheaves of groups given by

$$\begin{aligned} \mathcal{A}^{*,1,u}_{T^*M} \times \underline{\mathbb{C}}_{T^*M} &\longrightarrow \mathcal{A}ut(\mathcal{A}_{T^*M}, *, \nu) \\ (P, c) &\mapsto (\mathrm{Ad}_P, -c + \hbar ad_{\hat{t}\partial_t}(P) P^{-1}). \end{aligned} \tag{A.8}$$

Note that every pair $(P, c) \in \mathcal{A}^{*,1,u}_{T^*M} \times \underline{\mathbb{C}}_{T^*M}$ induces a 2-morphism in $\mathfrak{DQ}^{can}_{X,\omega}$ by $(P, 1+\epsilon c)$ and the above morphism can be regarded as the target morphism from a collection of 2-morphisms to 1-morphisms.

In fact, to prove the contractibility, it is sufficient to show that the 2-groupoid $\mathfrak{DQ}^{can}_{X,\omega}$ is weakly contractible, i.e. the homotopy groups $\pi_i(\mathfrak{DQ}^{can}_{X,\omega}), i = 0, 1, 2$ are all trivial. Let us denote the stack of autoequivalences of a canonical DQ algebroid $(\mathcal{A}_X, *, \mathrm{cl}, \nu)$ on X by

$$\mathfrak{Aut}(\mathcal{A}_X, *, \mathrm{cl}, \nu),$$

which is a stack of 2-groups. In the local model of Example A.3, $\mathfrak{Aut}(\mathcal{A}_X, *, \mathrm{cl}, \nu)$ is isomorphic to the associated stack of 2-groups of the sheaf of crossed modules

$$[\mathcal{A}^{*,1,u}_{T^*M} \times \underline{\mathbb{C}}_{T^*M} \longrightarrow \mathcal{A}ut(\mathcal{A}_{T^*M}, *, \nu)]$$

given in (A.8), which is a trivial stack of 2-groups according to Lemma A.7. Hence the stack of 2-groups $\mathfrak{Aut}(\mathcal{A}_X, *, \mathrm{cl}, \nu)$ is actually trivial globally. Using the cohomology of stacks of 2-groups used in [Po2, §4], we have

$$\pi_0(\mathfrak{DQ}^{can}_{X,\omega}) = H^1(X, \mathfrak{Aut}(\mathcal{A}_X, *, \mathrm{cl}, \nu)) = 0, \ \pi_1(\mathfrak{DQ}^{can}_{X,\omega}) = H^0(X, \mathfrak{Aut}(\mathcal{A}_X, *, \mathrm{cl}, \nu)) = 0.$$

For $\pi_2(\mathfrak{DQ}^{can}_{X,\omega})$, it is also trivial since the morphism in (A.8) is injective. □

Note that it follows from the first part of the proof of Proposition A.6 that the canonical isomorphism class of DQ algebroids considered in Construction 2.10 is exactly given by the contractible 2-groupoid $\mathfrak{DQ}^{can}_{X,\omega}$ after forgetting the additional data $(*, \mathrm{cl}, \nu)$. Therefore, this contractible 2-groupoid $\mathfrak{DQ}^{can}_{X,\omega}$ gives the categorical characterization of the canonical isomorphism class of DQ algebroids.

Remark A.9. In this section, we have been considering the strict canonical DQ algebroids (Definition A.4), whose additional data are all strict in the categorical sense, for simplicity of statements and proofs. One can also consider a weak version of canonical DQ algebroids with weak additional data. In fact, following an argument similar to the proof of [Po1, Theorem 3.3], one can show that the 2-groupoid of weak canonical DQ algebroids is equivalent to the 2-groupoid of strict canonical DQ algebroids and hence is also contractible.

A.2. **Categorical Characterization of Canonical DQ Pairs.** Given a strict canonical DQ algebroid $(\mathcal{A}_X, *, \mathrm{cl}, \nu)$, we will consider a simple holonomic DQ module $\mathcal{M}_L$ supported on a Lagrangian submanifold L over $\mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(L)}$, which will serve as a quantization of the Lagrangian L.

Definition A.10 (Microdifferential Structure)**.** A *microdifferential structure* μ on $\mathcal{M}_L$ with respect to ν is a morphism of modules

$$\mu : \mathcal{M}_L \longrightarrow \mathcal{M}_L \otimes_{\mathbb{C}} \mathbb{C}[\epsilon]/\epsilon^2$$

under the morphism of algebroids

$$\nu : \mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(L)} \longrightarrow \mathcal{A}_X \otimes_{\mathbb{C}} \mathbb{C}_{w_2(L)} \otimes_{\mathbb{C}} \mathbb{C}[\epsilon]/\epsilon^2$$

satisfying

$$/\epsilon \circ \mu = Id_{M_L},$$

where

$$/\epsilon : \mathcal{M}_L \otimes_{\mathbb{C}} \mathbb{C}[\epsilon]/\epsilon^2 \longrightarrow \mathcal{M}_L$$

is the quotient morphism by ϵ. Note that we have omitted the twist of $\mathbb{C}[\epsilon]/\epsilon^2$ by the symplectic class $[\omega]$ since it is always trivial near a Lagrangian.[6]

We call the morphism μ in Definition A.10 a microdifferential structure because it is equivalent to the action of the sheaf of formal microdifferentials on the DQ module $\mathcal{M}_L$. In fact, on the local model 2.8 and A.3, the local model of a simple holonomic DQ module is $K_M^{1/2}[\![\hbar]\!]$. Then a microdifferential structure μ is just given by

$$\begin{aligned} \mu : K_M^{1/2}[\![\hbar]\!] &\longrightarrow K_M^{1/2}[\![\hbar]\!] \otimes_{\mathbb{C}} \mathbb{C}[\epsilon]/\epsilon^2 \\ m &\mapsto m + \epsilon\beta(m), \end{aligned}$$

where β satisfies

$$\beta(a \cdot m) = [\hat{t}, a] \cdot m + a \cdot \beta(m), \ a \in \mathcal{A}_{T^*M}, \ m \in K_M^{1/2}[\![\hbar]\!].$$

Therefore, we can identify β with the action of the $\hat{t}$-operator in $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$, i.e. $\beta(m) = \hat{t} \cdot m$. As a result, with a microdifferential structure μ, we can enhance the DQ module $K_M^{1/2}[\![\hbar]\!]$ to a microdifferential module.

However, the sheaves of formal microdifferentials, or the formal microdifferential algebroids, do not make sense globally on general holomorphic symplectic manifolds. Therefore, we need to localize to a neighborhood of the Lagrangian L and use the following notion of contactification of a Lagrangian submanifold L.

[6] In fact, Proposition A.12 can be enhanced to the fact that on a Lagrangian L, there is a contractible gerbe of contactifications of L. By forgetting the Legendrian lifts, this contractible gerbe has a functor to the restriction of the gerbe of contactifications of the ambient symplectic manifold X. As a result, on the Lagrangian L, this forgetful functor will induce a canonical trivialization of the gerbe of contactifications of the symplectic manifold X, and hence a canonical trivialization of $(\mathbb{C}[\epsilon]/\epsilon^2)_\omega$.

Definition A.11 (Contactification of Lagrangian Submanifolds)**.** Let V be an open neighborhood of a Lagrangian submanifold L in a holomorphic symplectic manifold (X, ω). Let $\rho : \tilde{V} \longrightarrow V$ be a contactification of V, i.e. ρ is locally isomorphic to the map ρ in 2.8. Let $\tilde{i} : L \hookrightarrow \tilde{V}$ be a lift of the Lagrangian embedding $i : L \hookrightarrow V$ such that $\tilde{i}$ is a Legendrian embedding. Then the tuple $(V, \rho, \tilde{i})$ is called a *contactification* of the Lagrangian submanifold L.

The isomorphism of two contactifications $(V, \rho : \tilde{V} \longrightarrow V, \tilde{i})$ and $(V, \rho' : \tilde{V}' \longrightarrow V, \tilde{i}')$ on the neighborhood V is given by a contact transformation $\tilde{\Psi} : \tilde{V} \longrightarrow \tilde{V}'$ satisfying $\rho = \rho' \circ \tilde{\Psi}$, $\tilde{i}' = \tilde{\Psi} \circ \tilde{i}$. Let $Cont_V(L)$ be the collection of isomorphism classes of contactifications of L on V. An equivalence class in $\varinjlim_{V \supseteq L} Cont_V(L)$ is called a *germ of contactification* of L.

Proposition A.12 (Existence and Uniqueness of Germ of Contactification [DS, Lemma 8.4], [DK, Lemma 2.5.1], [GS6, Proposition 1.12])**.** *For any holomorphic Lagrangian submanifold L, there exists a unique germ of contactification of L.*

On the contactification $\tilde{V}$ of L, there exists a canonical formal microdifferential algebroid $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V}}(0)$ which serves as a quantization of the contact manifold $\tilde{V}$ according to [Ka1, Theorem 2]. Also, the 2-groupoid of canonical formal microdifferential algebroids with classical limits and formal adjoint operators is also contractible (see [Po1, Theorem 3.3]). Moreover, the canonical formal microdifferential algebroid on $\tilde{V}$ induces a strict canonical DQ algebroid on V denoted by $(\rho_* \widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V},\partial_t}(0), *, \mathrm{cl}, \nu)$, where $\rho_* \widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V},\partial_t}(0)$ is given by the subalgebroid commuting with ∂_t and $*, \mathrm{cl}, \nu$ are given by the local model in Example A.3. Due to the contractibility of the 2-groupoid of strict canonical DQ algebroids, we can canonically identify the restriction of the strict canonical DQ algebroid to V with $(\rho_* \widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V},\partial_t}(0), *, \mathrm{cl}, \nu)$.

Let $\rho_* \widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V},\partial_t}(0) - mod^L_{sh,\nu}$ be the category of simple holonomic DQ modules together with microdifferential structures with respect to ν supported on L and $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V}}(0) - mod^{\tilde{L}}_{sh}$ be the category of simple holonomic microdifferential modules supported on $\tilde{L}$. Then, due to the identification of the microdifferential structure μ with the action of the formal microdifferential algebroid, we have the following equivalence of categories.

Proposition A.13. *There is an equivalence of categories*

$$\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V}}(0) - mod^{\tilde{L}}_{sh} \cong \rho_* \widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V},\partial_t}(0) - mod^L_{sh,\nu}$$

According to [GS6, Theorem 3.16], given a choice of spin structure $K_L^{1/2}$, there is a unique simple holonomic microdifferential module $\widetilde{\mathcal{M}}^{can}_{L,\bar{\lambda}}$ over $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V}}$ of certain order $\bar{\lambda} \in \mathbb{C}/\mathbb{Z}$ and with microlocalization $K_L^{1/2}$ ([GS6, Definition 3.6]). This module is called the canonical microdifferential module in [GS6] of order $\bar{\lambda} \in \mathbb{C}/\mathbb{Z}$.

In fact, we can enhance this result in the context of simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde{V}}(0)$-modules, and the distinction is that now the order λ is required to be in $\mathbb{C}$ rather than $\mathbb{C}/\mathbb{Z}$. More specifically, first observe that, in the local model of simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-modules of order $\bar{\lambda} \in \mathbb{C}/\mathbb{Z}$, there exists a unique simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-lattice of order $\lambda \in \mathbb{C}$ for any lift λ of $\bar{\lambda}$.

Proposition A.14. *Consider the simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-module $\tau^\lambda K_M^{1/2}((\tau^{-1})), \lambda \in \mathbb{C}$. Note that shifting λ by an integer gives an isomorphic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-module. Then for any $m \in \mathbb{Z}$, there exists a unique simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-submodule of order $\lambda+m \in \mathbb{C}$ which generates*

$\tau^\lambda K_M^{1/2}((\tau^{-1}))$. *This module is called a* simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-lattice *of order* $\lambda+m$ *and always takes the form* $\tau^{\lambda+m}K_M^{1/2}[\![\tau^{-1}]\!]$.

Proof. Let u be a local generator of $\tau^{\lambda+m}K_M^{1/2}[\![\tau^{-1}]\!]$. Consider another simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-lattice of order $\lambda+m\in\mathbb{C}$ with local generator u'. Since u',u are both simple generators of $\tau^\lambda K_M^{1/2}((\tau^{-1}))$, locally there exists an invertible operator $P\in\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$ such that $u'=Pu$ ([KK, Theorem 4.1], [Ka2, Theorem 8.20] for a proof). Since u',u are of the same order, we know that P is of order zero. Therefore the $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-submodules generated by u',u are the same. □

Given a lift $\lambda\in\mathbb{C}$ of $\bar\lambda\in\mathbb{C}/\mathbb{Z}$, since the principal symbols and orders of generators are preserved under $\hat\tau^{-1},*$-preserving quantized contact transformations, we know that the simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}(0)$-lattice of order $\lambda\in\mathbb{C}$ is preserved under any isomorphism of $\widehat{\mathcal{E}}^{\sqrt{v}}_{M\times\mathbb{C}}$-modules above a $\hat\tau^{-1},*$-preserving quantized contact transformation. Therefore, we can globalize the simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde V}(0)$-lattice of certain order $\lambda\in\mathbb{C}$.

Proposition A.15. *For any simple holonomic* $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde V}$*-module of order* $\bar\lambda\in\mathbb{C}/\mathbb{Z}$ *and any lift* $\lambda\in\mathbb{C}$*, there exists a unique simple holonomic* $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde V}(0)$*-lattice of order* λ*.*

As a result, given a choice of spin structure $K_L^{1/2}$, up to isomorphism, there is a unique tuple

$$(\mathcal{A}_X,*,\mathrm{cl},\nu,\mathcal{M}_L,\mu)$$

such that

(1) $(\mathcal{A}_X,*,\mathrm{cl},\nu)$ is a strict canonical DQ algebroid as defined in Definition A.4,
(2) $(\mathcal{M}_L,\mu)$ is a simple holonomic DQ module supported on L with a microdifferential structure μ with respect to ν
(3) Under the equivalence in Proposition A.13, $(\mathcal{M}_L,\mu)$ gives the simple holonomic $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde V}(0)$-module of order $\frac{1}{2}$ such that the microlocalization of the base change $\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde V}\otimes_{\widehat{\mathcal{E}}^{\sqrt{v}}_{\tilde V}(0)}\mathcal{M}_L$ is $K_L^{1/2}$ ([GS6, Definition 3.6]).

The canonical pair $(\mathcal{A}_X^c,\mathcal{M}_L^c)$ defined in Definition 2.27 is just the canonical isomorphism class of DQ pairs obtained by forgetting the additional data $(*,\mathrm{cl},\nu,\mu)$ in the tuple above. This gives the categorical characterization of the canonical DQ pairs in Definition 2.27.

References

[ABK] Mina Aganagic, Vincent Bouchard, and Albrecht Klemm. "Topological Strings and (Almost) Modular Forms". In: *Commun. Math. Phys.* 277 (2008), pp. 771–819. DOI: 10.1007/s00220-007-0383-3. arXiv: hep-th/0607100.

[AC] Dan Abramovich and Qile Chen. "Stable logarithmic maps to Deligne–Faltings pairs II". In: *Asian Journal of Mathematics* 18.3 (2014), pp. 465–488. DOI: 10.4310/AJM.2014.v18.n3.a5. arXiv: 1102.4531 [math.AG].

[ACDKV] Mina Aganagic, Miranda C. N. Cheng, Robbert Dijkgraaf, Daniel Krefl, and Cumrun Vafa. "Quantum geometry of refined topological strings". In: *Journal of High Energy Physics* 2012.11 (Nov. 2012). ISSN: 1029-8479. DOI: 10.1007/jhep11(2012)019. URL: http://dx.doi.org/10.1007/JHEP11(2012)019.

[ACGS] Dan Abramovich, Qile Chen, Mark Gross, and Bernd Siebert. *Punctured Logarithmic Maps*. Vol. 15. Memoirs of the European Mathematical Society. Berlin: EMS Press, 2025. ISBN: 978-3-98547-086-0. DOI: 10.4171/mems/15. URL: https://ems.press/books/mems/300.

[AKKT] Takashi Aoki, Takahiro Kawai, Tatsuya Koike, and Yoshitsugu Takei. "On the exact WKB analysis of operators admitting infinitely many phases". In: *Adv. Math.* 181.1 (2004), pp. 165–189. ISSN: 0001-8708,1090-2082. DOI: 10.1016/S0001-8708(03)00049-5. URL: https://doi.org/10.1016/S0001-8708(03)00049-5.

[AKV] Mina Aganagic, Albrecht Klemm, and Cumrun Vafa. "Disk Instantons, Mirror Symmetry and the Duality Web". In: *Zeitschrift für Naturforschung A* 57.1–2 (Feb. 2002), pp. 1–28. ISSN: 0932-0784. DOI: 10.1515/zna-2002-1-201. URL: http://dx.doi.org/10.1515/zna-2002-1-201.

[Au] Denis Auroux. "Mirror symmetry and T-duality in the complement of an anticanonical divisor". In: *Journal of Gökova Geometry and Topology* 1 (2007), pp. 51–91. URL: https://www.gokovagt.org/journal/2007/jggt07-auroux.pdf.

[AV] Mina Aganagic and Cumrun Vafa. *Mirror Symmetry, D-Branes and Counting Holomorphic Discs*. 2000. arXiv: hep-th/0012041 [hep-th]. URL: https://arxiv.org/abs/hep-th/0012041.

[Ba] Salvatore Baldino. *WKB Methods for Finite Difference Schrodinger Equations*. 2024. arXiv: 2410.14628 [math-ph]. URL: https://arxiv.org/abs/2410.14628.

[BBG] Pierrick Bousseau, Andrea Brini, and Michel van Garrel. "On the log–local principle for the toric boundary". In: *Bulletin of the London Mathematical Society* 54.1 (2022), pp. 161–181. DOI: 10.1112/blms.12566.

[BCOV1] M. Bershadsky, S. Cecotti, H. Ooguri, and C. Vafa. "Holomorphic anomalies in topological field theories". In: *Nuclear Physics B* 405.2 (1993), pp. 279–304. ISSN: 0550-3213. DOI: https://doi.org/10.1016/0550-3213(93)90548-4. URL: https://www.sciencedirect.com/science/article/pii/0550321393905484.

[BCOV2] M. Bershadsky, S. Cecotti, H. Ooguri, and C. Vafa. "Kodaira-Spencer theory of gravity and exact results for quantum string amplitudes". In: *Communications in Mathematical Physics* 165.2 (1994), pp. 311–427. DOI: 10.1007/BF02099774. URL: https://doi.org/10.1007/BF02099774.

[BFGW] Pierrick Bousseau, Honglu Fan, Shuai Guo, and Longting Wu. "Holomorphic anomaly equation for $(\mathbb{P}^2, E)$ and the Nekrasov-Shatashvili limit of local $\mathbb{P}^2$". In: *Forum Math. Pi* 9 (2021), Paper No. e3, 57. ISSN: 2050-5086. DOI: 10.1017/fmp.2021.3. URL: https://doi.org/10.1017/fmp.2021.3.

[BGKP] Vladimir Baranovsky, Victor Ginzburg, Dmitry Kaledin, and Jeremy Pecharich. "Quantization of line bundles on lagrangian subvarieties". In: *Selecta Math. (N.S.)* 22.1 (2016), pp. 1–25. ISSN: 1022-1824,1420-9020. DOI: 10.1007/s00029-015-0181-2. URL: https://doi.org/10.1007/s00029-015-0181-2.

[BK] R. Bezrukavnikov and D. Kaledin. "Fedosov quantization in algebraic context". In: *Mosc. Math. J.* 4.3 (2004), pp. 559–592, 782. ISSN: 1609-3321,1609-4514. DOI: 10.17323/1609-4514-2004-4-3-559-592. URL: https://doi.org/10.17323/1609-4514-2004-4-3-559-592.

[BKMP] Vincent Bouchard, Albrecht Klemm, Marcos Mariño, and Sara Pasquetti. "Remodeling the B-Model". In: *Communications in Mathematical Physics* 287.1 (2009), pp. 117–178. DOI: 10.1007/s00220-008-0620-4. URL: https://doi.org/10.1007/s00220-008-0620-4.

[Bo1] Pierrick Bousseau. "Tropical refined curve counting from higher genera and lambda classes". In: *Inventiones mathematicae* 215.1 (2019), pp. 1–79. DOI: 10.1007/s00222-018-0823-z. URL: https://doi.org/10.1007/s00222-018-0823-z.

[Bo2] Pierrick Bousseau. "Quantum mirrors of log Calabi–Yau surfaces and higher-genus curve counting". In: *Compositio Mathematica* 156.2 (Jan. 2020), pp. 360–411. ISSN: 1570-5846. DOI: 10.1112/s0010437x19007760. URL: http://dx.doi.org/10.1112/S0010437X19007760.

[Bo3] Pierrick Bousseau. "The quantum tropical vertex". In: *Geometry & amp; Topology* 24.3 (Sept. 2020), pp. 1297–1379. ISSN: 1465-3060. DOI: 10.2140/gt.2020.24.1297. URL: http://dx.doi.org/10.2140/gt.2020.24.1297.

[Bo4] Pierrick Bousseau. "A proof of N. Takahashi's conjecture for $(\mathbb{P}^2, E)$ and a refined sheaves/Gromov–Witten correspondence". In: *Duke Mathematical Journal* 172.15 (2023), pp. 2895–2955. DOI: 10.1215/00127094-2022-0095. URL: https://doi.org/10.1215/00127094-2022-0095.

[BR] Paul Bressler and Juan Diego Rojas. "On the classification of symplectic DQ-algebroids". In: *Theory and Applications of Categories* (2023). URL: https://api.semanticscholar.org/CorpusID:258041320.

[BS] Andrea Brini and Yannik Schuler. *Refined Gromov-Witten invariants*. 2024. arXiv: 2410.00118 [math.AG]. URL: https://arxiv.org/abs/2410.00118.

[BW] Sean Bates and Alan Weinstein. *Lectures on the Geometry of Quantization*. Vol. 8. Berkeley Mathematics Lecture Notes. American Mathematical Society, 1997. ISBN: 9780821807989. URL: https://math.berkeley.edu/~alanw/GofQ.pdf.

[CCGGK] Alessio Corti, Tom Coates, Sergey Galkin, Vasily Golyshev, and Alexander Kasprzyk. "Mirror Symmetry and Fano Manifolds". In: *European Congress of Mathematics (Kraków, 2012)*. European Mathematical Society, 2014, pp. 285–300. DOI: 10.4171/120-1/16.

[CJL] Tristan C. Collins, Adam Jacob, and Yu-Shen Lin. "Special Lagrangian submanifolds of log Calabi–Yau manifolds". In: *Duke Mathematical Journal* 170.7 (2021), pp. 1291–1375. DOI: 10.1215/00127094-2021-0012.

[CKK] Jinwon Choi, Sheldon Katz, and Albrecht Klemm. "The refined BPS index from stable pair invariants". In: *Comm. Math. Phys.* 328.3 (2014), pp. 903–954. ISSN: 0010-3616,1432-0916. DOI: 10.1007/s00220-014-1978-0. URL: https://doi.org/10.1007/s00220-014-1978-0.

[CKYZ] Ti-ming Chiang, Albrecht Klemm, Shing-Tung Yau, and Eric Zaslow. "Local mirror symmetry: Calculations and interpretations". In: *Advances in Theoretical and Mathematical Physics* 3 (1999), pp. 495–565. URL: https://api.semanticscholar.org/CorpusID:15870294.

[CPS] Michael Carl, Max Pumperla, and Bernd Siebert. "A Tropical View on Landau–Ginzburg Models". In: *Acta Mathematica Sinica, English Series* 40.1 (Jan. 2024), pp. 329–382. ISSN: 1439-7617. DOI: 10.1007/s10114-024-2295-y. URL: http://dx.doi.org/10.1007/s10114-024-2295-y.

[DK] Andrea D'Agnolo and Masaki Kashiwara. "On Quantization of Complex Symplectic Manifolds". In: *Communications in Mathematical Physics* 308.1 (2011), pp. 81–113. DOI: 10.1007/s00220-011-1325-7. URL: https://doi.org/10.1007/s00220-011-1325-7.

[DS] Andrea D'Agnolo and Pierre Schapira. "Quantization of complex Lagrangian submanifolds". In: *Advances in Mathematics* 213.1 (Aug. 2007), pp. 358–379. ISSN: 0001-8708. DOI: 10.1016/j.aim.2006.12.009. URL: http://dx.doi.org/10.1016/j.aim.2006.12.009.

[EO1] Bertrand Eynard and Nicolas Orantin. "Invariants of algebraic curves and topological expansion". In: *Communications in Number Theory and Physics* 1.2 (2007), pp. 347–452. arXiv: math-ph/0702045 [math-ph].

[EO2] Bertrand Eynard and Nicolas Orantin. "Computation of Open Gromov–Witten Invariants for Toric Calabi–Yau 3-Folds by Topological Recursion, a Proof of the BKMP Conjecture". In: *Commun. Math. Phys.* 337.2 (2015), pp. 483–567. DOI: 10.1007/s00220-015-2361-5. arXiv: 1205.1103 [math-ph].

[ES] Tobias Ekholm and Vivek Shende. *Skeins on Branes*. 2025. arXiv: 1901.08027 [math.SG]. URL: https://arxiv.org/abs/1901.08027.

[FL] Bohan Fang and Chiu-Chu Melissa Liu. "Open Gromov-Witten Invariants of Toric Calabi-Yau 3-Folds". In: *Communications in Mathematical Physics* 323.1 (2013), pp. 285–328. DOI: 10.1007/s00220-013-1771-5. URL: https://doi.org/10.1007/s00220-013-1771-5.

[FLYZ] Bohan Fang, Chiu-Chu Melissa Liu, Song Yu, and Zhengyu Zong. "Remodeling conjecture with descendants". In: *arXiv preprint arXiv:2504.15696* (2025).

[FLZ] Bohan Fang, Chiu-Chu Melissa Liu, and Zhengyu Zong. "On the Remodeling Conjecture for Toric Calabi-Yau 3-Orbifolds". In: *J. Am. Math. Soc.* 33.1 (2020), pp. 135–222. DOI: 10.1090/jams/934. arXiv: 1604.07123 [math.AG].

[GGR] Michel van Garrel, Tom Graber, and Helge Ruddat. "Local Gromov–Witten invariants are log invariants". In: *Advances in Mathematics* 350 (2019), pp. 860–876. DOI: 10.1016/j.aim.2019.04.063.

[GHS] Mark Gross, Paul Hacking, and Bernd Siebert. "Theta functions on varieties with effective anti-canonical class". In: *Memoirs of the American Mathematical Society* 278.1367 (2022).

[Gr1] Tim Graefnitz. *Counting (tropical) curves via scattering.sage*. 2022. arXiv: 2210.10455 [math.AG]. URL: https://arxiv.org/abs/2210.10455.

[Gr2] Tim Graefnitz. "Tropical correspondence for smooth del Pezzo log Calabi-Yau pairs". In: *Journal of Algebraic Geometry* 31 (2022), pp. 687–749. DOI: 10.1090/jag/794.

[Gr3] Tim Gräfnitz. "Theta functions, broken lines and 2-marked log Gromov-Witten invariants". In: *manuscripta mathematica* 176.4 (2025), p. 41. DOI: 10.1007/s00229-025-01640-z. URL: https://doi.org/10.1007/s00229-025-01640-z.

[GRS1] Michel van Garrel, Helge Ruddat, and Bernd Siebert. "Intrinsic Enumerative Mirror Symmetry: Takahashi's Log Mirror Symmetry for $(\mathbb{P}^2, E)$ Revisited". In: *Journal of Mathematical Study* (2025). URL: https://api.semanticscholar.org/CorpusID:278959432.

[GRS2] Michel van Garrel, Helge Ruddat, and Bernd Siebert. "Intrinsic local enumerative mirror symmetry". in progress.

[GRZ] Tim Gräfnitz, Helge Ruddat, and Eric Zaslow. "The proper Landau-Ginzburg potential is the open mirror map". In: *Advances in Mathematics* 447 (2024), p. 109639. ISSN: 0001-8708. DOI: https://doi.org/10.1016/j.aim.2024.109639. URL: https://www.sciencedirect.com/science/article/pii/S0001870824001543.

[GRZZ] Tim Gräfnitz, Helge Ruddat, Eric Zaslow, and Benjamin Zhou. "Enumerative geometry of quantum periods". In: *Advances in Mathematics* 499 (2026), p. 111063. ISSN: 0001-8708. DOI: https://doi.org/10.1016/j.aim.2026.111063. URL: https://www.sciencedirect.com/science/article/pii/S0001870826002859.

[GS1] Michel van Garrel and Bernd Siebert. *From wall structures to closed mirror symmetry. The case of $K_{\mathbb{P}^2}$: Renormalized periods over the positive real locus, closed*

Gromov-Witten invariants from wall functions, and tropical enumeration. 2026. arXiv: 2610.04252 [math.AG]. URL: https://arxiv.org/abs/2610.04252.

[GS2] Mark Gross and Bernd Siebert. "From real affine geometry to complex geometry". In: *Annals of Mathematics* 174.3 (2011), pp. 1301–1428. DOI: 10.4007/annals.2011.174.3.1.

[GS3] Mark Gross and Bernd Siebert. "Logarithmic Gromov-Witten invariants". In: *Journal of the American Mathematical Society* 26 (2013), pp. 451–510. DOI: 10.1090/S0894-0347-2012-00757-7. URL: https://www.ams.org/journals/jams/2013-26-02/S0894-0347-2012-00757-7/.

[GS4] Mark Gross and Bernd Siebert. "The canonical wall structure and intrinsic mirror symmetry". In: *Inventiones mathematicae* 229.3 (2022), pp. 1101–1202. DOI: 10.1007/s00222-022-01126-9. URL: https://doi.org/10.1007/s00222-022-01126-9.

[GS5] Mark Gross and Bernd Siebert. "Intrinsic mirror symmetry". In: *Journal of the American Mathematical Society* 39.2 (2026), pp. 313–451. DOI: 10.1090/jams/1066.

[GS6] Sam Gunningham and Pavel Safronov. "DEFORMATION QUANTIZATION AND PERVERSE SHEAVES". In: *Duke Mathematical Journal* 175.6 (2026), pp. 1067–1161. DOI: 10.1215/00127094-2025-0041. URL: https://doi.org/10.1215/00127094-2025-0041.

[HKR] Gerhard P. Hochschild, Bertram Kostant, and Alex Rosenberg. "Differential Forms on Regular Affine Algebras". In: *Transactions of the American Mathematical Society* 102 (1962), pp. 383–408. DOI: 10.1090/S0002-9947-1962-0142598-8.

[HN] Lotte Hollands and Andrew Neitzke. "Exact WKB and Abelianization for the T_3 Equation". In: *Communications in Mathematical Physics* 380.1 (2020), pp. 131–186. DOI: 10.1007/s00220-020-03875-1. URL: https://doi.org/10.1007/s00220-020-03875-1.

[HS] Mingyuan Hu and Vivek Shende. *Skein-valued mirror curves for toric CY3 strips*. 2025. arXiv: 2512.07762 [math.SG]. URL: https://arxiv.org/abs/2512.07762.

[HSZ] Mingyuan Hu, Gus Schrader, and Eric Zaslow. "Skeins, Clusters and Wavefunctions". In: *Advances in Theoretical and Mathematical Physics* 29.6 (2025), pp. 1695–1771. DOI: 10.4310/ATMP.251118002906.

[HV] Kentaro Hori and Cumrun Vafa. *Mirror Symmetry*. 2000. arXiv: hep-th/0002222 [hep-th]. URL: https://arxiv.org/abs/hep-th/0002222.

[HW] Min-xin Huang and Xian-fu Wang. "Topological strings and quantum spectral problems". In: *J. High Energy Phys.* 9 (2014), 150, front matter+46. ISSN: 1126-6708,1029-8479. DOI: 10.1007/JHEP09(2014)150. URL: https://doi.org/10.1007/JHEP09(2014)150.

[IKV] Amer Iqbal, Can Kozçaz, and Cumrun Vafa. "The refined topological vertex". In: *Journal of High Energy Physics* 2009.10 (Oct. 2009), p. 069. DOI: 10.1088/1126-6708/2009/10/069. URL: https://doi.org/10.1088/1126-6708/2009/10/069.

[Ka1] Masaki Kashiwara. "Quantization of contact manifolds". In: *Publ. Res. Inst. Math. Sci.* 32.1 (Mar. 1996), pp. 1–7. ISSN: 0034-5318. DOI: 10.2977/prims/1195163179. URL: https://doi.org/10.2977/prims/1195163179.

[Ka2] Masaki Kashiwara. *D-modules and Microlocal Calculus*. Trans. by Mutsumi Saito. Vol. 217. Translations of Mathematical Monographs. Providence, R.I.: American Mathematical Society, 2003. ISBN: 9780821827666.

[KK] Masaki Kashiwara and Takahiro Kawai. "Microlocal Analysis". In: *Publications of the Research Institute for Mathematical Sciences* 19.3 (1983), pp. 1003–1032. DOI: 10.2977/PRIMS/1195182018.

[KS] Masaki Kashiwara and Pierre Schapira. *Deformation quantization modules*. en. Astérisque 345. Société mathématique de France, 2012. URL: https://www.numdam.org/item/AST_2012__345__R1_0/.

[Li1] Jun Li. "Stable morphisms to singular schemes and relative stable morphisms". In: *Journal of Differential Geometry* 57.3 (2001), pp. 509–578. DOI: 10.4310/jdg/1090348132. arXiv: math/0009097 [math.AG].

[Li2] Jun Li. "A degeneration formula of GW-invariants". In: *Journal of Differential Geometry* 60.2 (2002), pp. 199–293. DOI: 10.4310/jdg/1090351102. arXiv: math/0110113 [math.AG].

[Li3] Yu-Shen Lin. "Open Gromov–Witten Invariants on Elliptic K3 Surfaces and Wall-Crossing". In: *Communications in Mathematical Physics* 349.1 (2017), pp. 109–164. DOI: 10.1007/s00220-016-2754-0. URL: https://doi.org/10.1007/s00220-016-2754-0.

[Li4] Yu-Shen Lin. "Enumerative geometry of del Pezzo surfaces". In: *Transactions of the American Mathematical Society* 377 (2024), pp. 3015–3054. DOI: 10.1090/tran/8817. URL: https://doi.org/10.1090/tran/8817.

[LLL] Siu-Cheong Lau, Tsung-Ju Lee, and Yu-Shen Lin. "On the complex affine structures of SYZ fibration of del Pezzo surfaces". In: *Advances in Theoretical and Mathematical Physics* 26.6 (2022), pp. 1837–1872. DOI: 10.4310/ATMP.2022.v26.n6.a7.

[ML] Fabrizio Del Monte and Pietro Longhi. *Monodromies of Second Order q-difference Equations from the WKB Approximation*. 2024. arXiv: 2406.00175 [math-ph]. URL: https://arxiv.org/abs/2406.00175.

[MT] Davesh Maulik and Yukinobu Toda. "Gopakumar-Vafa invariants via vanishing cycles". In: *Invent. Math.* 213.3 (2018), pp. 1017–1097. ISSN: 0020-9910,1432-1297. DOI: 10.1007/s00222-018-0800-6. URL: https://doi.org/10.1007/s00222-018-0800-6.

[Po1] Pietro Polesello. *Uniqueness of quantization of complex contact manifolds*. 2005. arXiv: math/0512501 [math.QA]. URL: https://arxiv.org/abs/math/0512501.

[Po2] Pietro Polesello. "Classification of deformation quantization algebroids on complex symplectic manifolds". In: *Publications of the Research Institute for Mathematical Sciences* 44.3 (2008), pp. 725–748. DOI: 10.2977/prims/1216238303. URL: https://ems.press/journals/prims/articles/2158.

[PS] Pietro Polesello and Pierre Schapira. "Stacks of quantization-deformation modules on complex symplectic manifolds". In: *International Mathematics Research Notices* 2004.49 (Jan. 2004), pp. 2637–2664. ISSN: 1073-7928. DOI: 10.1155/S1073792804132819. eprint: https://academic.oup.com/imrn/article-pdf/2004/49/2637/2422822/2004-49-2637.pdf. URL: https://doi.org/10.1155/S1073792804132819.

[RS] Helge Ruddat and Bernd Siebert. "Period integrals from wall structures via tropical cycles, canonical coordinates in mirror symmetry and analyticity of toric degenerations". In: *Publ. Math. Inst. Hautes Études Sci.* 132 (2020), pp. 1–82. ISSN: 0073-8301,1618-1913. DOI: 10.1007/s10240-020-00116-y. URL: https://doi.org/10.1007/s10240-020-00116-y.

[SKK] M. Sato, T. Kawai, and M. Kashiwara. *Microfunctions and Pseudo-differential Equations*. Lecture Notes in Math. 287. Springer, 1973, pp. 265–523.

[SSZ] Gus Schrader, Linhui Shen, and Eric Zaslow. "The Chromatic Lagrangian: Wavefunctions and Open Gromov-Witten Conjectures". In: *Advances in Theoretical and Mathematical Physics* 28 (2024), pp. 1781–1879. DOI: 10.4310/ATMP.241031004509.

[Ta] Nobuyoshi Takahashi. "Log Mirror Symmetry and Local Mirror Symmetry". In: *Communications in Mathematical Physics* 220.2 (2001), pp. 293–299. DOI: 10.1007/PL00005567. URL: https://doi.org/10.1007/PL00005567.

[TZ] David Treumann and Eric Zaslow. "Cubic Planar Graphs and Legendrian Surface Theory". In: *Adv. Theor. Math. Phys.* 22 (2018), pp. 1289–1345. DOI: 10.4310/ATMP.2018.v22.n5.a5. arXiv: 1609.04892 [math.SG].

University of Oxford, Mathematical Institute, Oxford, OX2 6GG, UK
Email address: pierrick.bousseau@maths.ox.ac.uk

The University of Texas at Austin, Department of Mathematics, AUSTIN, TX 78712, USA
Email address: siebert@math.utexas.edu

The University of Texas at Austin, Department of Mathematics, AUSTIN, TX 78712, USA
Email address: wangyao@utexas.edu